ClaudeForth

An ANS Forth System for the Motorola 6809

Target: the Minimalist Eurocard Board (MECB), 6809 version, configured with 16kB of ROM.

Author: Richard Rothwell

Date: July 11, 2026

Generated with: Claude Sonnet 5

Design & code generation duration: 14 hours

Documentation generation duration: 4 hours

Duration addressing AI generated task list, debugging, assembler fixes, memory map resolution: 16 hours

Table of Contents

1. Introduction

1.1 Overview

ClaudeForth is an ANS Forth implementation targeting the Motorola 6809 microprocessor. It was designed and built incrementally, decision by decision, across a single extended conversation: the inner-interpreter model was chosen first, then the cell size, character set, memory map, and every word in the Core and Core Extension word sets, followed by double-precision arithmetic, string handling, exception handling, and the Tools word set. This document describes the resulting architecture, the reasoning behind its major design choices, and provides a complete glossary of every word implemented.

1.2 Inner-Interpreter Model: Subroutine Threading

ClaudeForth uses subroutine-threaded code (STC) rather than indirect or direct threading. A colon definition compiles as a flat sequence of 6809 JSR instructions terminated by a compiled RTS. This means the 6809's own JSR/RTS mechanism is the inner interpreter — there is no separate NEXT dispatch loop for straight-line execution. This was chosen because the 6809's hardware stack (register S) is naturally suited to serving as both the CPU's call stack and Forth's return stack simultaneously, and because STC is generally the fastest threading model available on this processor, at the cost of slightly larger compiled code than indirect threading would produce (each call is 3 bytes — a JSR opcode plus a 16-bit address — rather than a 2-byte threaded pointer).

1.3 Register Assignment

U is the data (parameter) stack pointer, manipulated with PSHU/PULU. S is the return stack pointer, using the CPU's native hardware stack directly, so JSR/RTS work as the inner interpreter with no extra bookkeeping. X and Y are general-purpose working registers, used throughout for scanning, indexing, and address computation. D (the combined A:B accumulator) holds top-of-stack values in flight and is the primary vehicle for 16-bit arithmetic, matching the cell size described below.

1.4 Cell Size and Character Set

One cell is 16 bits (two bytes), matching the 6809's native D, X, Y, U, and S registers exactly. This gives a single-cell signed range of -32768 to 32767, unsigned 0 to 65535, and means a cell can always hold a full address across the 6809's entire 64K address space. Because the 6809 has no alignment restriction, ALIGN and ALIGNED are true no-ops. Characters are single-byte 7-bit ASCII (values 0-127), so CHARS and CHAR+ reduce to a no-op and a plain +1 respectively, while CELLS requires an actual multiply-by-two.

1.5 The Three-Region Memory Split

The single most distinctive architectural decision in ClaudeForth is that dictionary headers, compiled code, and user data are tracked as three independent, separately growing regions, each with its own allocation pointer (DPHERE, CODEHERE, and VARHERE respectively) rather than the single unified dictionary most Forth systems use. This departs from the ANS assumption of one HERE; in this system HERE reports CODEHERE specifically, and a parallel word family (V, / VC, / VALLOT / VHERE) exists to manage the variable region directly. The practical consequence is a genuine distinction between immutable data (CONSTANT values, compiled literals, and string literals, all living in code space) and mutable data (VARIABLE, VALUE, and BUFFER: contents, living in variable space) that most single-region Forths do not enforce architecturally.

The intent behind this split is to provide for a ROMable basic Forth system: BASECODE and BASEDICT (Section 2) hold the entire Core and Core Extension implementation as fixed, burned-in ROM content, needing no RAM at all to boot to a working interactive prompt. The RAM regions — APPCODE, APPDICT, and APPVARS — exist specifically to support interactive user application development on top of that fixed base: new words are compiled, tested, and revised freely in RAM without touching the ROM image underneath them. Once an application is complete, the same compiled RAM content (APPCODE and APPDICT together) is intended to be burned into an expanded ROM alongside the base system, turning a finished, debugged application into a second fixed, non-volatile layer rather than something that must be re-compiled from source into RAM on every power-up.

A feature under consideration, not yet designed or implemented, is the ability to strip the dictionary — discarding header (name, LINK, flags) information for some or all ROM or application words once development is finished, while leaving their code intact and callable. A stripped system would trade FIND-ability and interactive extensibility (WORDS, ', and ordinary text interpretation of a word by name) for a smaller ROM footprint and a correspondingly larger RAM variable area, since the dictionary and variable regions currently share the same fixed page-zero and BASEDICT budgets described in Section 2.

1.6 Exception Handling

CATCH and THROW are implemented as direct manipulation of the return stack (register S) rather than a linked list of handler records: CATCH saves the current data-stack pointer and the previous HANDLER value as a small frame on S, and a non-local THROW unwinds directly to that frame with a single TFR instruction, discarding every intervening call level in one step. QUIT wraps INTERPRET in a CATCH and, on a caught error, rolls back all three region pointers (and LATEST) to their state at the start of the failing input line, reclaiming space that an uncaught error would otherwise abandon.

1.7 What This Document Covers

Section 2 gives the full memory map. Section 3 is the glossary of every word implemented, organized by topic, each with its stack-effect notation and any side effects. Section 4 covers five practical development topics: serial I/O configuration and handshaking, a brief introduction to Forth syntax and stack operations, worked examples of the Tools word set, dictionary management with MARKER, and why the Search-Order word set is not included. Section 5 is a checklist of every item still open, resolved, or deliberately deferred during the build. Section 6 covers build instructions: which files to assemble and the assembler this source targets. Section 8 is the complete assembler source listing, including the ROM base dictionary (Section 8.28). Section 9 is an alphabetical index of every word with a page reference back into the glossary.

2. Memory Map

64K address space, addresses shown high to low. Three independent allocation pointers (DPHERE, CODEHERE, VARHERE) track the dictionary, code, and mutable-variable regions respectively; none of the three boundaries below is enforced by a runtime bounds check.

Address Range	Size/bytes	Region	Contents
$FFF0–$FFFF	16	**VECTORS**	Hardware vectors
$FFA2–$FFEF	78	**INITCODE**	COLDSTRT / WARM entry code - was INIT ($FFC0, then $FFA0)
$DFF4–$FFA1	8110	**BASECODE**	ROM primitives and compiled system words, 8110-byte nominal budget — ends exactly one byte below INITCODE's start
$D83F–$DFF3	1973	**BASEDICT**	ROM dictionary headers, exact fit (1973 bytes, zero padding)
$C000–$C0FF	256	**INOUT**	Input/output device block, sits directly below USROMSTRT; 6850 ACIA registers live at INOUT+8, not the block base
$BD00–$BFFF	768	**RSTACK**	Return stack (S), grows downward — exactly contiguous with DSTACK below, no gap
$B900–$BCFF	1024	**DSTACK**	Data stack (U), grows downward — occupied bottom matches CODETOP exactly
$7000–$B8FF	18688	**APPCODE**	Application code (CODEHERE) — nominal ceiling (CODETOP) correctly bounded by DSTACK's true range
$2000–$6EA4	20133	**APPDICT**	Application dictionary (DPHERE), 20133 bytes
$021B–$1FFF	7653	**APPVARS**	Mutable variable space (VARHERE), 7653 bytes usable — APPVARSEND self-adjusts to APPDICT-1
$01FB–$021A	32	**SIBUF**	S" interpretation-mode string buffer
$01DA–$01FA	33	**WORDBUF**	WORD's counted-string scratch buffer
$018A–$01D9	80	**TIBBUF**	Terminal input buffer (80 bytes)
$014A–$0189	64	**OUTBUF**	Output ring buffer (64 bytes)
$010A–$0149	64	**INBUF**	Input ring buffer (64 bytes)
$0106–$0109	4	**SERBUF idx**	Serial ring buffer head/tail indices
$0100–$0105	6	**MVSCRATCH**	Shared scratch: MVCNT/MVDST/MVSRC, aliased by FILLCNT/HSLEN, FILLADDR/HSADDR, MRESULT/FILLCHR
$0000–$00FF	256	**GLOBALS**	Shared system globals — Direct Page = $00

ROM Size Required

VECTORS, INITCODE, BASECODE, and BASEDICT are the four regions that must actually live in ROM; everything else in the table is RAM or the ACIA hardware itself, memory-mapped rather than stored. USROMSTRT ($C100) and USROMEND ($FFEF, one address before VECTORS' start) define the outer boundary these four must fall within, and all four are genuinely contained within it, checked numerically. A full pairwise sweep across every region in the memory map, not just these four, confirms zero overlaps anywhere: BASECODE's nominal end ($FFA1) lands exactly one byte below INITCODE's start ($FFA2).

Using the most precise count available (BASECODE's real size confirmed by a full instruction-by-instruction pass at 7984 bytes), VECTORS + INITCODE + BASECODE + BASEDICT comes to 10051 bytes (about 9.82 KB), which is why the ROM part is a 16K×8 EPROM (a 27128 or equivalent) rather than an 8K×8 part — comfortable headroom for real usage.

The INOUT block shadows 256 bytes in the ROM block: a 16K×8 EPROM occupying the natural $C000–$FFFF footprint that size implies would include $C000–$C0FF, but INOUT claims that same 256-byte range for the ACIA and other memory-mapped I/O. This is why USROMSTRT starts at $C100, not $C000 — the bottom 256 bytes of the chip's raw address footprint are never available for ROM content at all, regardless of how much of the chip is otherwise unused.

ROM unused byte count: a 16K×8 EPROM (16384 bytes) minus INOUT's 256-byte shadow minus the 10051 bytes of real VECTORS + INITCODE + BASECODE + BASEDICT content leaves 6077 bytes genuinely unused — about 37% of the chip's total capacity still free. This figure inherits the same caveat as the content totals it's built from: BASECODE's contribution is a careful manual count, not a real assembler's output, so treat 6077 as the best current estimate rather than a guaranteed figure.

Dictionary Entry Layout

Every word in BASEDICT (and every application word compiled into APPDICT) shares this same four-field layout, packed with no padding, in this order:

Code	Size	Name	Comments
LEN/FL	`1 byte`	Length + Flags	bit 7 = IMMEDIATE, bit 6 = SMUDGE, bits 4-0 = name length (0-31)
NAME	`n bytes`	Name Field	n raw ASCII characters, no terminator - n comes from LEN/FL's low bits
LINK	`2 bytes`	Link Field	address of the previous header in the chain, or 0 if this is the oldest entry
CFA	`2 bytes`	Code Field Address	address FIND returns - the word's own code, or a DODOES trampoline for a CREATEd word

3. Glossary

Words are grouped by topic below. Each entry gives the stack-effect notation, a short description, and any side effects beyond the stack transformation shown.

3.1 System / Console I/O

KEY
`( -- char )`
Blocking read of one character from the input ring buffer.
Side effects: *Spins with interrupts enabled until a character is available.*

KEY?
`( -- flag )`
Non-blocking test for a waiting input character.
Side effects: *Read-only; never blocks.*

EMIT
`( char -- )`
Queue one character for transmission.
Side effects: *Enables the ACIA transmit interrupt if not already on.*

ACCEPT
`( addr n -- n' )`
Line-edit up to n characters into addr; returns actual count.
Side effects: *Echoes input; handles backspace/delete.*

EXPECT
`( addr n -- )`
Obsolescent line input; same editing as ACCEPT.
Side effects: *Reports its count through SPAN rather than the stack.*

QUERY
`( -- )`
Refill the terminal input buffer and reset >IN.
Side effects: *Sets SRCADDR/SRCLEN/SRCID for the terminal source.*

TYPE
`( addr len -- )`
Output len characters starting at addr.
Side effects: *None.*

CR
`( -- )`
Output a newline (CR then LF).
Side effects: *None.*

SPACE
`( -- )`
Output one space character.
Side effects: *None.*

SPACES
`( n -- )`

Output n spaces.
Side effects: *No output if n <= 0.*

ABORT
`( -- ) ( R: ... -- )`
Clear the data stack and enter QUIT. Never returns.
Side effects: *Full return-stack reset; never executes RTS.*

QUIT
`( -- ) ( R: ... -- )`
The outer interpreter's top-level loop. Never returns.
Side effects: *Resets the return stack; wraps INTERPRET in CATCH.*

3.2 Stack Manipulation — Data Stack

DUP
`( n -- n n )`
Duplicate the top stack item.
Side effects: *None.*

DROP
`( n -- )`
Discard the top stack item.
Side effects: *None.*

SWAP
`( n1 n2 -- n2 n1 )`
Exchange the top two items.
Side effects: *None.*

OVER
`( n1 n2 -- n1 n2 n1 )`
Copy the second item to the top.
Side effects: *None.*

ROT
`( n1 n2 n3 -- n2 n3 n1 )`
Rotate the third item to the top.
Side effects: *None.*

?DUP
`( n -- n n | 0 )`
Duplicate top only if it is nonzero.
Side effects: *None.*

DEPTH
`( -- n )`
Push the number of items on the data stack.
Side effects: *Computed relative to SP0.*

2DUP
`( x1 x2 -- x1 x2 x1 x2 )`
Duplicate the top cell pair.
Side effects: *None.*

2DROP
`( x1 x2 -- )`
Discard the top cell pair.
Side effects: *None.*

2SWAP
`( x1 x2 x3 x4 -- x3 x4 x1 x2 )`
Exchange the top two cell pairs.
Side effects: *None.*

2OVER
`( x1 x2 x3 x4 -- x1 x2 x3 x4 x1 x2 )`
Copy the second cell pair to the top.
Side effects: *None.*

NIP
`( x1 x2 -- x2 )`
Discard the second item.
Side effects: *None.*

TUCK
`( x1 x2 -- x2 x1 x2 )`
Copy the top item under the second.
Side effects: *None.*

PICK
`( xu ... x0 u -- xu ... x0 xu )`
Copy the item u cells deep to the top.
Side effects: *No bounds check against actual stack depth.*

ROLL
`( xu ... x0 u -- xu-1 ... x0 xu )`
Remove the item u cells deep and move it to the top.
Side effects: *Cost scales with u; no bounds check.*

2ROT
`( x1..x6 -- x3 x4 x5 x6 x1 x2 )`
Rotate the third cell pair to the top.
Side effects: *None.*

3.3 Stack Manipulation — Return Stack

>R
`( x -- ) ( R: -- x )`
Move top of data stack to the return stack.
Side effects: *Must be balanced by R> within the same definition.*

R>
`( -- x ) ( R: x -- )`
Move top of return stack to the data stack.
Side effects: *None.*

R@
`( -- x ) ( R: x -- x )`
Copy top of return stack to the data stack.

Side effects: *Non-destructive.*

2>R

`( x1 x2 -- ) ( R: -- x1 x2 )`

Move top cell pair to the return stack.

Side effects: *Must be balanced by 2R> in the same definition.*

2R>

`( -- x1 x2 ) ( R: x1 x2 -- )`

Move top cell pair back to the data stack.

Side effects: *None.*

2R@

`( -- x1 x2 ) ( R: x1 x2 -- x1 x2 )`

Copy top cell pair from the return stack.

Side effects: *Non-destructive.*

3.4 Arithmetic — Single Cell

+

`( n1 n2 -- sum )`

Add.

Side effects: *None.*

-

`( n1 n2 -- diff )`

Subtract n2 from n1.

Side effects: *None.*

*****

`( n1 n2 -- prod )`

Signed multiply, truncated to one cell.

Side effects: *None.*

/

`( n1 n2 -- quot )`

Signed symmetric division.

Side effects: *Throws -10 if n2 is zero.*

MOD

`( n1 n2 -- rem )`

Signed symmetric remainder.

Side effects: *Throws -10 if n2 is zero.*

/MOD

`( n1 n2 -- rem quot )`

Quotient and remainder together.

Side effects: *Throws -10 if n2 is zero.*

NEGATE

`( n -- -n )`

Two's-complement negate.

Side effects: *None.*

ABS

`( n -- |n| )`
Absolute value.
Side effects: *None.*

`MIN`
`( n1 n2 -- min )`
Signed minimum.
Side effects: *None.*

`MAX`
`( n1 n2 -- max )`
Signed maximum.
Side effects: *None.*

`1+`
`( n -- n+1 )`
Add one.
Side effects: *None.*

`1-`
`( n -- n-1 )`
Subtract one.
Side effects: *None.*

`2+`
`( n -- n+2 )`
Add two. Not an ANS-standard word.
Side effects: *None.*

`2*`
`( n -- n*2 )`
Arithmetic shift left one bit.
Side effects: *None.*

`2/`
`( n -- n/2 )`
Arithmetic shift right one bit, sign-preserving.
Side effects: *None.*

`*/`
`( n1 n2 n3 -- quot )`
Multiply then divide via a double-cell intermediate.
Side effects: *Throws -10 if n3 is zero.*

`*/MOD`
`( n1 n2 n3 -- rem quot )`
As */ but yields remainder and quotient.
Side effects: *Throws -10 if n3 is zero.*

3.5 Arithmetic — Mixed & Double Precision

`UM*`
`( u1 u2 -- ud )`
Unsigned single by single multiply, double result.
Side effects: *None.*

UM/MOD
`( ud u1 -- u2 u3 )`
Unsigned double by single divide.
Side effects: *Throws -10 if u1 is zero.*

M*
`( n1 n2 -- d )`
Signed single by single multiply, double result.
Side effects: *None.*

FM/MOD
`( d n1 -- n2 n3 )`
Floored double by single division.
Side effects: *Throws -10 if n1 is zero.*

SM/REM
`( d n1 -- n2 n3 )`
Symmetric double by single division.
Side effects: *Throws -10 if n1 is zero.*

D+
`( d1 d2 -- d3 )`
Double-cell add.
Side effects: *None.*

D-
`( d1 d2 -- d3 )`
Double-cell subtract.
Side effects: *None.*

DNEGATE
`( d1 -- d2 )`
Double-cell two's-complement negate.
Side effects: *None.*

DABS
`( d1 -- d2 )`
Double-cell absolute value.
Side effects: *None.*

M+
`( d1 n -- d2 )`
Add a single-cell value into a double.
Side effects: *None.*

S>D
`( n -- d )`
Sign-extend a single cell to double.
Side effects: *None.*

D>S
`( d -- n )`
Narrow a double to a single cell.
Side effects: *Implementation-defined if d does not fit; the high cell is dropped.*

DMAX
`( d1 d2 -- d3 )`
Double-cell signed maximum.
Side effects: *None.*

DMIN
`( d1 d2 -- d3 )`
Double-cell signed minimum.
Side effects: *None.*

3.6 Logic, Shifts & Address Arithmetic

AND
`( n1 n2 -- n1&n2 )`
Bitwise AND.
Side effects: *None.*

OR
`( n1 n2 -- n1|n2 )`
Bitwise OR.
Side effects: *None.*

XOR
`( n1 n2 -- n1^n2 )`
Bitwise exclusive OR.
Side effects: *None.*

INVERT
`( n -- ~n )`
Ones'-complement.
Side effects: *None.*

LSHIFT
`( x1 u -- x2 )`
Logical shift left u bits, zero-fill.
Side effects: *Cost scales with u; no early-out for large u.*

RSHIFT
`( x1 u -- x2 )`
Logical shift right u bits, zero-fill.
Side effects: *Cost scales with u; no early-out for large u.*

CELLS
`( n -- n*2 )`
Convert a cell count to a byte offset.
Side effects: *None.*

CELL+
`( addr -- addr+2 )`
Add the size of one cell.
Side effects: *None.*

CHARS
`( n -- n )`
Convert a character count to a byte offset. No-op on this system.

Side effects: None.

CHAR+
`( addr -- addr+1 )`
Add the size of one character.
Side effects: None.

ALIGN
`( -- )`
Align HERE to a cell boundary. No-op on the 6809.
Side effects: None.

ALIGNED
`( addr -- addr )`
Align a given address. No-op on the 6809.
Side effects: None.

3.7 Comparison

=
`( n1 n2 -- flag )`
True if equal.
Side effects: None.

<
`( n1 n2 -- flag )`
True if n1 signed less than n2.
Side effects: None.

>
`( n1 n2 -- flag )`
True if n1 signed greater than n2.
Side effects: None.

0=
`( n -- flag )`
True if n is zero.
Side effects: None.

0<
`( n -- flag )`
True if n is negative.
Side effects: None.

U<
`( u1 u2 -- flag )`
True if u1 unsigned less than u2.
Side effects: None.

<>
`( n1 n2 -- flag )`
True if not equal.
Side effects: None.

0<>

`( n -- flag )`
True if n is not zero.
Side effects: *None.*

0>
`( n -- flag )`
True if n is greater than zero.
Side effects: *None.*

U>
`( u1 u2 -- flag )`
True if u1 unsigned greater than u2.
Side effects: *None.*

WITHIN
`( n1 n2 n3 -- flag )`
True if n2 <= n1 < n3.
Side effects: *Computed via unsigned comparison of offsets, handles wraparound ranges correctly.*

D=
`( d1 d2 -- flag )`
Double-cell equal.
Side effects: *None.*

D<
`( d1 d2 -- flag )`
Double-cell signed less than.
Side effects: *Compares high cells (signed) first, low cells (unsigned) only if equal.*

DU<
`( d1 d2 -- flag )`
Double-cell unsigned less than.
Side effects: *Both tiers compared unsigned.*

3.8 Control Flow — Conditionals & Loops

IF
`Compile: ( -- ) Run: ( flag -- )`
Begin a conditional. IMMEDIATE, compile-only.
Side effects: *Compiles ZBRANCH and a forward-patch field.*

THEN
`( -- )`
Resolve an IF or ELSE forward branch. IMMEDIATE, compile-only.
Side effects: *Throws -22 if the control-flow tag does not match.*

ELSE
`( -- )`
Alternate branch of IF. IMMEDIATE, compile-only.
Side effects: *Throws -22 if the control-flow tag does not match.*

BEGIN
`( -- )`
Mark a loop target. IMMEDIATE, compile-only.
Side effects: *None.*

UNTIL

```
( flag -- )
```

Branch back to BEGIN while flag is false. IMMEDIATE, compile-only.

Side effects: *Throws -22 on a tag mismatch.*

AGAIN

```
( -- )
```

Unconditional branch back to BEGIN. IMMEDIATE, compile-only.

Side effects: *Throws -22 on a tag mismatch.*

WHILE

```
( flag -- )
```

Conditional exit within a BEGIN loop. IMMEDIATE, compile-only.

Side effects: *Leaves a second pending forward-patch field.*

REPEAT

```
( -- )
```

Close a BEGIN…WHILE loop. IMMEDIATE, compile-only.

Side effects: *Patches both the back-edge and WHILE's forward exit.*

RECURSE

```
( -- )
```

Compile a call to the word currently being defined. IMMEDIATE, compile-only.

Side effects: *Bypasses FIND's smudge check; reads LATEST's own CFA directly.*

DO

```
Compile: ( -- ) Run: ( n1|u1 n2|u2 -- ) ( R: -- loop-sys )
```

Begin a counted loop. IMMEDIATE, compile-only.

Side effects: *Builds a 4-cell loop frame on the return stack; runs at least once even if limit=index.*

?DO

```
Compile: ( -- ) Run: ( n1|u1 n2|u2 -- ) ( R: -- loop-sys )
```

As DO, but skips the loop entirely if limit=index. IMMEDIATE, compile-only.

Side effects: *Builds the same loop frame; adds a forward skip branch.*

LOOP

```
( -- ) ( R: loop-sys -- | loop-sys )
```

Increment index by 1; loop until index reaches limit. IMMEDIATE, compile-only.

Side effects: *Discards the loop frame on exit.*

+LOOP

```
( n -- ) ( R: loop-sys -- | loop-sys )
```

Increment index by n; exits when the step crosses limit. IMMEDIATE, compile-only.

Side effects: *Handles positive or negative step; discards the frame on exit.*

I

```
( -- n ) ( R: loop-sys -- loop-sys )
```

Fetch the innermost loop's index.

Side effects: *Non-destructive read of the current loop frame.*

J

```
( -- n ) ( R: loop-sys1 loop-sys2 -- loop-sys1 loop-sys2 )
```

Fetch the next-outer loop's index.

Side effects: *Only reaches one level of nesting out; no J2 equivalent exists.*

LEAVE

```
( -- ) ( R: loop-sys -- loop-sys )
```

Force the innermost loop to exit at its next LOOP/+LOOP.

Side effects: *Assumes it is textually inside the loop it affects; unverified across an intervening word call.*

UNLOOP

```
( -- ) ( R: loop-sys -- )
```

No-op. Previously discarded the innermost loop frame directly; EXIT's own frame-unwinding (via a compile-time count of enclosing DO/?DO frames) already discards it correctly and automatically on every path, including with no enclosing loop at all. Calling UNLOOP immediately before EXIT used to discard the frame twice - once here, once more in EXIT's own unwind - corrupting the return stack. UNLOOP now does nothing, so both "EXIT" alone and "UNLOOP EXIT" are safe.

Side effects: none. Kept in the dictionary for source compatibility with code written for systems where EXIT does not auto-unwind and UNLOOP is genuinely required first.

EXIT

```
( -- )
```

Return from the current definition immediately. IMMEDIATE, compile-only in the sense that it must be used inside a definition.

Side effects: Counts and discards any open DO/?DO frames at compile time before compiling its own return. This unwinding is automatic and correct on its own - UNLOOP is not required beforehand (see UNLOOP, above, which is a no-op for exactly this reason).

CASE

```
( -- )
```

Begin a multi-way selection. IMMEDIATE, compile-only.

Side effects: *None.*

OF

```
( x n -- | x )
```

Begin one CASE clause's test. IMMEDIATE, compile-only.

Side effects: *Drops the selector only on a matching clause.*

ENDOF

```
( -- )
```

End one CASE clause. IMMEDIATE, compile-only.

Side effects: *Throws -22 on a tag mismatch.*

ENDCASE

```
( x -- )
```

Close a CASE structure, discarding the selector.

Side effects: *Throws -22 if the control-flow stack is malformed.*

3.9 Defining Words

:

```
( "name" -- )
```

Begin a colon definition.

Side effects: *Header is smudged until ; snapshot control-flow depth for later checking.*

;

```
( -- )
```

End a colon definition. IMMEDIATE.

Side effects: *Unsmudges the header; throws -22 if a control structure is left open.*

CREATE
`( "name" -- )`
Build a header and a DOES>-ready trampoline.
Side effects: *PFA defaults to CODEHERE (immutable space).*

DOES>
`( -- )`
Attach runtime behavior to a CREATEd word. IMMEDIATE, compile-only.
Side effects: *Patches the trampoline's BEHAVIOR field at run time via a double return.*

VARIABLE
`( "name" -- )`
Create a one-cell mutable variable.
Side effects: *PFA reserved in VARHERE (mutable space); initialized to zero.*

CONSTANT
`( x "name" -- )`
Create a word that returns a fixed value.
Side effects: *Value stored in CODEHERE (immutable space).*

VALUE
`( x "name" -- )`
Create a reassignable named value.
Side effects: *PFA in VARHERE (mutable space); reassigned via TO.*

TO
`( x "name" -- )`
Assign a new value to a VALUE. IMMEDIATE.
Side effects: *Behaves differently interpreting vs. compiling.*

2VARIABLE
`( "name" -- )`
Create a two-cell mutable variable.
Side effects: *PFA in VARHERE; both cells initialized to zero.*

2CONSTANT
`( x1 x2 "name" -- )`
Create a word returning a fixed double value.
Side effects: *Stores x1 at the lower address, x2 at the higher, matching 2@/2!.*

BUFFER:
`( u "name" -- )`
Create a word returning the address of u reserved, uninitialized bytes.
Side effects: *PFA in VARHERE.*

DEFER
`( "name" -- )`
Create a word whose action can be set later.
Side effects: *Default action throws -21 until IS/DEFER! is used.*

DEFER@
`( xt-defer -- xt )`
Fetch a deferred word's current target.
Side effects: *None.*

DEFER!
`( xt xt-defer -- )`
Set a deferred word's target.
Side effects: *None.*

IS
`( xt "name" -- )`
Set a deferred word's target by name. IMMEDIATE.
Side effects: *Behaves differently interpreting vs. compiling.*

ACTION-OF
`( "name" -- xt )`
Fetch a deferred word's current target by name. IMMEDIATE.
Side effects: *Behaves differently interpreting vs. compiling.*

MARKER
`( "name" -- )`
Create a word that, when executed, forgets itself and everything defined after it.
Side effects: *Restores all three HERE pointers and LATEST from a snapshot.*

3.10 Compiling Words

IMMEDIATE
`( -- )`
Mark the most recently defined word as IMMEDIATE.
Side effects: *Sets a header flag bit.*

STATE
`( -- addr )`
Variable: nonzero while compiling.
Side effects: *None.*

[
`( -- )`
Enter interpret state. IMMEDIATE.
Side effects: *None.*

]
`( -- )`
Enter compile state.
Side effects: *None.*

'
`( "name" -- xt )`
Look up a word's execution token.
Side effects: *Throws -13 if not found. Works identically interpreting or compiling.*

COMPILE,
`( xt -- )`
Compile a call to the given execution token.
Side effects: *None.*

LITERAL
`( x -- )`
Compile x as a literal. IMMEDIATE, compile-only.

Side effects: *None.*

[']

```
( "name" -- )
```

Compile-time tick: compile a word's xt as a literal. IMMEDIATE, compile-only.

Side effects: *Throws -14 if used while interpreting; -13 if the name is not found.*

POSTPONE

```
( "name" -- )
```

Defer a word's compiling behavior into the current definition. IMMEDIATE, compile-only.

Side effects: *Throws -14 if used while interpreting; -13 if the name is not found.*

>BODY

```
( xt -- pfa )
```

Recover a CREATEd word's parameter field address.

Side effects: *Assumes the standard trampoline layout; fixed offset arithmetic.*

EXECUTE

```
( xt -- )
```

Execute the word with the given execution token.

Side effects: *None.*

SLITERAL

```
( addr len -- )
```

Compile a runtime string as a literal. IMMEDIATE, compile-only.

Side effects: *Throws -14 if used while interpreting.*

ABORT"

```
Compile: ( "msg" -- ) Run: ( flag -- )
```

Conditionally type a message and abort. IMMEDIATE, compile-only.

Side effects: *On true flag: types the message, throws -2.*

3.11 Memory

@

```
( addr -- x )
```

Fetch a cell.

Side effects: *None.*

!

```
( x addr -- )
```

Store a cell.

Side effects: *None.*

C@

```
( addr -- char )
```

Fetch a byte, zero-extended.

Side effects: *None.*

C!

```
( char addr -- )
```

Store the low byte only.

Side effects: *None.*

+!

`( n addr -- )`
Add n into the cell at addr.
Side effects: *None.*

2@
`( addr -- x1 x2 )`
Fetch a double cell.
Side effects: *Lower address is x1, higher address is x2.*

2!
`( x1 x2 addr -- )`
Store a double cell.
Side effects: *x1 stored at the lower address, x2 at the higher.*

,
`( n -- )`
Append one cell to the immutable/code region (CODEHERE).
Side effects: *Advances CODEHERE.*

C,
`( char -- )`
Append one byte to the immutable/code region.
Side effects: *Advances CODEHERE.*

ALLOT
`( n -- )`
Reserve (or release, if negative) bytes in the immutable/code region.
Side effects: *No bounds check against adjoining regions.*

HERE
`( -- addr )`
Current allocation point of the immutable/code region.
Side effects: *Reports CODEHERE.*

V,
`( n -- )`
Append one cell to the mutable/variable region (VARHERE).
Side effects: *Advances VARHERE.*

VC,
`( char -- )`
Append one byte to the mutable/variable region.
Side effects: *Advances VARHERE.*

VALLOT
`( n -- )`
Reserve (or release) bytes in the mutable/variable region.
Side effects: *No bounds check against adjoining regions.*

VHERE
`( -- addr )`
Current allocation point of the mutable/variable region.
Side effects: *Reports VARHERE.*

PAD

```
( -- addr )
```

A scratch buffer address, offset above CODEHERE by PADMINSIZE (84 characters, matching the ANS minimum). Floats relative to CODEHERE, not a separately reserved region - its address changes as CODEHERE advances.

***Side effects:** Also used by interpreted S" (see S", below) for its own string storage, per ANS's allowance for non-standard words to use PAD as long as the use is documented. This means PAD and interpreted S" text now share the same space - using PAD directly right after an interpreted S" call (or vice versa) will overwrite one with the other.*

UNUSED

```
( -- u )
```

Bytes remaining in the region HERE addresses.

***Side effects:** Reports distance to the return/data-stack boundary.*

VUNUSED

```
( -- u )
```

Bytes remaining in the mutable/variable region.

***Side effects:** Reports distance to APPCODE's start; not itself enforced.*

MOVE

```
( src dst u -- )
```

Copy u bytes, overlap-safe.

***Side effects:** Chooses direction automatically based on address order.*

FILL

```
( addr u char -- )
```

Fill u bytes with char.

***Side effects:** None.*

ERASE

```
( addr u -- )
```

Fill u bytes with zero.

***Side effects:** None.*

CMOVE

```
( src dst u -- )
```

Copy u bytes low address to high.

***Side effects:** Not overlap-safe in the high-over-low direction.*

CMOVE>

```
( src dst u -- )
```

Copy u bytes high address to low.

***Side effects:** Not overlap-safe in the low-over-high direction.*

3.12 Strings & Parsing

COUNT

```
( c-addr -- addr len )
```

Convert a counted string to address/length form.

***Side effects:** None.*

WORD

```
( char -- c-addr )
```

Parse a delimited token into a counted string.

***Side effects:** Hard 31-character cap; skips leading delimiters.*

CHAR

`( "name" -- char )`

Parse the next word, leave its first character.

Side effects: *Inherits WORD's 31-character cap (irrelevant here).*

[CHAR]

`( "name" -- )`

Compile-time CHAR: compiles the character as a literal. IMMEDIATE, compile-only.

Side effects: *Throws -14 if used while interpreting.*

PARSE

`( char "ccc<char>" -- addr len )`

Parse a delimited token, no length cap.

Side effects: *Does not skip leading delimiters, unlike WORD.*

PARSE-NAME

`( "<spaces>name<space>" -- addr len )`

Parse a space-delimited token, no length cap.

Side effects: *Skips leading spaces.*

S"

`Compile: ( "str" -- ) Run: ( -- addr len )`

Parse and compile, or stage, a string literal.

Side effects: Compiling: appends inline to CODEHERE - the definition's own permanent storage, unaffected by anything below. Interpreting: copies into PAD (up to PADMINSIZE - 84 - characters, up from a fixed 32-character buffer previously). PAD's address is computed fresh from CODEHERE on every call, but if nothing is compiled or allocated between two interpreted S" calls, CODEHERE hasn't moved, so PAD hasn't either - a second S" call before the first string is actually used will still silently overwrite it. For any use needing more than one string to coexist (e.g. REPLACES's two arguments), wrap each in its own colon definition (: name S" text" ;) instead - compiled S" never touches PAD at all, giving each string its own permanent storage.

."

`( "str" -- )`

Compile a string that types itself at run time. IMMEDIATE, compile-only.

Side effects: *No interpretation semantics, per ANS.*

COMPARE

`( addr1 len1 addr2 len2 -- n )`

Compare two strings, signed three-way result.

Side effects: *None.*

SEARCH

`( addr1 len1 addr2 len2 -- addr3 len3 flag )`

Find a substring within a string.

Side effects: *Empty needle reports not-found (an implementation choice).*

-TRAILING

`( addr len1 -- addr len2 )`

Trim trailing spaces from a string's reported length.

Side effects: *None.*

/STRING

`( addr1 len1 n -- addr2 len2 )`

Trim n characters from the front of a string.

Side effects: *None.*

REPLACES

```
( addr1 len1 addr2 len2 -- )
```

Register a name/value substitution pair for SUBSTITUTE - addr1/len1 is the replacement text, addr2/len2 is the substitution name (matched later between % delimiters).

Side effects: Single-slot only, not a table; a second call overwrites the first. Both strings must come from stable storage, not from S" typed directly at the interpreter prompt - interpreted S" writes into PAD (see PAD and S", above), whose address only changes when something is compiled or allocated in between, so a later S" call before an earlier string is actually used (including the one for SUBSTITUTE's own source string) will still silently overwrite it. Wrap each string in its own colon definition (: name S" text" ;) for stable, independent storage instead - compiled S" never touches PAD.

SUBSTITUTE

```
( addr1 len1 addr2 len2 -- addr2 len3 n )
```

Scan addr1/len1 for text between '%' delimiter pairs. A pair enclosing the name registered via REPLACES has the whole %name% span - delimiters included - replaced by the registered substitution text; %% collapses to a single %; an unmatched name, or a trailing % with no closing delimiter, is passed through unchanged. Result is stored at addr2, length len3.

Side effects: n is the number of substitutions made (0 or positive on success). Throws -1 on destination overflow; single-slot only (see REPLACES).

SNAME

```
( xt -- addr len )
```

Reverse dictionary lookup: xt to name. Non-standard.

Side effects: *Returns zero length if not found.*

UNESCAPE

```
( addr len -- addr len' )
```

Process backslash escapes in place. Non-standard.

Side effects: *Result length is always <= input length.*

3.13 Numeric Output

<#

```
( -- )
```

Begin pictured numeric output conversion.

Side effects: *Sets HLD to PAD's address.*

#

```
( ud1 -- ud2 )
```

Convert one digit.

Side effects: *Uses the current BASE.*

#S

```
( ud1 -- ud2 )
```

Convert remaining digits until the value is zero.

Side effects: *Uses the current BASE.*

#>

```
( ud -- addr len )
```

End pictured conversion, discarding ud.

Side effects: *None.*

HOLD

```
( char -- )
```

Insert one character into the pictured output buffer.

Side effects: *Buffer grows downward from PAD; no bound enforced.*

HOLDS
`( addr len -- )`
Insert an entire string into the pictured output buffer.
Side effects: *Depends on HOLD's exact decrement-by-one behavior.*

SIGN
`( n -- )`
Insert a minus sign if n is negative.
Side effects: *None.*

.
`( n -- )`
Print signed, trailing space.
Side effects: *None.*

U.
`( u -- )`
Print unsigned, trailing space.
Side effects: *None.*

.R
`( n width -- )`
Print signed, right-justified in a field.
Side effects: *No trailing space; width is a minimum, not a cap.*

U.R
`( u width -- )`
Print unsigned, right-justified in a field.
Side effects: *No trailing space; width is a minimum, not a cap.*

?
`( addr -- )`
Fetch and print the cell at addr, signed.
Side effects: *None.*

D.
`( d -- )`
Print signed double, trailing space.
Side effects: *None.*

D.R
`( d width -- )`
Print signed double, right-justified.
Side effects: *No trailing space.*

3.14 Base / Radix Control

BASE
`( -- addr )`
Variable holding the current number base.
Side effects: *None.*

DECIMAL
`( -- )`
Set BASE to 10.

Side effects: *None.*

HEX

`( -- )`

Set BASE to 16.

Side effects: *None.*

BINARY

`( -- )`

Set BASE to 2. Not an ANS-standard word.

Side effects: *None.*

3.15 Exception Handling

CATCH

`( xt -- 0 | n )`

Execute xt, trapping any THROW.

Side effects: *Restores data stack depth and HANDLER on either path.*

THROW

`( n -- )`

Raise exception n, or do nothing if n is zero.

Side effects: *Nonzero: non-local exit to the matching CATCH, or JMP ABORT if none is active.*

3.16 Comments

(

`( "ccc<paren>" -- )`

Discard input up to the next close paren. IMMEDIATE.

Side effects: *Works interpreting or compiling.*

****

`( "ccc<eol>" -- )`

Discard the remainder of the current input line. IMMEDIATE.

Side effects: *Follows SOURCE, so it operates correctly inside EVALUATE.*

3.17 Environmental & System Queries

ENVIRONMENT?

`( addr len -- false | value true )`

Query implementation-defined characteristics.

Side effects: *Table is incomplete: missing /HOLD, /PAD, MAX-D, MAX-UD, WORDLISTS, FLOORED.*

SOURCE

`( -- addr len )`

The current input source.

Side effects: *Follows terminal input or an active EVALUATE string.*

SOURCE-ID

`( -- 0 | -1 )`

Identify the current input source. 0 = terminal, -1 = string.

Side effects: *None.*

REFILL

`( -- flag )`

Attempt to refill the input source.
Side effects: *Fails (false) if the current source is a string, per ANS.*

EVALUATE
`( addr len -- )`
Interpret a string as Forth source.
Side effects: *Saves and restores the whole input-source state around the call.*

TIB
`( -- addr )`
The fixed terminal input buffer address.
Side effects: *Always the terminal buffer, unaffected by EVALUATE.*

#TIB
`( -- addr )`
Variable: count of valid characters in TIB.
Side effects: *Always the terminal buffer's count.*

>IN
`( -- addr )`
Variable: scan offset into the current input source.
Side effects: *None.*

SPAN
`( -- addr )`
Variable: EXPECT's returned character count.
Side effects: *None.*

BL
`( -- char )`
The blank/space character, 32.
Side effects: *None.*

3.18 Tools

.S
`( -- )`
Print the data stack, top to bottom.
Side effects: *Non-destructive.*

WORDS
`( -- )`
List every word in the dictionary.
Side effects: *Walks the full ROM+RAM chain from LATEST.*

DUMP
`( addr u -- )`
Hex and ASCII memory dump, 16 bytes per line.
Side effects: *None.*

4. Forth Development Notes

This section is written for the audience who will program the completed system: users developing a Forth application on top of ClaudeForth, not those building or modifying the system itself. It assumes the base system already runs and focuses on what's needed to work productively within it day to day.

This section covers five practical aspects of working with the system that fall outside the word-by-word glossary: how serial communication is configured, a brief introduction to Forth syntax and stack operations, worked examples of the Tools word set, how to reclaim dictionary space during interactive development using MARKER, and why the optional Search-Order word set is not part of this implementation.

4.1 Serial I/O and Handshaking

Serial communication parameters — baud-rate divisor, word length, parity, and stop bits — are fixed by the hardware, not selectable from Forth. The 6850 ACIA's control register is written once, at COLDSTRT, to a constant value (CR_RXON, or CR_RXTX when the output ring has data queued) encoding a ÷16 clock, 8 data bits, no parity, 1 stop bit, and receive interrupts enabled. There is no word in this system that changes these settings at run time; the terminal or host at the other end of the link must be configured to match this fixed set of parameters independently.

Hardware (RTS/CTS) handshaking is implemented; software (XON/XOFF) is not.

RTS — this system's output telling the remote device when to pause sending — is toggled automatically based on the input ring buffer's fill level. IRQH's receive path calls RTSCHECKHI after storing each incoming byte; once the ring reaches INHIWATER (48 of its 64 bytes), RTS is driven high by writing CR_RTSHI to the ACIA control register, and RTS stays high until KEY, on the mainline side, drains the ring back down to INLOWATER (16 bytes) and calls RTSCHECKLO to drop it low again. The 32-byte gap between the two marks is deliberate hysteresis, avoiding RTS toggling right at a single threshold.

CTS — the remote device's own flow-control signal telling this system when it may transmit — needs no firmware logic at all: the 6850 ACIA inhibits TDRE in hardware whenever CTS is deasserted, and every transmit path in this system (EMIT, IRQH's TXCHK) already gates on TDRE, so CTS is honored automatically with no code changes required.

One real ACIA constraint shapes this design directly: the 6850's control register ties RTS state and transmit-interrupt-enable to the same two bits, so RTS-high and TX-interrupt-enabled can never be selected simultaneously. While RTS is asserted high, EMIT still queues outgoing bytes into the output ring but deliberately does not enable the transmit interrupt to drain them — RTSCHECKLO re-enables it once RTS drops low, restoring CR_RXTX if output is still queued at that point. A second, genuine race is worth naming rather than glossing over: RTSCHECKHI runs from inside the interrupt handler, where hardware has already masked further interrupts, but RTSCHECKLO runs from ordinary mainline code (KEY) and could otherwise be interrupted mid-decision by IRQH's own TXOFF path, which also writes the control register — RTSCHECKLO masks IRQ around its own critical section specifically to prevent that.

Software (XON/XOFF) handshaking remains unimplemented: this system still neither transmits XON/XOFF nor recognizes those bytes if a remote device sends them — an incoming $11 or $13 is treated as an ordinary character and queued like any other. On a genuine 3-wire serial link with no RTS/CTS wiring at all, that link would still have no flow control of any kind; the hardware path described above only helps when RTS/CTS are physically connected.

4.2 Forth Syntax and Stack Operations

Forth has no operators, no operator precedence, and no infix expressions. A line of input is simply a sequence of whitespace-delimited words, read and acted on strictly left to right. Each word is either a number, which is pushed onto the data stack, or a name that is looked up in the dictionary and executed immediately. Arithmetic and every other operation work by taking their arguments off the top of the stack and pushing their result back onto it — this is postfix, or reverse Polish, notation: the operator always comes after its operands, never between or before them.

A numerical example:

Computing (3 + 4) × 2 — 14 — is written:

```
3 4 + 2 * .
14   ok
```

Read left to right, tracking the data stack after each word (shown deepest item first, top of stack last):

```
3       ->  ( 3 )             push 3
4       ->  ( 3 4 )           push 4
+       ->  ( 7 )             pop 4 and 3, push their sum
2       ->  ( 7 2 )           push 2
*       ->  ( 14 )            pop 2 and 7, push their product
.       ->  ( )               pop and print: 14
```

The stack-effect notation used throughout the Glossary (Section 3) — for example, +'s (n1 n2 -- sum) — describes exactly this: the items expected on the stack before a word runs, to the left of "--", and what it leaves in their place afterward, to the right. Reading a word's stack effect this way is enough to know how to use it correctly without needing to know anything about how it is implemented internally.

This system does not require building up a vocabulary from nothing: a large set of predefined words — arithmetic, stack manipulation, comparison, control flow, string and memory operations, and more — already exists and is documented word by word in the Glossary. At the interactive prompt, executing WORDS lists every one of these preexisting words currently in the dictionary, from the most recently defined word back through every built-in primitive. See Section 4.3 for a worked example of WORDS and the other Tools words.

4.3 Tools Word Set: Explanation and Examples

The three Tools words in the Glossary — .S, WORDS, and DUMP — are interactive aids rather than words a program typically calls itself. The examples below show plausible session transcripts. One real characteristic of this system worth knowing before reading them: output from a word is written immediately, with no leading newline of its own — the newline seen before "ok" in a normal session comes from QUIT, not from the word that ran. The transcripts below add a line break before each result for readability; on the actual wire, output from .S, WORDS, or DUMP appears immediately after the typed command, on the same line.

.S — print the data stack, top to bottom, non-destructively

Pushes nothing and removes nothing; it only reads the stack from the current top down to SP0 and prints each cell with the same signed formatting as DOT. This makes it safe to run at any point without disturbing whatever a definition is part-way through building on the stack.

```
1 2 3 .S
3 2 1   ok

\ top of stack (3, pushed last) prints first, deepest
\ item (1, pushed first) prints last
```

WORDS — list every word in the dictionary

Walks the LATEST chain from the most recently defined application word all the way back through every ROM primitive, printing each name separated by a space. With roughly 150 application-level words already in the Glossary plus every ROM-resident primitive behind them, real output is long — the excerpt below shows the shape of it rather than a complete transcript.

```
WORDS
SQUARE FENCE VALUE TO VARIABLE CONSTANT DOES>
CREATE ... DUP DROP SWAP OVER ROT DEPTH ...
+ - * / MOD ... IF THEN ELSE BEGIN ... ok

\ most-recently-defined application words appear first,
\ since the walk follows LATEST backward through LINK;
\ ROM primitives make up the remainder of the list
```

DUMP — hex and ASCII memory dump, 16 bytes per line

Takes an address and a byte count, and prints that many bytes, 16 per line: each byte as two hex digits, then a matching ASCII column with unprintable bytes shown as a period. Worth noting explicitly since it differs from many DUMP implementations: this one does not print a leading address on each line — only the hex bytes and their ASCII rendering — so the caller must track which address a given line corresponds to separately if that matters.

```
HEX
7000 10 DUMP
BD E4 32 8E 00 00 39 00 00 00 00 00 00 00 00 00  ........9.......

\ 16 bytes shown starting at $7000 (the start of APPCODE);
\ actual bytes depend entirely on what has been compiled
\ into code space by the time DUMP is run
```

4.4 Dictionary Management with MARKER

ANS Forth provides no general "undefine" word; instead, the standard's mechanism for reclaiming dictionary space is MARKER. Executing MARKER with a name creates an ordinary dictionary entry that, when later executed itself, forgets its own definition and every definition made after it — restoring the dictionary to exactly the state it was in at the moment MARKER was invoked. This is the standard way to experiment interactively without permanently accumulating discarded or superseded definitions.

In this implementation, a word created by MARKER stores a snapshot of all three region pointers — DPHERE (dictionary), CODEHERE (code), and VARHERE (mutable variable space) — plus LATEST, at the moment of its own creation. Executing the marker word (DOMARKER) simply writes those four saved values back, in one step. Nothing needs to be erased or unwound word by word: every allocation word in the system (comma, ALLOT, HEADER, and their VARHERE counterparts) only ever appends at the current pointer, so rolling the pointer back alone is sufficient to make everything defined afterward unreachable and its space available again for reuse.

Example:

```
MARKER FENCE
: SQUARE  DUP * ;
5 SQUARE .            \ prints 25

FENCE                 \ forgets FENCE itself, and SQUARE

5 SQUARE .            \ throws -13, undefined word: SQUARE
```

MARKER FENCE creates FENCE and captures the current DPHERE/CODEHERE/VARHERE/LATEST as its own PFA contents. SQUARE is then compiled normally, advancing all three pointers past that snapshot. Executing FENCE restores the four saved values, which puts LATEST back to whatever it was immediately before FENCE was defined — so SQUARE is no longer reachable by FIND, and FENCE itself is gone too, along with the dictionary, code, and variable space both of them occupied. A second MARKER call after this point would need a new name, since FENCE no longer exists to be reused.

Caveat: *as noted in the glossary, a MARKER's snapshot only covers the single dictionary chain this system currently maintains. It is not defined against the optional Search-Order word set, which this implementation does not include — see Section 4.5.*

4.5 The Search-Order Word Set Is Not Included

ANS Forth's optional Search-Order word set (FORTH-WORDLIST, WORDLIST, GET-CURRENT, SET-CURRENT, GET-ORDER, SET-ORDER, ALSO, ONLY, PREVIOUS, DEFINITIONS, and SEARCH-WORDLIST) is not implemented in this system. Every one of the roughly 150 words in the Glossary is resolved through a single, unified dictionary chain rooted at LATEST — there is only ever one wordlist, and new definitions always link into it. Search-Order's whole purpose is letting a program maintain several independent wordlists (for example, to keep an application's vocabulary separate from the words used only to build it) and to search a chosen, ordered subset of them; without it, every word ever defined is always visible to FIND, in one flat namespace, for the lifetime of the system.

This was a foundational decision, not an oversight discovered late: the single-chain design was fixed from the very first system-initialization code, where LATEST is set to the top of the ROM dictionary at cold start and every application word links onto that same chain afterward. Adding Search-Order now would not be a small addition — it would touch code that already exists throughout the system, not just introduce new words. FIND itself would need restructuring into an outer loop over an active search order, trying each wordlist in turn; HEADER, which every defining word (colon definitions, CREATE, VARIABLE, CONSTANT, VALUE, and the rest) already funnels through, would need to link new definitions into a selectable "current" wordlist rather than unconditionally into LATEST; and a few words that deliberately read LATEST directly today — RECURSE, the unsmudging step in semicolon, and WORDS — would each need the same substitution or they would silently keep operating on the wrong wordlist once more than one existed. MARKER's snapshot would also need to widen to cover every wordlist's head cell, not just LATEST, with the further complication that ANS itself leaves a marker's interaction with wordlists created after it as implementation-defined, not fully specified.

In practice, this means every program written against this system lives in one shared, flat namespace — there is no mechanism to hide a helper word's name from being visible everywhere else, and no way to have two differently-scoped words share the same name. For the scope of a single-user, single-application embedded system, this is a reasonable simplification rather than a defect; it is flagged here specifically so it is not mistaken for an accidental gap in ANS coverage.

5. Open Items Checklist

Everything below was explicitly flagged during the build as incomplete, unverified, or deliberately deferred. Nothing here is a surprise — each item was named at the point it came up. This is a consolidated list to work from, not a new set of findings. Regenerated to reflect fixes and design decisions made since the original version.

5.1 Known bugs — resolved since the original checklist

☑ **DUMP's partial-final-line bug** — fixed via DUVALID tracking (`25_tools_word_set.asm`). The ASCII column no longer reads past the valid bytes on a short final line.

☑ **SUBSTITUTE** — completed. It now performs a real bounds-checked copy (prefix / replacement / suffix) via a shared SUBCOPY helper, reusing SEARCHW for the match. Still scoped to a single registered name/value pair, not a full table — see "Open design questions" below; that scope limit is a deliberate simplification, not a bug.

☑ **VALUE's PFA** — moved from CODEHERE (immutable space) to VARHERE (mutable space), matching VARIABLE's pattern, so every TO-driven update now writes into the correct region instead of into code space. Not on the original checklist (found and fixed after this list was first generated); logged here for the record.

☑ **Every scratch/global cell now has a real RMB-assigned address in page zero** (`00_memory_map_and_globals.asm`), applied rather than left as a placeholder. Two real findings came out of doing this: the budget is **253 of 256 bytes used — only 3 bytes of headroom** in the GLOBALS page, and SNEND (documented in the original placeholder list) turned out to be genuinely dead — never read or written anywhere — and was dropped rather than given an address. Any future scratch cell added to this system will need to fit in that remaining 3 bytes or the page-zero fast-addressing property (DP = $00) stops covering it.

☑ **The ROM base dictionary is now real** (`27_forth_dictionary.asm`) — every primitive with actual code got a real header, chained via LINK, CFA pointing straight at its code label. ABORTHDR's LINK field, a placeholder 0 since it was first built, is now resolved to the chain's newest entry. Building this surfaced two real findings, not just mechanical work: the original 1024-byte BASEDICT could not hold the header table (ultimately 1954 bytes needed, once DOES> was added — see below), so it was resized to 2048 bytes, taking the space from BASECODE (now 6080 bytes, was 7104) — a resize based purely on the header-table budget, not on any actual measurement of BASECODE's assembled size, which has never been checked with a real 6809 assembler. Generating this table also surfaced that **DOES> had no corresponding code anywhere in the file** — resolved in a follow-up pass, see below.

☑ **DOES> — resolved.** SETDOES (the patching runtime) was added beside DODOES/DOESRT0, and DOES> itself (code label DOESGT, since a literal ">" is not a valid 6809 assembler label) was added right after CREATE. DOESBEH was added to the GLOBALS layout to support it, using 2 of the page's last 3 free bytes — only 1 byte of headroom now remains in page zero. DOES> also has a real dictionary header now (H_DOESGT, the chain's newest entry); ABORTHDR's LINK was updated again to point at it.

☑ **Four internal loop-label names were reused across unrelated routines: FLOOP, UDLOOP, DRPOS, CMLOOP.** Resolved — each pair renamed to a distinct name scoped to its own routine: FIND's FLOOP → FFLOOP; FILLW's FLOOP → FILLOOP; UDIV16's UDLOOP → UD16LOOP; UDDIGIT's

UDLOOP → UDDLOOP; DIVCOMMON's DRPOS → DCRPOS; DDOTR's DRPOS → DDRPOS; CMOVEW's CMLOOP → CMVLOOP; COMPAREW's CMLOOP → CMPLOOP. Verified zero duplicate labels and zero stray references to any of the old names remain anywhere in the file.

✅ **Hardware (RTS) serial handshaking — implemented.** IRQH's receive path and KEY now toggle RTS based on the input ring's fill level (INFILL, RTSCHECKHI, RTSCHECKLO against INHIWATER=48/INLOWATER=16, with hysteresis between them) via a new CR_RTSHI control byte and RTSSTATE flag — the last free byte in the GLOBALS page, which is now fully packed at 256/256. CTS (the other handshaking direction) needed no firmware logic at all: confirmed against the 6850 datasheet that TDRE is automatically inhibited while CTS is deasserted, so this system's existing TDRE-gated transmit logic already respects it. A real race was identified and closed: mainline code (KEY, via RTSCHECKLO) and the ISR (IRQH's TXOFF) can both decide to write the control register, so RTSCHECKLO masks IRQ around its critical section and TXOFF checks RTSSTATE before writing. Software (XON/XOFF) handshaking remains unimplemented — see below.

✅ **TRUE and FALSE — implemented.** Surfaced while organizing the ANS test suite: neither existed as an actual dictionary word, only as the internal TRUEV/FALSEV assembler constants. Added as `CONSTANT TRUE -1` / `CONSTANT FALSE 0` (TRUEBODY/ FALSEBODY, section 26; headers H_TRUE/H_FALSE, chain's two newest entries, section 27) — the first CONSTANT-pattern ROM-resident words in this system. Every other ROM word's CFA is a plain code label; a CONSTANT's CFA is the DODOES-trampoline pattern instead, built here with fixed, assemble-time addresses rather than a CODEHERE snapshot, since there is no interactive CREATE/CONSTANT phase for ROM content.

✅ **One genuine 6309-only instruction (TSTD) was in use — fixed.** Found by auditing every mnemonic in the file against the real 6809 instruction set, rather than trusting the Build Instructions section's earlier claim that no 6309 extensions were used (that claim was wrong until this fix). TRYNUM (section 9) used TSTD with no operand to test whether D was zero after `PULU D` — PULU/PULS don't affect condition codes on genuine 6809 hardware, unlike a load, so this relied on a 6309-only convenience instruction. Replaced with `CMPD #0`, a real 6809 instruction with identical behavior for this purpose. A full mnemonic audit after the fix found zero remaining non-6809 instructions anywhere in the file.

✅ **BASELATEST was an undefined symbol — a real assembly-breaking bug, not a placeholder.** COLD (section 4) has referenced `LDD #BASELATEST` to initialize LATEST since this file's earliest version, and the SECTION 27 header comment has long asserted "BASELATEST remains QUITHDR" - but no EQU or label actually named BASELATEST existed anywhere in the file. This would have failed to assemble outright. Fixed by adding `BASELATEST EQU QUITHDR` directly after QUITHDR's own definition (section 26), matching what the comment always said it should equal. Verified: exactly one definition, no duplicate symbols, and COLD's forward reference to it resolves under standard two-pass assembly.

✅ **JSR CR (bare, undefined) appeared at five call sites — a real assembly-breaking bug, not a naming inconsistency.** The actual routine has always been labeled CRW (section 22), following this file's own convention of giving short/symbolic ANS names a distinct code label. ABORT, QUIT's error-report path, QUIT's ok-prompt path, WORDS (WWDONE), and DUMP (DULEND) all called the bare, undefined CR instead. Fixed at each site to `JSR CRW`, preserving each call site's original spacing exactly. Verified: CRW defined exactly once, referenced 7 times total, zero remaining bare-CR instruction operands anywhere in the file (the two harmless bare "CR" occurrences that remain are a

section-title comment and the dictionary header's FCC "CR" name string, neither of which is a bug).

☑ **Two ALU instructions used invalid register-to-register syntax (ADDA B, EORA B) — the 6809 has no such addressing mode, so the assembler read B as an undefined direct-page symbol.** Found in the single-cell multiply routine (ADDA B, twice) and DOPLUSTEST's sign comparison for +LOOP (EORA B). Fixed with the standard 6809 idiom for combining two registers: PSHS B followed by ADDA ,S+ / EORA ,S+ (push B, then operate through the auto-incrementing stack-indexed operand). Verified: a full sweep of every ALU instruction (ADDA/ADDB/SUBA/SUBB/ ANDA/ANDB/ORA/ORB/EORA/EORB/CMPA/CMPB/ADCA/ ADCB/SBCA/SBCB/BITA/BITB) against a bare A/B operand found zero remaining instances; every other bare B in the file (in STA/LDA/LEAX ...,X and PSHS B) is genuine, valid 6809 accumulator-offset indexed addressing or register-list syntax.

☑ **Nine scratch symbols (MRESULT, MVCNT, MVDST, MVSRC, FILLCHR, FILLCNT, FILLADDR, HSLEN, HSADDR) were used throughout MOVE/CMOVE/CMOVE>, FILL, HOLDS, and the single-cell multiply routine but never declared anywhere — a real, assembly-breaking gap.** Auditing every one of the 142 already-declared GLOBALS cells for actual use found zero dead cells to reclaim this time (unlike SNEND earlier), and the GLOBALS page was independently confirmed at exactly 256/256 bytes with no headroom. Since MOVE-family, FILL, HOLDS, and the multiply routine never call each other or run concurrently in this single-threaded interpreter, they now share physical storage: three new cells (MVCNT, MVDST, MVSRC) hold the real storage, and FILLCNT/HSLEN alias MVCNT, FILLADDR/HSADDR alias MVDST, and MRESULT/FILLCHR alias MVSRC, all via EQU. These three new cells could not fit in the full GLOBALS page, so they live at $0100, carved from the front of USER0 (shrunk from 128 to 122 bytes, now starting at $0106) — a real, documented tradeoff: these three cells use ordinary extended addressing, not direct-page, including inside CMOVEW's per-byte copy loop.

☑ **JSR SPACE (bare, undefined) — same class of bug as CR, fixed the same way.** WORDS called the bare, undefined SPACE instead of the actual routine label SPACEW (section 22). Fixed to JSR SPACEW. Checked SPACES for the same pattern (none found — SPACESW correctly calls EMIT directly) and swept the file for any other bare SPACE instruction reference (none remain; the two harmless bare "SPACE" occurrences left are a section-title comment and the dictionary header's FCC "SPACE" name string).

☑ **Three short branches (BHS, BEQ, BNE) overflowed the 8-bit short-branch range (±127 bytes) — a real assembler error, not a style issue.** SUBCOPY's overflow check (BHS SUBOVERFLOW, ~87 source lines to target), DUMPW's per-line loop test (BEQ DUDONE, ~76 lines), and DUMPW's loop-back (BNE DULINE, ~76 lines) all spanned too much code for an 8-bit displacement. Fixed by converting each to its long-branch equivalent (LBHS/LBEQ/LBNE), which uses a 16-bit displacement (±32767 bytes) - functionally identical, just not range-limited.

□ **Ten more short branches have a large (41-51 source line) span to their target and were not individually confirmed safe.** Found by a heuristic sweep (line-count as a rough proxy for byte distance, since no real assembler is available in this environment to get an exact count) after fixing the three confirmed overflows above, all of which spanned 76+ lines - these ten are meaningfully shorter but not verified within range: QUERY's QLOOP (three branches, lines 579/582/589), FIND's NOTFOUND/FFLOOP (lines 1210/1256), ACCEPT's ALOOP/ADONE (lines 1501/1462), UNESCAPE's UEDONE/UELOOP (lines 3888/3931), and EMPTY (line 1153). None of these were reported as failing

and none have been changed; if a real assembler flags any of them, the fix is the same: convert to the matching LBxx long-branch form.

☑ **`ORG BASECODE` was missing entirely — a fundamental placement bug, not a cosmetic gap.** VECTORS (`$FFF0`) and `INIT` (`$FFC0`) both had their own `ORG`, but `SECTION 3` (`IRQH`) — and every routine through `SECTION 26`, i.e. nearly the entire interpreter — had no `ORG` placing it in BASECODE at all. Without it, this code would have continued growing from wherever `INIT`'s WARM message left the location counter, inside `INIT`'s own 48-byte `$FFC0-$FFEF` budget, overflowing directly into VECTORS rather than landing in BASECODE (`$E800-$FFBF`) anywhere. Fixed by adding `ORG BASECODE` immediately before `SECTION 3` begins.

□ **Adding the missing ORG makes BASECODE's byte budget concretely checkable for the first time — and a rough estimate suggests it may not fit.** This was previously flagged only as "unverified" (no real assembler has ever been run against this file); with a real ORG boundary now in place, a heuristic per-instruction byte count (inherent/direct-page/indexed/ immediate/extended opcode sizes, correctly distinguishing direct-page GLOBALS operands from extended ones) over every instruction from `SECTION 3` through `SECTION 26` estimates roughly **8660 bytes against a 6080-byte budget — about 2580 bytes over**. This is a rough approximation, not a real assembler's output, and could be wrong in either direction, but it's a strong enough signal to treat as a real, likely problem rather than a formality. Not resolved here — resizing BASECODE/BASEDICT again would cascade into the memory map, the documentation, and the SVG diagram, and a real LWASM pass would give a far more reliable number than this estimate before deciding how much room is actually needed.

☑ **`USER0` and `USER1` (250 bytes of unallocated reserve space, never read or written by any code) were deleted, and the region from `MVSCRATCH` through `APPDICT` was made fully contiguous.** `SERBUF`, `INBUF`, `OUTBUF`, `TIBBUF`, `WORDBUF`, and `SIBUF` all shifted down to sit immediately after `MVSCRATCH` (`$0106`) with no gaps between any of them. This also closed a separate, pre-existing 11-byte gap between `WORDBUF` and `SIBUF` (`$02F5-$02FF`) that had nothing to do with `USER0`/`USER1` but was caught while verifying the region was genuinely contiguous end to end. `APPDICT` was extended downward to close the resulting gap too (new start `$021B`, was `$0320`) — its end (`$6EFF`) is unchanged, so this is a pure 261-byte gain in application dictionary space, not a resize of its budget. This was an interpretation of "contiguous," not something explicitly asked for beyond the 6 buffers themselves; flagged here in case a fixed `APPDICT` start was actually intended instead. Verified: zero duplicate symbols, zero remaining references to `USER0`/`USER1` anywhere in the code, and the full `GLOBALS->MVSCRATCH->buffers->APPDICT` span checked programmatically for zero gaps.

☑ **`APPVARS` and `APPDICT` swapped positions - `APPVARS` now sits below `APPDICT`.** `APPDICT` moved from `$021B` to `$031B` (still ending at `$6FFF`, directly below APPCODE); APPVARS moved from `$6F00` down to `$021B` (same 256-byte size, now sitting directly above the buffers instead of directly below APPCODE). Checked before making the change: only two code references exist for either symbol (`COLD` initializing VARHERE/DPHERE to each region's start) and neither depends on their relative order, so the swap was safe. Verified: zero duplicate symbols, the whole region from `GLOBALS` through APPCODE's start remains contiguous with zero gaps, dictionary chain unaffected (219 entries, since this change doesn't touch it at all).

☑ **`APPVARS` grown from 256 to 8000 bytes, taking the space directly from `APPDICT`.** APPVARS now spans `$021B-$215A` (start address unchanged); `APPDICT` shrank by the same 7744 bytes to `$215B-$6FFF` (end unchanged, still directly below APPCODE), giving 20133 bytes of application dictionary space, down from 27877. Checked before making the change: still only two code

references to either symbol (COLD's VARHERE/ DPHERE initialization), neither size-dependent. Verified: zero duplicate symbols, APPVARS size is exactly 8000 bytes, the whole region stays contiguous end to end, dictionary chain unaffected.

✅ **ACIA (the 256-byte memory-mapped I/O block) renamed to INOUT, and the actual 6850 ACIA chip's registers moved to INOUT+8 instead of the block's base.** The block still spans the same 256 bytes ($DF00-$DFFF); it's now modeled as a general I/O region with the ACIA chip occupying one small part of it (ACIACR/ACIASR = INOUT+8 = $DF08, and the data register ACIADR at $DF09), leaving INOUT+0..INOUT+7 free for other memory-mapped devices sharing the block. Doing this rename surfaced a real, previously-hidden bug: COLDSTRT's ACIA reset sequence used STA ACIA (the block's base) instead of STA ACIACR (the control register) in two places - this only ever worked because ACIA and ACIACR happened to be the same address in the old scheme. Once ACIACR moved to INOUT+8, writing to the bare block base would have silently stopped resetting the ACIA at all. Fixed both to STA ACIACR. Verified: INOUT defined once, the bare ACIA symbol (at the time) no longer existed anywhere, and every other ACIA register access in the file already used the correctly-named register symbols rather than the bare block name.

✅ **ACIA reintroduced as an explicit base symbol (ACIA EQU INOUT+8), with ACIACR/ACIASR now defined relative to it (EQU ACIA) rather than directly to INOUT+8.** The data register was briefly renamed to ACIRDR in the same pass, then reverted back to ACIADR immediately afterward once flagged as a typo - both code sites (IRQH's receive and transmit paths) were updated each time the name changed. Verified: each of ACIA/ACIACR/ACIASR/ACIADR defined exactly once, resolving to the expected values (ACIA = INOUT+8, ACIACR/ACIASR = ACIA, ACIADR = ACIA+1), zero remaining references to ACIRDR anywhere, zero duplicate symbols, dictionary chain unaffected.

✅ **INITEND/INITSIZE landmark constants added at the true end of the INIT block** (INITEND EQU *, INITSIZE EQU *-COLDSTRT), right after WARMMSGL. The request referenced a COLDSTART label, which doesn't exist in this file - used the actual existing label COLDSTRT instead of introducing a new undefined symbol.

✅ **INITEND's comment now states the invariant explicitly ("value should match vector ORG") - and checking it confirms it currently holds.** The precise instruction-by-instruction byte count from when the VECTORS collision was first found (78 bytes exactly, COLDSTRT through WARMMSGL) gives INITCODE start ($FFA2) + 78 = $FFF0, exactly matching VECTORS' ORG. Zero gap, zero overlap - unlike the still-open INITCODE/BASECODE overlap at the other end of this same region, which remains a real problem. Same caveat as always: this is a manual count, not a real assembler's output, so INITSIZE (computed automatically at assembly time) is the figure to trust once this file is actually assembled.

✅ **BASECODESTRT/BASECODEEND/BASECODESIZE and BASEDICTSTRT/BASEDICTEND/BASEDICTSIZE landmark constants added, matching the INITEND/INITSIZE pattern - and checking the two stated invariants gives one clean confirmation and one confirmation of an already-known problem.** BASEDICTEND ($D85D + the exact 1973-byte dictionary content = $E012) matches ORG BASECODE ($E012) precisely - this boundary is genuinely correct, not just close. BASECODEEND, using the same rough heuristic estimate flagged much earlier (~8660 bytes, never confirmed with a real assembler) starting from BASECODESTRT ($E012), lands around $101E6 - not only failing to match ORG INITCODE ($FFA2) as the comment expects, but overflowing past $FFFF entirely on that estimate. The exact amount of room actually available between BASECODESTRT and INITCODE ($FFA2 - $E012 = 8080

bytes) is itself less than the ~8660-byte estimate, meaning even filling every available byte up to INITCODE with zero gap would still likely fall short. This is the same underlying problem already tracked elsewhere (BASECODE possibly not fitting its budget, and separately the INITCODE/BASECODE overlap), now independently reachable via these new landmarks once the file is actually assembled, rather than a new, different issue.

- □ **Follow-up: a real instruction-by-instruction count (not the rough heuristic above) puts BASECODE's actual size at 7984 bytes ($1F30), 96 bytes (1.2%) short of a target assembler- listing value of $1F90 (8080 bytes) given for comparison.** Built a mnemonic-and-addressing-mode-aware counter distinguishing inherent/immediate/direct/extended/indexed forms, correct prefix requirements ($10/$11 for LDY/STY/LDS/STS/CMPD/ CMPY/CMPU/CMPS), and true direct-page symbols (only $0000-$00FF, matching DP) - a meaningfully more rigorous pass than the original ~8660-byte estimate. Every one of the 3751 instructions in the region was classified (zero "unrecognized" remaining after fixing early misses like LSLB/ LSLA as ASLB/ASLA synonyms). Checked for likely sources of the remaining 96-byte gap before accepting it: zero indexed operands use an offset outside the 5-bit range that would need extra encoding bytes (all 125 numeric-offset operands found are small, e.g. 1,X/-1,Y), and the two apparent "indirect [ ]" matches were a false positive (a section-title comment, not real indirect addressing). The target value $1F90 is notable in its own right: BASECODE + $1F90 = $FFA2 exactly, meaning if the real assembled size actually matched it, BASECODE and INITCODE would be perfectly contiguous with zero gap and zero overlap - which would also resolve the separate, still-open INITCODE/BASECODE overlap noted elsewhere. My count doesn't confirm that, though it's close (1.2% off). This is still not a real assembler's output - the standing caveat throughout this file - and the gap could come from encoding edge cases a manual model can approximate but not guarantee against.

- ☑ **ROMSTRT/ROMEND added as the outer ROM boundary ($C100/VECTORS-1), and INITCODE/BASECODE/BASEDICT verified contained within it - all three pass.** ROMEND was initially defined as $10000 ("one past VECTORS's end," a symbolic value that doesn't fit as a real 16-bit address), then corrected to VECTORS-1 ($FFEF, one before VECTORS's start) - a genuine 16-bit address usable directly in comparisons or as a memory operand, not just symbolic arithmetic. Re-checked after the correction rather than assumed still valid: INITCODE ($FFA2-$FFEF), BASECODE ($E012-$FFBF), and BASEDICT ($D85D-$E011) are all still genuinely within ROMSTRT..ROMEND - INITCODE's own end now lands exactly on the new boundary, a tighter and more meaningful check than before, not a violation. This containment check passes cleanly, independent of the other open problems below.

- ☑ **ROMSTRT/ROMEND renamed to USROMSTRT/USROMEND, each with a short "Usable ROM start"/"Usable ROM end" comment.** Cosmetic, not functional - neither symbol was referenced by any code, only by nearby comments (three sites, all updated: the pair's own definitions and INOUT's cross-reference). Verified: zero remaining bare ROMSTRT/ROMEND references anywhere, zero duplicate symbols, dictionary chain unaffected.

- ☑ **VECTOREND/VECTORSIZE landmark constants added at the true end of the vector table, matching the INITEND/INITSIZE pattern - and checking the stated invariant confirms it exactly, not approximately.** Unlike INITEND/BASECODEEND (which needed a manual, approximate instruction-by-instruction byte count since real 6809 instructions have variable-length encoding), the vector table is pure FDB data with a fixed, unambiguous 2-byte width per entry - counting the 8 entries (VRESV through VRESET) gives exactly 16 bytes, matching the comment's stated $10 expectation precisely. Verified: zero duplicate symbols, dictionary chain unaffected.

✅ **BASECODESTRT removed as requested - but BASECODESIZE depended on it, so removing it alone would have left a genuine dangling undefined-symbol reference, the same bug class found and fixed several times earlier in this file.** Confirmed first that BASECODESTRT was numerically identical to BASECODE itself (only comments sit between `ORG BASECODE` and where BASECODESTRT was defined, zero emitted bytes), then fixed BASECODESIZE to read `BASECODEEND-BASECODE` directly instead - matching the same pattern just applied to `INITSIZE/INITCODE`, not a new approach invented for this case. Verified: zero remaining BASECODESTRT references anywhere, BASECODESIZE resolves cleanly, zero duplicate symbols, dictionary chain unaffected.

✅ **BASEDICTSTRT removed the same way, for the same reason - BASEDICTSIZE depended on it too.** Same fix pattern as BASECODESTRT immediately above, applied to its counterpart: confirmed BASEDICTSTRT was numerically identical to BASEDICT (only `ORG BASEDICT` sits before it, zero emitted bytes), removed it, and repointed BASEDICTSIZE to `BASEDICTEND-BASEDICT` directly. Verified: zero remaining BASEDICTSTRT references anywhere, BASEDICTSIZE resolves cleanly, zero duplicate symbols, dictionary chain unaffected.

✅ **`INOUT` moved from `$DF00` to `$C000` (with `INOUTEND EQU INOUT+$FF` added immediately after), and every ACIA-related address verified between the two - also passes.** ACIA/ACIACR/ACIASR (`$C008`) and ACIADR (`$C009`) all fall within `INOUT-INOUTEND` (`$C000-$C0FF`), checked numerically. Confirmed there is no other memory-mapped I/O device anywhere in this file besides the ACIA family - nothing else needed checking. This move has a real, positive side effect worth naming clearly: it resolves the INOUT portion of the three-way collision flagged when BASEDICT moved to `$D85D` two turns ago (`INOUT` no longer overlaps BASEDICT, since `$C0FF < $D85D`). The DSTACK and RSTACK portions of that same collision were untouched by this specific change, but have since been resolved separately - see below.

✅ **`INOUT`'s new collision with APPCODE is resolved, and so is the rest of the original BASEDICT collision - RSTACK, DSTACK, CODETOP, APPCODE, and APPDICT all moved down `$2000`.** RSTACK (`$BC00-$BEFF`) and DSTACK (`$B800-$BBFF`) no longer overlap BASEDICT at all - the three-way collision first found when BASEDICT moved to `$D85D` is now fully resolved (`INOUT` resolved it two turns ago, this resolves the remaining two thirds). APPCODE's move (`$5000-$B7FF`) also separately resolves its overlap with `INOUT` from the previous entry, as a side effect of the same shift, not a second fix. Re-verified with a full pairwise sweep after the move, not assumed: the only overlap remaining from the four found last turn is the original, unrelated `INITCODE/BASECODE` one (30 B) - `INITCODE/ BASECODE/BASEDICT/RSTACK/DSTACK/INOUT/APPCODE` are all now mutually clean.

✅ **`APPDICT`'s collision with APPVARS is now resolved - APPVARSEND changed from a static `APPVARS+8000` to `APPDICT-1`, making it self-derive from wherever `APPDICT` actually sits instead of a fixed size.** APPVARSEND is now `$1FFF` (was `$215B`), giving APPVARS 7653 bytes of actual usable space (down from the originally-intended 8000 - the 347-byte reduction exactly matches the overlap this closes). APPVARS's own EQU still starts at `$021B` and its comment still cites the historical "8000 bytes" intent, now explicitly marked as the original intent rather than the current usable size. Functionally, APPVARS's enforced range (`$021B-$1FFF`) no longer reaches APPDICT (`$2000-$6EA4`) at all - checked numerically, not assumed. This also makes the boundary self-correcting: if APPDICT moves again, APPVARSEND moves with it automatically, rather than needing another manual fix like the last several turns' worth of address changes required. VUNUSEDW's own code is unchanged - it already computed against APPVARSEND (see below), so

this fix took effect purely by changing what that symbol means. Verified: zero duplicate symbols, dictionary chain unaffected.

☑ **DSTACK moved up $200 (to $BCFF), resolving both the CODETOP/DSTACK mismatch and the RSTACK/DSTACK gap from last turn in one move.** DSTACK's new occupied bottom ($B900, from its $BCFF top and 1024-byte size) now matches CODETOP ($B900) exactly - APPCODE's nominal ceiling no longer overstates safe growth room, and the 512-byte overlap that created between APPCODE's nominal range and DSTACK's true range is gone. As a bonus, not something separately requested: DSTACK's new top ($BCFF) is now exactly contiguous with RSTACK's bottom ($BD00), closing the 512-byte gap that had opened between them too. Verified with a full pairwise sweep after the move: exactly two overlaps remain in the current memory map - the still-open APPDICT/APPVARS one (347 B) and the original, unrelated INITCODE/BASECODE overlap (30 B) - down from three last turn.

☑ **The VECTORS collision found by verifying INITEND is now resolved - INIT's ORG moved from $FFC0 to $FFA0.** Widened INIT from 48 to 80 bytes ($FFA0-$FFEF), comfortably covering the ~78 bytes COLDSTRT+WARM+WARMMSG actually needs (2 bytes of margin) - still an estimate from manual byte-counting, not a real assembler's output. INIT's own ORG was also converted from a literal $FFC0 to symbolic ORG INIT, matching BASECODE/BASEDICT's convention, so the EQU and the actual placement can no longer drift apart the way BASECODE's did before that was caught. One real, unresolved interaction worth naming: INIT's new start ($FFA0) is 32 bytes below the old documented BASECODE end ($FFBF), tightening the still-open BASECODE overflow risk below by that much further - not a new problem in kind, since that risk was already estimated at roughly 2580 bytes over budget, dwarfing this additional 32-byte reduction, but worth tracking alongside it rather than treated as independent.

□ **INIT moved again, from $FFA0 to $FFA2 (request contained a &FFA2 typo - used the correct $ hex prefix instead).** INIT is now exactly 78 bytes ($FFA2-$FFEF), matching the COLDSTRT+WARM+WARMMSG byte-count estimate with zero margin (was 80 bytes, with 2 bytes of slack). This is a genuinely tighter fit than before, worth naming as a real tradeoff: any inaccuracy in the manual byte count, or any future addition to COLDSTRT/WARM, now has zero room to absorb. The overlap with BASECODE noted above is **not resolved by this change, only reduced** - from 32 bytes to 30 bytes ($FFA2-$FFBF). This was a real, pre-existing problem before this turn touched anything ($FFA0 already overlapped BASECODE's declared end by 32 bytes), not something newly introduced. Not fixed here, in either direction (shrinking BASECODE further or moving INIT above it instead of below): both are bigger architectural decisions than "change one EQU," and a real assembler run would settle the actual BASECODE byte count this depends on.

☑ **INIT renamed to INITCODE** (EQU and its ORG reference, both in the memory-map constants). The section-title comment ("SECTION 2: INIT CODE") and one historical narrative comment describing a past bug scenario (with the byte figures true at that time, not today) were deliberately left referring to "INIT" as plain English/history, not the symbol - renaming a symbol doesn't obligate rewriting prose that correctly describes what was true before the rename existed. Doing this rename surfaced that the documentation's "ROM Size Required" section (Section 2) had gone stale independent of the rename itself: it still asserted the four ROM regions were contiguous at exactly 8192 bytes, which stopped being true as soon as INIT moved to $FFA2 last turn and started overlapping BASECODE. Rewritten to state the overlap plainly instead of the now-false claim. Verified: INITCODE defined exactly once, zero duplicate symbols, dictionary chain unaffected.

☑ **BASEDICT and BASECODE addresses applied exactly as requested, and the documented ROM requirement widened to a 16K part - but this surfaced a severe, unresolved collision, not a minor tradeoff.** BASEDICT moved from $E000 to $D85D - verified before making the change that $E012 - $D85D = 1973 bytes exactly matches SECTION 27's real, measured dictionary content, a genuine zero-padding exact fit (was 2048 bytes, 75 bytes of slack). BASECODE moved from $E800 down to $E012 to match; its own upper bound was not given and stayed at $FFBF, growing it from 6080 to 8110 bytes.

☐ **BASEDICT's new range ($D85D-$E011) entirely overlaps three other live regions: DSTACK (931 of its 1024 bytes, $D85D-$DBFF), RSTACK (all 768 bytes, $DC00-$DEFF), and INOUT (all 256 bytes, $DF00-$DFFF).** Found by checking the new address against every other region's boundaries before updating the documentation - not caught until then. This is a fundamental, system-breaking conflict, not a byte-budget tightness issue like the INITCODE/BASECODE overlap noted elsewhere: as configured, the ROM dictionary, the data stack, the return stack, and the ACIA's registers would all be mapped to the same physical addresses simultaneously, which cannot work on real hardware without bank switching (which this system does not have). The requested EQU values were applied exactly as given, since that's what was asked; resolving the collision itself was not attempted here, since it requires a decision this response can't make alone - moving DSTACK/RSTACK/INOUT elsewhere, choosing a smaller BASEDICT/BASECODE split that doesn't reach down this far, or reconsidering whether $D85D was the address actually intended. The documented ROM part was still widened from 8K to 16K per the request (real total usage of BASEDICT+BASECODE+INITCODE+VECTORS alone, ignoring the collision, is about 10.15K, leaving roughly 6.2K of spare capacity within a 16K×8 EPROM), but that figure is secondary to the collision above. Verified: zero duplicate symbols, dictionary chain unaffected (219 entries, since this doesn't touch it).

☑ **The INITCODE/BASECODE overlap - open since it was first found, mentioned in at least five separate entries above across several turns - is finally resolved. BASECODE and BASEDICT both shifted down exactly 30 bytes** (BASECODE: $E012 -> $DFF4; BASEDICT: $D85D -> $D83F), chosen precisely: 30 bytes is exactly the overlap amount, so BASECODE's nominal end ($FFA1, unchanged 8110-byte budget) now lands exactly one byte below INITCODE's start ($FFA2) - zero gap, zero overlap. BASEDICT shifted the same amount to stay perfectly contiguous with BASECODE's new start, preserving its own exact, zero-padding fit. Verified with a full pairwise sweep across every region in the memory map, not just the two that moved: **zero overlaps anywhere** - the first time that's been true since this whole sequence of address changes began. USROMSTRT/USROMEND containment re-checked and still passes for all three ROM regions. Also cleaned up the accumulated resolved-issue narrative in the top-of-file note (previously ~40 lines chronicling the BASEDICT/DSTACK/RSTACK/INOUT/CODETOP/ APPDICT saga turn by turn, including one claim - the APPDICT/APPVARS overlap being open - that was already stale before this edit) down to a short, current-state summary, matching the same cleanup already applied to the documentation.

5.2 Structural duplication (identified, some resolved, some not)

☑ :/CREATE/VARIABLE's header-building — resolved via HEADER (section 7).

☑ COMMA/CODECOMMA/CCOMMA/VCOMMA/VCCOMMA/CCOMMA1 — resolved via APPENDCELL/APPENDBYTE (section 6).

☑ <#/#> recomputing PAD's address inline — resolved, both now call PADW (section 20 / section 6).

- □ No further known duplication, but the codebase was never given a full pass specifically hunting for more.

5.3 Real, load-bearing gaps

- □ **Software (XON/XOFF) handshaking is still not implemented.** Now that hardware RTS/CTS handshaking is in place, this is the one remaining flow-control gap: this system still never transmits XON/XOFF and would not recognize either byte as a control signal if the remote device sent them — an incoming `$11`/`$13` is just queued as an ordinary character. Only relevant for links with no RTS/CTS wiring (plain 3-wire serial); would need a byte- interception layer between the input ring and KEY's caller.
- □ **No hard boundary checks** anywhere DPHERE/CODEHERE/VARHERE grow toward each other or toward the stacks. UNUSED and VUNUSED both correctly *report* the distance to their boundary now (`CODETOP` and the newly-added APPVARSEND respectively), but nothing enforces either — a runaway compile can silently corrupt an adjacent region.
- ☑ **`VUNUSEDW` (the `VUNUSED` word) computed a meaningless number, not "how much APPVARS space remains" - a real, distinct bug, not just the absence of a check. Fixed.** `LDD #APPCODE / SUBD VARHERE` computed `APPCODE - VARHERE` (roughly `$7000` minus wherever VARHERE currently sits within APPVARS), which had nothing to do with APPVARS's own boundary - looked like a copy-paste slip from `UNUSEDW` immediately above it (`LDD #CODETOP / SUBD CODEHERE`, correct for `CODEHERE`'s own boundary). Surfaced directly by a question about how APPVARS's size is actually set: it wasn't - `APPVARS EQU $021B` defined only its start; there was no end/size constant anywhere. Fixed by adding `APPVARSEND EQU APPVARS+8000` (an exclusive upper bound, `$215B`, matching `CODETOP`'s own convention for APPCODE) and changing `VUNUSEDW` to `LDD #APPVARSEND / SUBD VARHERE`. Verified: at cold start (VARHERE=APPVARS), this computes exactly 8000, matching APPVARS's documented size precisely; zero duplicate symbols; dictionary chain unaffected. APPVARSEND initially fell within `APPDICT`'s current range - expected at the time, the same, already-tracked APPDICT/ APPVARS overlap, not something this fix caused. Since resolved by redefining APPVARSEND as `APPDICT-1` - see above.
- □ **ENVTABLE is incomplete.** Missing entries: `/HOLD`, `/PAD` (this build never fixed a capacity for either — filling them in would mean deciding a real bound first, not just picking a number), `MAX-D`, `MAX-UD` (need the dispatcher extended to push *two* cells for double-cell answers — current ENVQUERY only handles one), `WORDLISTS`/`FLOORED` (need the dispatcher to be able to report a recognized-but-false answer, distinct from "unrecognized name" — no such path exists yet).
- □ **`CATCH`-wrapped `QUIT`/`INTERPRET` rollback is scoped to one input line only.** A colon definition spanning multiple lines that fails partway through a later line only rolls back that line's contribution, not the whole definition back to `:`. A complete fix needs the region-pointer snapshot taken once at `:` and held until `;` or an error, not refreshed every `QLOOP` pass.
- □ **`ABORTHDR`'s `LINK` field is a placeholder `0`** in the consolidated/split source — must be set to the real prior LATEST value once final ROM layout/assembly order is fixed.

5.4 Never resolved after being explicitly raised

- □ **J doesn't generalize past one level of loop nesting.** No J2/deeper equivalent exists. A triple-nested loop's innermost body has no built-in way to reach the outermost index without manually replicating J's offset arithmetic one frame further.
- □ **LEAVE's correctness depends on always being textually inside the loop it affects** — never verified against a LEAVE called from a separately-defined word invoked from within a loop (which would read/set the wrong stack frame, or crash).
- □ **WORD's 31-character cap is now inconsistent within the system.** PARSE/PARSE-NAME were deliberately rewritten to not share it, but WORD itself — and everything still built on it (CHAR, [CHAR], header names, FIND) — still has it.
- □ **The optional Search-Order word set is not implemented.** Every word resolves through one unified dictionary chain rooted at LATEST; there is no multi-wordlist support (WORDLIST, GET-ORDER/SET-ORDER, ALSO/ONLY, etc.). This was a foundational decision fixed since COLDSTRT's first version, not an oversight — now documented explicitly (ClaudeForth documentation, Section 4.5) along with what adding it later would actually require: restructuring FIND into a loop over an active search order, changing HEADER to link into a selectable "current" wordlist instead of unconditionally into LATEST, and fixing the few words (RECURSE, ;'s unsmudge step, WORDS) that read LATEST directly today.

5.5 Open design questions (not defects — deliberate unresolved choices)

- □ Whether ALLOT/VALLOT/PICK/ROLL/BASE should range-check their inputs against actual stack/region bounds. Currently none of them do, consistently, by choice rather than oversight.
- □ Whether any additional distinction like TIB/SOURCE (fixed terminal buffer vs. current input source) is needed anywhere else in the system where EVALUATE's redirection might still cause a mismatch.
- □ SWIH's throw code (-99 in the consolidated source) was never assigned a real, deliberate value — it's a placeholder for "some hardware trap occurred," not a considered ANS-style code.
- □ REPLACES/SUBSTITUTE remain single-slot (one registered name/value pair, overwritten by the next REPLACES call) rather than a true multi-entry table. A real table needs its own storage layout and lookup structure — a deliberate scope decision made when SUBSTITUTE was completed, not an oversight.

5.6 What's solid (for reference, not action)

Stack manipulation (Core + Core Ext + return-stack transfer), all arithmetic (single + double + mixed precision), logic, comparison (single + double), full control flow including CASE, all defining words including DEFER/MARKER/VALUE/TO (VALUE now correctly targeting mutable space), memory and string operations (including a completed SUBSTITUTE), SOURCE/EVALUATE/REFILL mechanics, and the Tools word set (.S/WORDS/DUMP, with DUMP's edge-case bug fixed) are complete and were traced/verified against concrete cases during the build, not merely asserted correct.

6. Build Instructions

Which Files to Assemble

The buildable target is the single consolidated source file, forth6809.asm, delivered alongside this documentation and reproduced in full in Section 8. It is self-contained: all 27 numbered sections, the memory-map constants, and the applied GLOBALS page-zero layout live in this one file, assembled top to bottom with no external dependencies.

The per-section split files (00_memory_map_and_globals.asm through 27_forth_dictionary.asm, plus OPEN_ITEMS_CHECKLIST.md) are reference copies only, generated by splitting forth6809.asm along its own section markers for easier reading and review. They are not wired together with INCLUDE directives and are not themselves a multi-file build target — assembling them individually, or as a set, is not supported. Any change made to one must be reflected back into forth6809.asm to actually take effect.

Assembler

This source follows standard Motorola-style 6809 assembler syntax throughout (EQU, ORG, RMB, FCB, FCC, FDB) and targets LWASM, part of the LWTools toolchain — an actively maintained, open-source cross-assembler for the Motorola 6809 and Hitachi 6309, available at https://www.lwtools.ca/. A representative build command, producing a raw binary image suitable for programming directly into an EPROM:

```
lwasm --6809 --format=raw --output=forth6809.bin forth6809.asm
```

The --6809 flag restricts LWASM to genuine 6809 instructions. This is a real constraint, not a formality: an audit of every mnemonic in this file against the genuine 6809 instruction set found one 6309-only instruction in actual use — TSTD, used to test whether D was zero after a PULU (which, unlike a load, does not itself affect condition codes on real 6809 hardware). It has been replaced with CMPD #0, a genuine 6809 instruction with identical behavior for this purpose. --format=raw honors this source's own ORG directives directly, producing a flat binary image rather than a relocatable object file, matching the fixed, non-relocated memory map described in Section 2.

Testing

Initial testing is performed using MAME, configured with the existing mecb_6809 profile already present in MAME. The command line used is:

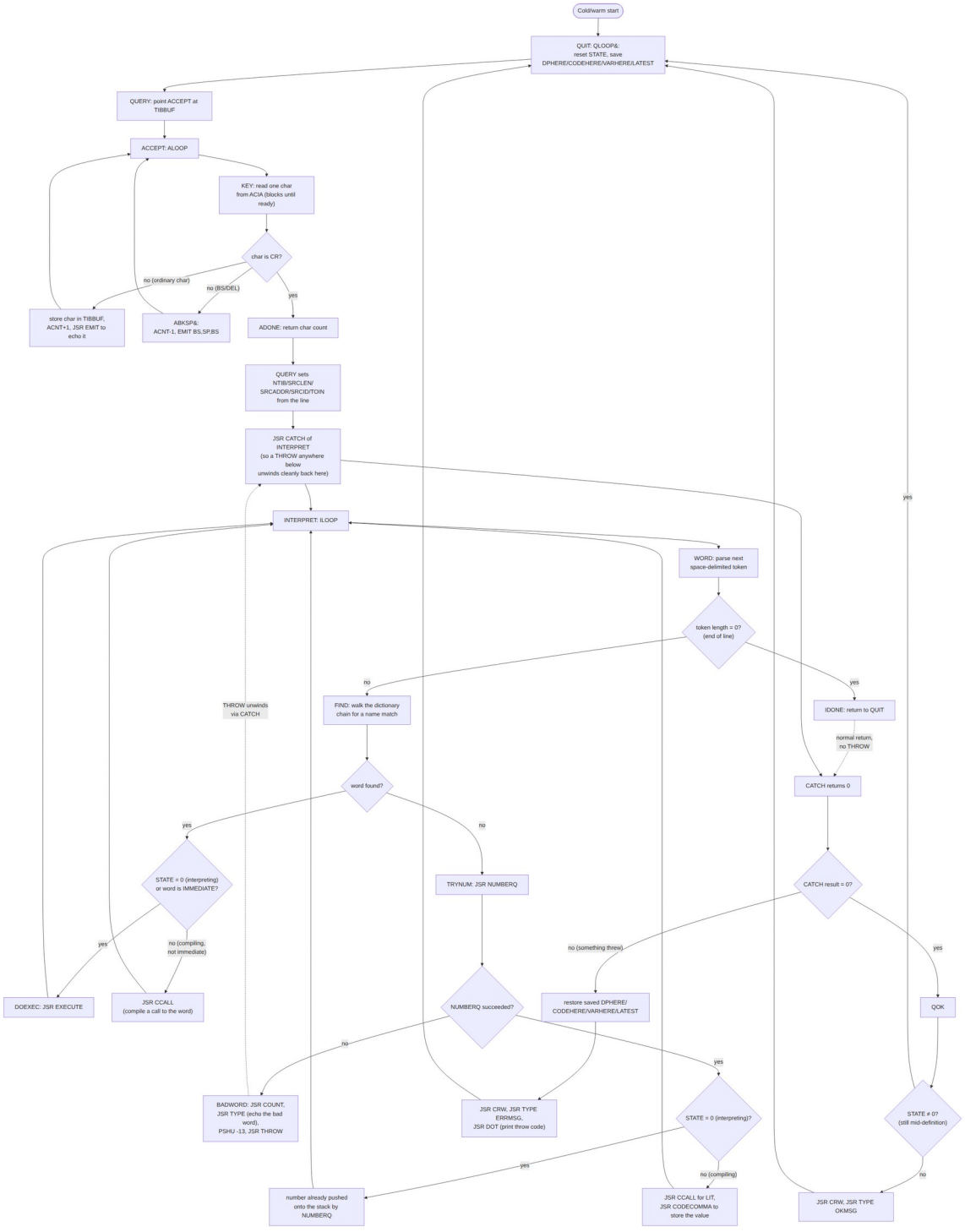
```
mame mecb_6809 -io_bank c0 -rom_size 16k -cart1 "/path/to/your/rom_image.bin"
-window
```

The rom_image.bin file is already produced by the assembler.

The emulated serial output will target the terminals, though a serial comms application can also act in the interactive role.

7. Architecture

7.1 (Outer) Interpreter



ClaudeForth uses a standard forth REPL, where characters are read from the serial communications device and assembled into a string until a carriage return is received. The dictionary is then searched for

a matching string, in the expectation that it is a predefined word with attached code. If it’s not found, an attempt is made to interpret the string as a number. Otherwise an exception is thrown.
Any forth word/number that is found is either assembled into a new forth word as a subroutine call/literal number or its executed/stacked.
The diagram above shows a UML activity diagram of the cooperating routines.

7.2 Dictionary Operations

Structure

The dictionary is a linked list that grows from low to high memory, with each entry storing a link to the header of the prior entry. The first created entry has a link field of zero. The last created entry’s header is pointed to by an external link stored in a variable called LATEST.

Initialization

Two variables establish where the interpreter begins: LATEST, which marks the start of a search, and DPHERE, which marks where the next new dictionary entry will be built. Both are set once, by COLD at cold start: LATEST is set to BASELATEST, the address of the newest entry already built into the ROM dictionary, so a search immediately has a valid chain to walk; DPHERE is set to APPDICT, the start of the application dictionary's RAM area, so the first word a user defines is built starting at a known, fixed address.

Word Search

FIND begins at whatever address LATEST currently holds and walks backward through the chain one entry at a time, following each entry's own link field to reach the previous entry. At each entry, it first checks the length/flags byte: an entry whose smudge bit is set is skipped outright, and an entry whose name length doesn't match the search string is also skipped before a single character is ever compared, since a length mismatch alone is enough to rule it out. Only when the length matches are the name's characters actually compared. The search ends one of two ways: a full name match, in which case the entry's code-field address is returned; or the link field read is zero, meaning the chain's original first entry has been reached with no match, which signals the word was not found.

Word Append

HEADER builds a new entry by first capturing the current value of DPHERE as the address the new entry will occupy. It writes the length/flags byte and the name's characters starting there, advancing a working pointer as it goes. It then writes the current value of LATEST — the address of whatever is still the newest entry at this point — into the new entry's own link field, so the new entry points back to the one before it. A code-field address follows the link field. Only once the whole entry has been written are the two pointers updated: DPHERE is advanced past the new entry, so the next word will be built after it, and LATEST is set to the new entry's own address, so it becomes the newest entry — the one a search will now start from, and the one the next appended entry's link field will point back to.

8. Assembler Source

The complete 6809 assembly source for this system, as consolidated and verified across the design conversation, including every correction made along the way (the corrected VALUE PFA, the applied GLOBALS page-zero layout, the DUMP and SUBSTITUTE fixes, and every other fix described in Section 5's

checklist). This matches the forth6809.asm file delivered alongside this documentation; the section breakdown below follows the same numbered sections used in that file and in its per-section split.

8.1 Memory Map, Constants & GLOBALS Layout

```
; ============================================================
; 6809 ANS FORTH - consolidated source
; Assembled from the full design conversation.
;
; GLOBALS LAYOUT: applied. Every scratch/global cell now has a
; fixed RMB-assigned address in page zero ($0000-$00FF, DP=$00
; at reset) - see the GLOBALS section below for the full layout
; and its byte budget (256 of 256 bytes used, 0 free - the page
; is now fully packed; any future scratch cell will need to
; reuse an existing one or move a cell out of page zero).
;
; SERIAL HANDSHAKING: RTS (hardware, output) is implemented -
; IRQH/KEY toggle it based on the input ring's fill level
; against INHIWATER/INLOWATER (see SECTION 3). CTS (input) needs
; no firmware logic at all: the 6850 hardware automatically
; inhibits TDRE while CTS is deasserted, and this system's
; existing TDRE-gated transmit logic already respects that with
; no code changes. Software (XON/XOFF) handshaking remains not
; implemented - this system still neither transmits nor
; recognizes those bytes.
;
; DICTIONARY: applied (SECTION 27), 219 entries (215 original +
; DOES> + TRUE + FALSE, added in later passes - see below).
; Every primitive with a real code label now has a real ROM
; header, chained via LINK, CFA pointing directly at its code.
; Building this surfaced two real findings, not just mechanical
; work: (1) the original 1024-byte BASEDICT could not hold the
; header table (1954 bytes needed) - BASEDICT was resized to
; 2048 bytes ($E000-$E7FF), taking the space from BASECODE (then
; $E800-$FFBF, 6080 bytes, was 7104). This resize was based on
; the header-table budget alone; the actual assembled byte size
; of BASECODE's code was never measured with a real 6809
; assembler, so whether 6080 bytes was enough for all ~530
; routine labels in this file was never verified before BASECODE
; moved again (below). (2) DOES>
; initially had no corresponding code anywhere in the file -
; SETDOES and the DOES> compiling word (code label DOESGT, since
; a literal ">" is not a valid 6809 assembler label) were added
; in a follow-up pass, alongside DODOES/DOESRT0, closing that
; gap; DOESBEH was added to the GLOBALS layout to support it,
; using 2 of the page's last 3 free bytes (1 now remains). TRUE
; and FALSE were added in a still later pass: CONSTANT TRUE -1
; and CONSTANT FALSE 0, the first CONSTANT-pattern ROM-resident
; words in this system (TRUEBODY/FALSEBODY, section 26) - every
; other ROM word's CFA is a plain code label, but a CONSTANT's
; CFA is the DODOES-trampoline pattern, built by hand here with
; fixed, assemble-time addresses since there is no interactive
; CREATE/CONSTANT phase for ROM content.
;
; BASEDICT now holds 1992 bytes ($D83F-$E006) - grown from the
; earlier exact-fit 1973 bytes when H_ABORT/H_QUIT (formerly
; ABORTHDR/QUITHDR) and BASELATEST moved in from their own
; section so every dictionary header lives in one contiguous
; block. BASECODE's start ($DFF4) did not move to match, since
; it's a fixed address, not derived from BASEDICT's real size -
; this reintroduces a real overlap, 19 bytes ($DFF4-$E006),
; between the two. Not resolved here; see the open-items
; checklist. USROMSTRT/USROMEND, INOUT, RSTACK, DSTACK, CODETOP,
; APPCODE, APPDICT, and APPVARS remain mutually consistent
; otherwise, with no other overlaps anywhere in the current
; memory map. See the ROM Size Required section of the
; documentation and the open-items checklist for real content
; totals and remaining margin.
;
```

```
; This file preserves the code exactly as derived and verified
; turn-by-turn in the conversation, including the corrected
; versions of every bug that was caught and fixed in place.
; SUBSTITUTE is complete but deliberately scoped to a single
; registered name/value pair, not a full table (see REPLACES).
; ENVTABLE's /HOLD and /PAD entries, and the DPHERE/CODEHERE/
; VARHERE boundary checks, remain explicitly incomplete/absent -
; see the inline notes preserved from that discussion, and the
; open-items checklist.
; ============================================================

; ------------------------------------------------------------
; MEMORY MAP
; ------------------------------------------------------------
ROMSTRT  EQU $C000     ; true physical start of the 16K EPROM's own
                       ; address decode (distinct from USROMSTRT
                       ; below, which excludes INOUT's 256-byte
                       ; shadow) - see the ROM Size Required section
                       ; of the documentation, and the ROM: padding
                       ; block right before SECTION 27
USROMSTRT EQU $C100     ; Usable ROM start. Beginning of the usable
                        ; EPROM address range. INITCODE, BASECODE,
                        ; and BASEDICT must all fall within
                        ; USROMSTRT..USROMEND - see the verification
                        ; note below each one's EQU.
USROMEND EQU  VECTORS-1 ; Usable ROM end. Corrected: 1 before VECTORS'
                        ; start ($FFEF), not one past VECTORS' end as
                        ; originally defined - this is a real 16-bit
                        ; address (the last byte available before the
                        ; reserved vector table), usable directly in
                        ; comparisons or as a memory operand, unlike
                        ; the previous $10000 definition
VECTORS  EQU  $FFF0
INITCODE EQU  $FFA9     ; was $FFA2 - shifted up 7 bytes to reduce the
                        ; overlap with BASECODE's nominal end ($FFB4)
                        ; from 19 bytes to 12 - improved, not resolved.
                        ; CORRECTED: INITCODE's real content is 71
                        ; bytes ($47), confirmed by an actual assembler
                        ; run - not the 78-byte manual estimate relied
                        ; on for several turns, which was wrong by 7
                        ; bytes. At $FFA9, real content now ends at
                        ; $FFEF, exactly one byte below VECTORS - a
                        ; genuine, assembler-confirmed exact fit, zero
                        ; gap, zero overlap. A prior turn claimed this
                        ; shift created a new 7-byte VECTORS overlap;
                        ; that was based on the incorrect 78-byte
                        ; estimate and was wrong - retracted here. The
                        ; BASECODE overlap (12 bytes against its
                        ; nominal budget) is separate and unaffected
                        ; by this correction; not resolved. See the
                        ; open-items checklist.
BASECODE EQU  $E02A     ; was $E007 - shifted up 35 bytes. Two effects:
                        ; (1) opens a new 35-byte GAP between
                        ; BASEDICT's real end ($E006, 1992 bytes) and
                        ; this new start - previously exactly
                        ; contiguous (zero gap). Not itself broken,
                        ; just unused address space where there wasn't
                        ; any before. (2) WORSENS the existing overlap
                        ; with INITCODE: BASECODE's nominal size (8110)
                        ; is unchanged, so its nominal end also shifted
                        ; up 35 bytes, from $FFB4 to $FFD7 - INITCODE
                        ; (currently $FFA9) is now overlapped by 47
                        ; bytes, up from 12. Applied exactly as
                        ; requested; neither effect resolved here. See
                        ; the open-items checklist.
BASEDICT EQU  $D83F     ; was $D85D - shifted down the same 30 bytes as
                        ; BASECODE, preserving the exact, zero-padding
                        ; fit for its real 1973-byte dictionary content
                        ; (SECTION 27) - contiguous with BASECODE's
                        ; start at the time. Since grown to 1992 bytes
                        ; (H_ABORT/H_QUIT/BASELATEST moved in), and
```

```
                          ; BASECODE has since moved again too, no longer
                          ; to match - a 35-byte gap now sits between
                          ; them; see BASECODE's own EQU.
INOUT    EQU  $C000      ; was $DF00 - moved so INOUT (256 B) sits
                          ; directly below USROMSTRT ($C100), contiguous,
                          ; no gap. This also resolves the INOUT portion
                          ; of the collision flagged when BASEDICT moved
                          ; to $D85D: INOUT no longer overlaps BASEDICT
                          ; ($D85D-$E011), since $C0FF < $D85D. The
                          ; DSTACK and RSTACK portions of that same
                          ; collision were NOT touched by this specific
                          ; change, but were resolved separately when
                          ; those two regions moved (see below). This
                          ; move ALSO overlapped APPCODE at the time
                          ; ($7000-$D7FF then) - since resolved too, when
                          ; APPCODE moved down $2000 (see below).
INOUTEND EQU  INOUT+$FF
RSTACK   EQU  $BFFF      ; was $BEFF - occupied range is $BD00-$BFFF
                          ; (768 bytes, unchanged size). RESOLVED: once a
                          ; 512-byte gap sat between this and DSTACK
                          ; below; DSTACK moving up $200 closed it - now
                          ; exactly contiguous, no gap
DSTACK   EQU  $BCFF      ; was $BAFF - moved up $200. RESOLVED (both):
                          ; occupied range is now $B900-$BCFF, which
                          ; exactly matches CODETOP ($B900) as its true
                          ; bottom - the 512-byte mismatch is gone - and
                          ; is now exactly contiguous with RSTACK's
                          ; bottom ($BD00), closing that gap too
CODETOP  EQU  $B900      ; was $B800 - code space ceiling (data stack
                          ; begins here). RESOLVED: this once no longer
                          ; matched DSTACK's true occupied bottom (was
                          ; $B700, a 512-byte mismatch) - now that DSTACK
                          ; moved up $200 to $B900, CODETOP matches it
                          ; exactly again

; ANS transient-region minimums (forth-standard.org/standard/usage,
; section 3.3.3.6 "Other transient regions"), verified against this
; system's actual layout and used to replace the bare "84" that used
; to sit directly inside PADW with a named, documented constant:
;   PADMINSIZE  - PAD's own scratch region, size in characters. WORD's
;                 own separate, fixed WORDBUF already meets its own
;                 33-character minimum exactly (WORDBUF's span up to
;                 where SIBUF used to begin, before SIBUF's later
;                 retirement below, was precisely 33 bytes, confirmed
;                 by address subtraction) - unrelated to this offset,
;                 since this system's WORD uses a dedicated fixed
;                 buffer rather than HERE-relative storage.
;   HOLDMINSIZE - pictured numeric output buffer, size in characters,
;                 (2 * n) + 2 where n = bits per cell (16 here) = 34.
;                 LTNUM ("<#") anchors HLD to PADW's own return value
;                 and HOLD decrements before storing, so this buffer
;                 grows downward from PAD into the CODEHERE-to-PAD
;                 gap, not into PAD's own region above it - confirmed
;                 by tracing LTNUM/HOLD. The gap must be at least
;                 HOLDMINSIZE wide for this to fit.
; PADOFFSET (below) satisfies both: it equals PADMINSIZE, which is
; itself well above HOLDMINSIZE (84 vs 34, 50 bytes of margin) - kept
; as a single, larger-than-either-strict-minimum offset rather than
; two separately-tuned numbers, matching "larger than minimum is
; satisfactory" for a small system.
PADMINSIZE  EQU 84
WORDMINSIZE EQU 33
HOLDMINSIZE EQU 34
PADOFFSET   EQU PADMINSIZE  ; CODEHERE-to-PAD gap; also governs how
                            ; much of PAD's own upward-growing region
                            ; (now used by both the user and, per this
                            ; change, interpreted-mode S" text) is
                            ; guaranteed available before something
                            ; else might claim it

APPCODE  EQU  $7000      ; was $5000 - back to its original address.
```

```
                        ; RESOLVED: this once overlapped DSTACK's true
                        ; range by 512 bytes ($B700-$B8FF), a direct
                        ; consequence of the CODETOP/DSTACK mismatch -
                        ; now that DSTACK moved and CODETOP matches it
                        ; again, APPCODE's nominal range (up to
                        ; CODETOP-1) no longer reaches into DSTACK
APPDICT  EQU  $2000     ; was $015B - moved up, size unchanged (20133
                        ; bytes, now $2000-$6EA4). REDUCED BUT NOT
                        ; RESOLVED: still overlaps APPVARS below by 347
                        ; bytes ($2000-$215A), though SIBUF/WORDBUF/
                        ; TIBBUF/OUTBUF (swallowed by the previous
                        ; APPDICT address) are now clear. See the
                        ; open-items checklist
APPVARS  EQU  $021B     ; grown from 256 to 8000 bytes (end now $215A,
                        ; was $031A), taking the space directly from
                        ; APPDICT above it; start address unchanged.
                        ; That 8000-byte figure describes the original
                        ; intent, not the current actual usable size -
                        ; see APPVARSEND below, which now tracks
                        ; APPDICT's real position instead
APPVARSEND EQU APPDICT-1 ; was APPVARS+8000 ($215B) - now derives
                        ; directly from wherever APPDICT actually
                        ; starts, currently $1FFF (7653 bytes usable,
                        ; down from the static 8000). Self-correcting:
                        ; this can no longer go stale if APPDICT moves
                        ; again, unlike the previous fixed-size
                        ; definition. VUNUSEDW (below) is unchanged -
                        ; it already computed against APPVARSEND
; SIBUF EQU $01FB retired - was interpreted-mode S"'s fixed, dedicated
; 32-byte scratch buffer; S" now writes into PAD instead (see SQINTERP,
; String Words section), which is both larger and correctly per-call
; rather than a single buffer shared by every S" call in a session.
; The $01FB-$021A range it used to occupy is left unclaimed rather
; than reassigned - APPVARS below still starts at its own, unchanged
; address ($021B) - to avoid any risk to the rest of this carefully-
; verified memory map for the sake of reclaiming 32 bytes on a system
; with headroom to spare; reclaiming it would need APPVARS to move,
; a separate, larger change not undertaken here.
WORDBUF  EQU  $01DA     ; was $02D4
TIBBUF   EQU  $018A     ; was $0284
TIBBUFL  EQU  80
SERBUF   EQU  $0106     ; was $0200 - USER0/USER1 removed entirely (see
                        ; below); the 5 buffers (SERBUF's 4-byte index
                        ; block, INBUF, OUTBUF, TIBBUF, WORDBUF) still
                        ; sit contiguously right after MVSCRATCH, with
                        ; no gap - SIBUF, which used to sit right after
                        ; WORDBUF closing what was once an 11-byte gap
                        ; ($02F5-$02FF) in an earlier address scheme,
                        ; is now retired (see above) rather than part
                        ; of this contiguous run
INHEAD   EQU  SERBUF
INTAIL   EQU  SERBUF+1
OUTHEAD  EQU  SERBUF+2
OUTTAIL  EQU  SERBUF+3
INBUFSZ  EQU  64
OUTBUFSZ EQU  64
INBUF    EQU  SERBUF+4
OUTBUF   EQU  SERBUF+4+INBUFSZ
GLOBALS  EQU  $0000

SP0      EQU  DSTACK+1
RP0      EQU  RSTACK+1

; ------------------------------------------------------------
; SERIALPOLL - conditional-assembly switch for serial I/O.
; 1 (default): KEY/KEYQ/EMIT poll ACIASR directly, no interrupts,
; no ring buffers, no RTS/CTS handshaking - IRQH is a harmless RTI
; stub, matching the other unused vectors (NMIH/FIRQH/SWI2H/
; SWI3H). 0: the original interrupt-driven implementation, with
; INBUF/OUTBUF ring buffers serviced by IRQH and RTS-based flow
; control via INFILL/RTSCHECKHI/RTSCHECKLO. Uses LWASM's IFEQ/
```

```
; ELSE/ENDC (a numeric-expression test, not IFDEF/IFNDEF, since
; this is a value to compare, not a symbol's mere presence).
; ------------------------------------------------------------
SERIALPOLL EQU 1

; ------------------------------------------------------------
; ACIA (6850) constants - the chip sits at INOUT+8, not at the
; base of the I/O block, leaving INOUT+0..INOUT+7 free for other
; memory-mapped devices sharing this 256-byte region
; ------------------------------------------------------------
ACIA     EQU  INOUT+8
ACIACR   EQU  ACIA
ACIASR   EQU  ACIA
ACIADR   EQU  ACIA+1
SR_RDRF  EQU  $01
SR_TDRE  EQU  $02
SR_IRQ   EQU  $80
CR_RESET EQU  $03
CR_RXON  EQU  $95
CR_RXTX  EQU  $B5
CR_POLL  EQU  $15     ; bit7=0 (RX interrupt disabled), bits6-5=00 (RTS
                      ; low, TX interrupt disabled) - CR_RXON ($95) with
                      ; only the RX-interrupt-enable bit cleared. Used
                      ; only when SERIALPOLL=1; RTS stays permanently
                      ; low (asserted), since polling mode has no ring
                      ; buffer to overflow and so needs no flow control
CR_RTSHI EQU  $D5     ; bits6-5=10: RTS high, TX int disabled, RX int enabled -
                      ; derived from CR_RXON ($95) with bits6-5 changed from
                      ; 00 to 10; the ACIA has no combination offering RTS
                      ; high AND TX interrupt enabled simultaneously (bits6-5
                      ; only has 00/01/10/11, and only 01 enables TX interrupt,
                      ; which always ties RTS low) - EMIT/IRQH's TXCHK must
                      ; respect this and defer transmission while RTS is high

INHIWATER EQU 48       ; input ring fill level (of 64) at/above which RTS is
                       ; asserted high, telling the remote device to pause
INLOWATER EQU 16       ; fill level at/below which RTS is reasserted low;
                       ; deliberately well below INHIWATER (hysteresis) so
                       ; RTS doesn't chatter right at a single threshold

; ------------------------------------------------------------
; Flag / opcode constants
; ------------------------------------------------------------
TRUEV    EQU  $FFFF
FALSEV   EQU  $0000
OPJSR    EQU  $BD
RTSOPC   EQU  $39

; ------------------------------------------------------------
; Control-flow compile-time tags
; ------------------------------------------------------------
TAGFWD   EQU  1
TAGBACK  EQU  2
TAGDO    EQU  3
TAGCASE  EQU  4
TAGOF    EQU  5
TAGENDOF EQU  6

; ============================================================
; GLOBALS - real layout, applied. Every scratch/state cell
; referenced across the whole build now has a fixed address
; in page zero (DP = $00 at reset, set in COLDSTRT), laid out
; in RMB order below. Total: 256 of 256 bytes used - the page
; is fully packed. (DOESBEH and RTSSTATE were added in later
; passes, after this comment was first written with 253/256;
; they used up the last 3 bytes of headroom entirely.)
;
; SNEND, which appeared in the original placeholder list
; (documented under S"/."/SLITERAL/ABORT"'s string runtime),
; was dropped here: a full pass over every reference confirmed
; it is never actually read or written anywhere - SCNT/SPTR
```

```
; alone carry that runtime. CSAVE was removed earlier (see the
; COMMA/CODECOMMA history) and was never part of this layout.
; ============================================================
          ORG   $0000        ; GLOBALS page - Direct Page (DP) set to $00 at reset
STATE       RMB   2   ; offset $00
BASE        RMB   2   ; offset $02
LATEST      RMB   2   ; offset $04
DPHERE      RMB   2   ; offset $06
CODEHERE    RMB   2   ; offset $08
VARHERE     RMB   2   ; offset $0A
HANDLER     RMB   2   ; offset $0C
THROWN      RMB   2   ; offset $0E
TOIN        RMB   2   ; offset $10
NTIB        RMB   2   ; offset $12
DELIM       RMB   1   ; offset $14
WSTART      RMB   2   ; offset $15
SLEN        RMB   1   ; offset $17
SNAMEP      RMB   2   ; offset $18
FNDPTR      RMB   2   ; offset $1A
HDRPTR      RMB   2   ; offset $1C
HDRFLAGS    RMB   1   ; offset $1E
CADDR       RMB   2   ; offset $1F
CNTREM      RMB   1   ; offset $21
NUMNEG      RMB   1   ; offset $22
NADDR       RMB   2   ; offset $23
NCNT        RMB   2   ; offset $25
MULBASE     RMB   1   ; offset $27
CARRY       RMB   1   ; offset $28
MSCR        RMB   2   ; offset $29    (shared: -, *, comparisons, WITHIN...)
MSCR2       RMB   2   ; offset $2B
MSCR3       RMB   2   ; offset $2D
MSCR4       RMB   2   ; offset $2F
HLD         RMB   2   ; offset $31
DEPTHTMP    RMB   2   ; offset $33
AMAX        RMB   2   ; offset $35
ABUFP       RMB   2   ; offset $37
ACNT        RMB   2   ; offset $39
ACH         RMB   1   ; offset $3B
EMITCH      RMB   1   ; offset $3C
NEWHDR      RMB   2   ; offset $3D
NAMEP       RMB   2   ; offset $3F
NAMELEN     RMB   1   ; offset $41
PTARGET     RMB   2   ; offset $42
PFIELD      RMB   2   ; offset $44
NEWFLD      RMB   2   ; offset $46
CSP         RMB   2   ; offset $48
EXITCNT     RMB   2   ; offset $4A
EXITPTR     RMB   2   ; offset $4C
HDRSMUDGE   RMB   1   ; offset $4E
SCNT        RMB   2   ; offset $4F
SPTR        RMB   2   ; offset $51
SAVEN       RMB   2   ; offset $53
DRWIDTH     RMB   2   ; offset $55
DRLEN       RMB   2   ; offset $57
DRADDR      RMB   2   ; offset $59
DRPAD       RMB   2   ; offset $5B
PRODHI      RMB   2   ; offset $5D
PRODLO      RMB   2   ; offset $5F
PSIGN       RMB   1   ; offset $61
DIVNUM      RMB   2   ; offset $62
DIVDEN      RMB   2   ; offset $64
DIVREM      RMB   2   ; offset $66
DIVCNT      RMB   1   ; offset $68
DNSIGN      RMB   1   ; offset $69
DVSIGN      RMB   1   ; offset $6A
DVOWNSIGN   RMB   1   ; offset $6B
MAHI        RMB   1   ; offset $6C
MALO        RMB   1   ; offset $6D
MBHI        RMB   1   ; offset $6E
MBLO        RMB   1   ; offset $6F
MSIGN       RMB   1   ; offset $70
```

```
REM         RMB   2   ; offset $71
DCNT        RMB   1   ; offset $73
UDHI        RMB   2   ; offset $74
UDLO        RMB   2   ; offset $76
PRSIGN      RMB   1   ; offset $78
R2A         RMB   2   ; offset $79
R2B         RMB   2   ; offset $7B
RDST        RMB   2   ; offset $7D
RVAL        RMB   2   ; offset $7F
TR1         RMB   2   ; offset $81
TR2         RMB   2   ; offset $83
SHCNT       RMB   1   ; offset $85
SHCNT2      RMB   2   ; offset $86
TYPECNT     RMB   2   ; offset $88
TYPEADDR    RMB   2   ; offset $8A
PDELIM      RMB   1   ; offset $8C
PSTART      RMB   2   ; offset $8D
PLEN        RMB   2   ; offset $8F
CMPA1       RMB   2   ; offset $91
CMPL1       RMB   2   ; offset $93
CMPA2       RMB   2   ; offset $95
CMPL2       RMB   2   ; offset $97
CMPMIN      RMB   2   ; offset $99
SRCH1       RMB   2   ; offset $9B
SRCH1L      RMB   2   ; offset $9D
SRCH2       RMB   2   ; offset $9F
SRCH2L      RMB   2   ; offset $A1
SRCHPOS     RMB   2   ; offset $A3
SRCHI       RMB   2   ; offset $A5
UEADDR      RMB   2   ; offset $A7
UESRCLEN    RMB   2   ; offset $A9
UEDST       RMB   2   ; offset $AB
UEOUTLEN    RMB   2   ; offset $AD
SNXT        RMB   2   ; offset $AF
SNTARGET    RMB   2   ; offset $B1
REPLNAME    RMB   2   ; offset $B3
REPLNLEN    RMB   2   ; offset $B5
REPLVAL     RMB   2   ; offset $B7
REPLVLEN    RMB   2   ; offset $B9
SUBDESTCAP  RMB   2   ; offset $BB
SUBDESTADR  RMB   2   ; offset $BD
SUBSRCADR   RMB   2   ; offset $BF
SUBSRCLEN   RMB   2   ; offset $C1
SUBOUTLEN   RMB   2   ; offset $C3
SUBWPTR     RMB   2   ; offset $C5
SUBCOPYCNT  RMB   2   ; offset $C7
SUBCOPYSRC  RMB   2   ; offset $C9
MKDP        RMB   2   ; offset $CB
MKCODE      RMB   2   ; offset $CD
MKVAR       RMB   2   ; offset $CF
MKLATEST    RMB   2   ; offset $D1
EVSAVEA     RMB   2   ; offset $D3
EVSAVEL     RMB   2   ; offset $D5
EVSAVEI     RMB   2   ; offset $D7
EVSAVET     RMB   2   ; offset $D9
SRCADDR     RMB   2   ; offset $DB
SRCLEN      RMB   2   ; offset $DD
SRCID       RMB   2   ; offset $DF
SPAN        RMB   2   ; offset $E1
DSPTMP      RMB   2   ; offset $E3
WWALK       RMB   2   ; offset $E5
DUMPADDR    RMB   2   ; offset $E7
DUMPCNT     RMB   2   ; offset $E9
DUMPCOL     RMB   1   ; offset $EB
HEXBUF      RMB   2   ; offset $EC
DUVALID     RMB   1   ; offset $EE
ENVLEN      RMB   2   ; offset $EF
ENVADDR     RMB   2   ; offset $F1
QSAVEDP     RMB   2   ; offset $F3
QSAVECODE   RMB   2   ; offset $F5
QSAVEVAR    RMB   2   ; offset $F7
```

```
QSAVELATEST RMB  2   ; offset $F9
QTHROWCODE RMB   2   ; offset $FB
DOESBEH    RMB   2   ; offset $FD - SETDOES scratch
RTSSTATE   RMB   1   ; offset $FF - 0 = RTS low (normal), nonzero = RTS
                      ; high (paused) - see CR_RTSHI. GLOBALS page is now
                      ; fully packed: 256 of 256 bytes used, 0 free.

GLOBALS_USED EQU 256  ; total bytes used, of 256 available - fully packed

; ------------------------------------------------------------
; MVSCRATCH - three cells shared, one at a time, by routine
; families that never call each other or run concurrently in
; this single-threaded interpreter: MOVE/CMOVE/CMOVE>, FILL, and
; HOLDS (plus the single-cell multiply routine). Sharing avoids
; needing 3x the physical storage for what is provably the same
; scratch need at different times; the tradeoff is that these
; three cells use ordinary extended addressing (3-byte LDD/STD),
; not direct-page (2-byte), since page zero has no room left.
;
;   MVCNT    - MOVE/CMOVE's remaining-byte count
;     HSLEN    EQU MVCNT   - HOLDS's remaining-char count
;   MVDST    - MOVE/CMOVE's destination address
;     HSADDR   EQU MVDST   - HOLDS's source address
;   MVSRC    - MOVE/CMOVE's source address
;     MRESULT  EQU MVSRC   - single-cell multiply's 16-bit result
;     FILLCHR  EQU MVSRC   - FILL's fill character (1 byte, uses
;                            MVSRC's first byte only)
; FILLCNT/FILLADDR (formerly aliasing MVCNT/MVDST) were removed once
; genuinely unused - FILLW now keeps its count and address directly
; in Y/X rather than round-tripping through memory each iteration.
; ------------------------------------------------------------
        ORG   $0100
MVCNT      RMB   2
MVDST      RMB   2
MVSRC      RMB   2
HSLEN    EQU  MVCNT
HSADDR   EQU  MVDST
MRESULT  EQU  MVSRC
FILLCHR  EQU  MVSRC

; Provide padding, to ensure the correct ROM & .bin file size (and
; opcode offsets) for the MAME emulation and flash memory burn.
        ORG USROMSTRT

; ============================================================
; UNIT TEST FRAMEWORK
;
; Self-checking assembly-level tests for this ROM's own
; primitives, gated by UNITTESTS below and run once at cold
; boot, right after INITSERIAL and before COLD (so U/S are
; already valid - COLDSTRT sets them at its very start - but
; nothing else has been initialized yet: APPVARS/DPHERE/
; CODEHERE/LATEST/BASE all still hold whatever COLD is about to
; set them to). Lives here, in previously-unused ROM space right
; after INOUT's shadow, since this was pure FILL padding before
; this existed - UNITTESTS=1 removes it entirely and this block
; reverts to exactly that padding, computed automatically below
; via the ROM label rather than a fixed byte count, so it's
; correct either way without needing to be hand-adjusted.
;
; Each test is independent by construction: it saves the data
; stack pointer (U) before touching anything, and unconditionally
; restores it at the end regardless of pass or fail - so one
; test's assertions failing can never leave stack residue for the
; next test to inherit. Test scratch variables live in the very
; start of APPVARS - safe only because tests run strictly before
; COLD initializes VARHERE to that same address; COLD immediately
; and correctly re-purposes that space afterward.
;
; Reporting: each test's name (a counted string, matching how
; BADWORD itself prints a failing word) is printed via COUNT+TYPE,
```

```
; followed by " OK" or " FAIL", followed by a CR - all via
; TSTREPORT, shared by every test rather than duplicated in each.
; ============================================================
UNITTESTS EQU 0   ; 0 = included (matches this file's established
                  ; IFEQ convention, e.g. SERIALPOLL); 1 = excluded
                  ; entirely, reverting to plain FILL padding.

         IFEQ  UNITTESTS  ; >>>>>>>>>>

TSTU0    EQU   APPVARS       ; saved U, before a test touches it
TSTUB4   EQU   APPVARS+2     ; U immediately before the op under test
TSTUAF   EQU   APPVARS+4     ; U immediately after the op under test
TSTFLAG  EQU   APPVARS+6     ; scratch for TSTREPORT's pass/fail arg

TSTGUARD EQU   $3C7A         ; sentinel value, pushed below the value
                              ; under test, to prove an operation
                              ; doesn't disturb what's beneath it
TSTVAL1  EQU   $59E1         ; the value under test itself - neither
                              ; constant is 0, 1, or -1, so a test
                              ; that only appears to pass due to a
                              ; trivial/special-cased value would be
                              ; caught rather than masked

; ------------------------------------------------------------
; TSTREPORT - ( testname-caddr passflag -- ) shared by every
; test. Prints the test's name (via COUNT+TYPE), then " OK" or
; " FAIL" depending on passflag (TRUEV = pass, FALSEV = fail),
; then a CR, readying the terminal for the next test's line.
; ------------------------------------------------------------
TSTREPORT: PULU  D
           STD   TSTFLAG
           JSR   COUNT
           JSR   TYPE
           LDD   TSTFLAG
           BEQ   TSTFAILR
           LDD   #TSTOKMSG
           PSHU  D
           LDD   #TSTOKMSGL
           PSHU  D
           BRA   TSTPRINT
TSTFAILR:  LDD   #TSTFAILMSG
           PSHU  D
           LDD   #TSTFAILMSGL
           PSHU  D
TSTPRINT:  JSR   TYPE
           JSR   CRW
           RTS

TSTOKMSG:    FCC " OK"
TSTOKMSGL    EQU  *-TSTOKMSG
TSTFAILMSG:  FCC " FAIL"
TSTFAILMSGL  EQU  *-TSTFAILMSG

; ------------------------------------------------------------
; TSTRUNNER - calls each test group in turn. Add new groups
; here as they're written.
; ------------------------------------------------------------
TSTRUNNER: JSR   TSTSTACK
           RTS

; ------------------------------------------------------------
; TSTSTACK - data stack operation tests. Add new tests here as
; they're written.
; ------------------------------------------------------------
TSTSTACK:  JSR   TSTDUP
           RTS

; ------------------------------------------------------------
; TSTDUP - unit test for DUP ( x -- x x ). Verifies both the
; stack's contents (the duplicate and the original both equal
; the pushed test value, and the guard beneath is undisturbed)
```

```
; and the data stack pointer's movement (exactly one cell, 2
; bytes - DUP's own net effect, not conflated with the two
; pushes that set the test up).
; ------------------------------------------------------------
TSTDUP:    STU   TSTU0

           LDD   #TSTGUARD
           PSHU  D
           LDD   #TSTVAL1
           PSHU  D
           STU   TSTUB4

           JSR   DUP

           STU   TSTUAF

           PULU  D
           CMPD  #TSTVAL1
           BNE   TDFAIL
           PULU  D
           CMPD  #TSTVAL1
           BNE   TDFAIL
           PULU  D
           CMPD  #TSTGUARD
           BNE   TDFAIL

           LDD   TSTUB4
           SUBD  TSTUAF
           CMPD  #2
           BNE   TDFAIL

           LDD   #TRUEV
           BRA   TDDONE
TDFAIL:    LDD   #FALSEV
TDDONE:    LDX   #TSTDUPNAME
           PSHU  X
           PSHU  D
           JSR   TSTREPORT

           LDU   TSTU0
           RTS

TSTDUPNAME: FCB  6
            FCC  "TSTDUP"

         ENDC  ; <<<<<<<<<<

ROM:
         FILL $FF,BASEDICT-ROM
```

8.2 Hardware Vector Table

```
; ============================================================
; SECTION 1: HARDWARE VECTOR TABLE
; ============================================================
         ORG   VECTORS       ; VECTORS is $FFF0
VRESV    FDB   $0000
VSWI3    FDB   SWI3H
VSWI2    FDB   SWI2H
VFIRQ    FDB   FIRQH
VIRQ     FDB   IRQH
VSWI     FDB   SWIH
VNMI     FDB   WARM            ; NMI -> warm restart
VRESET   FDB   COLDSTRT

VECTOREND  EQU   *          ; Verify vectors size, value should match $10.
VECTORSIZE EQU   VECTOREND-VECTORS
```

```
; ============================================================
; END OF CONSOLIDATED SOURCE
; ============================================================
```

8.3 Init Code (COLDSTRT / WARM)

```
; ============================================================
; SECTION 2: INIT CODE (COLDSTRT / WARM)
; ============================================================
         ORG   INITCODE       ; INITCODE is $FFA2 (was INIT/$FFA0, before that literal $FFC0)
COLDSTRT:
         ORCC  #$50
         LDS   #RSTACK+1
         LDU   #DSTACK+1
         CLRA
         TFR   A,DP

         LDX   #GLOBALS
         LDB   #0
CLRGLOB: CLR   ,X+
         DECB
         BNE   CLRGLOB

         LDX   #SERBUF
         CLR   ,X+
         CLR   ,X

         JSR   INITSERIAL

         JMP   COLD

WARM:    ORCC  #$50
         CLRA
         TFR   A,DP
         LDU   #SP0
         LDS   #RP0
         LDX   #WARMMSG
         PSHU  X
         LDD   #WARMMSGL
         PSHU  D
         JSR   TYPE
         ANDCC #$AF
         JMP   ABORT

WARMMSG: FCC   "  warm"
WARMMSGL EQU   *-WARMMSG

INITEND  EQU   *          ; Verify no collision with vectors, value should match vector ORG
INITSIZE EQU   INITEND-INITCODE
```

8.4 ACIA Interrupt Handler

```
; ============================================================
; SECTION 3: ACIA INTERRUPT HANDLER
; ============================================================
         ORG   BASECODE       ; BASECODE is $DFF4. This ORG was missing
                              ; entirely - every routine from here through
                              ; SECTION 26 (IRQH, COLD/ABORT/QUIT, and
                              ; every primitive) would otherwise have
                              ; continued growing from wherever SECTION 2's
                              ; WARM message left the location counter,
                              ; inside INIT's 48-byte $FFC0-$FFEF budget,
                              ; overflowing directly into VECTORS ($FFF0)
                              ; instead of landing in BASECODE at all

; ------------------------------------------------------------
; INITSERIAL - initializes the ACIA: master reset, then selects
```

```
; interrupt-driven or polling operation depending on SERIALPOLL.
; Extracted from COLDSTRT, which now just JSRs here - moved into
; this section so the ACIA's own init code sits next to the rest
; of its interrupt/polling logic rather than inline in COLDSTRT.
; ------------------------------------------------------------
INITSERIAL: LDA   #$03
         STA   ACIACR         ; was "STA ACIA" - only correct by
                               ; coincidence while ACIA and ACIACR were
                               ; the same address; now genuinely distinct
         IFEQ SERIALPOLL  ; >>>>>>>>>>
         LDA   #CR_RXON        ; interrupt-driven mode: RX interrupt on
         ELSE  ; <<<<<>>>>>
         LDA   #CR_POLL        ; polling mode: no interrupts, RTS held low
         ENDC  ; <<<<<<<<<<
         STA   ACIACR         ; was "STA ACIA" - same fix
         RTS

         IFEQ SERIALPOLL  ; >>>>>>>>>>
; ------------------------------------------------------------
; INFILL - ( -- A=fill level, 0-63 ) input ring's current fill
; level. INBUFSZ is a power of two, and both indices are always
; kept in 0..INBUFSZ-1, so a plain masked subtraction gives the
; true mod-64 distance even across the wrap point.
; ------------------------------------------------------------
INFILL:  LDA   INHEAD
         SUBA  INTAIL
         ANDA  #INBUFSZ-1
         RTS

; ------------------------------------------------------------
; RTSCHECKHI - called from IRQH's own RX path (interrupts
; already masked by hardware during ISR execution, so no
; explicit masking needed here). If the input ring has reached
; INHIWATER and RTS is not already asserted high, assert it -
; telling the remote device to pause sending. Per the 6850, RTS
; has no automatic tie to reception; this is ordinary firmware
; flow control, not a chip feature.
; ------------------------------------------------------------
RTSCHECKHI: JSR  INFILL
            CMPA #INHIWATER
            BLO  RTSCHIDONE
            TST  RTSSTATE
            BNE  RTSCHIDONE       ; already high - nothing to do
            LDA  #CR_RTSHI
            STA  ACIACR
            LDA  #1
            STA  RTSSTATE
RTSCHIDONE: RTS

; ------------------------------------------------------------
; RTSCHECKLO - called from mainline code (KEY), NOT from the
; ISR, so it must mask IRQ around its critical section: IRQH's
; own TXOFF path also writes ACIACR, and an interrupt landing
; mid-decision here could otherwise race it. If the ring has
; drained to INLOWATER or below and RTS is currently high,
; reassert RTS low - restoring TX-interrupt-enable too if
; output happens to be queued, since the ACIA has no control
; byte combination offering RTS-high with TX-interrupt-enabled
; simultaneously (see CR_RTSHI's comment).
; ------------------------------------------------------------
RTSCHECKLO: JSR  INFILL
            CMPA #INLOWATER
            BHI  RTSCLODONE
            TST  RTSSTATE
            BEQ  RTSCLODONE       ; already low - nothing to do
            ORCC #$10              ; mask IRQ for the critical section
            CLR  RTSSTATE
            LDB  OUTTAIL
            CMPB OUTHEAD
            BEQ  RTSCLONOTX
            LDA  #CR_RXTX
```

```
            STA  ACIACR
            BRA  RTSCLOUNMASK
RTSCLONOTX: LDA  #CR_RXON
            STA  ACIACR
RTSCLOUNMASK: ANDCC #$EF
RTSCLODONE: RTS

IRQH:    LDA   ACIASR
         BITA  #SR_IRQ
         BEQ   IRQDONE

         BITA  #SR_RDRF
         BEQ   TXCHK

         LDB   INHEAD
         LDA   ACIADR
         LDX   #INBUF
         STA   B,X
         INCB
         ANDB  #INBUFSZ-1
         CMPB  INTAIL
         BEQ   IRQDONE
         STB   INHEAD
         JSR   RTSCHECKHI
         BRA   IRQDONE

TXCHK:   LDB   OUTTAIL
         CMPB  OUTHEAD
         BEQ   TXOFF
         LDX   #OUTBUF
         LDA   B,X
         STA   ACIADR
         INCB
         ANDB  #OUTBUFSZ-1
         STB   OUTTAIL
         BRA   IRQDONE

TXOFF:   TST   RTSSTATE
         BNE   IRQDONE         ; RTS is asserted high - leave ACIACR alone,
                               ; or this would incorrectly drop it back low
         LDA   #CR_RXON
         STA   ACIACR

IRQDONE: RTI

         ELSE  ; <<<<<>>>>>
IRQH:    RTI                ; polling mode (SERIALPOLL=1) - ACIA
                            ; interrupts are never enabled (see
                            ; COLDSTRT's CR_POLL init), so this should
                            ; never fire; kept as a safe stub matching
                            ; the other unused vectors below
         ENDC  ; <<<<<<<<<<

SWI3H:   RTI
SWI2H:   RTI
FIRQH:   RTI
NMIH:    RTI                  ; unused now that NMI -> WARM
SWIH:    LDD   #-99            ; placeholder hardware-trap code; push and
         PSHU  D                ; JMP THROW, per the CATCH/THROW turn
         JMP   THROW
```

8.5 COLD / ABORT / QUIT

```
; ============================================================
; SECTION 4: COLD / ABORT / QUIT  (with CATCH-wrapped INTERPRET)
; ============================================================
COLD:    LDD   #APPVARS
         STD   VARHERE
```

```
          LDD   #APPCODE
          STD   CODEHERE
          LDD   #APPDICT
          STD   DPHERE
          LDD   #BASELATEST
          STD   LATEST

          LDD   #10
          STD   BASE

          LDD   #TIBBUF
          STD   SRCADDR
          LDD   #0
          STD   SRCLEN
          STD   SRCID

          LDX   #SIGNON
          PSHU  X
          LDD   #SIGNONL
          PSHU  D
          JSR   TYPE
          JSR   CRW
          ; falls through into ABORT

ABORT:    LDU   #SP0
          ; falls through into QUIT

QUIT:     LDS   #RP0

QLOOP:    LDD   #0
          STD   STATE

          JSR   QUERY

          LDD   DPHERE
          STD   QSAVEDP
          LDD   CODEHERE
          STD   QSAVECODE
          LDD   VARHERE
          STD   QSAVEVAR
          LDD   LATEST
          STD   QSAVELATEST

          LDD   #INTERPRET
          PSHU  D
          JSR   CATCH
          PULU  D
          STD   QTHROWCODE
          CMPD  #0
          BEQ   QOK

          LDD   QSAVEDP
          STD   DPHERE
          LDD   QSAVECODE
          STD   CODEHERE
          LDD   QSAVEVAR
          STD   VARHERE
          LDD   QSAVELATEST
          STD   LATEST
          LDU   #SP0

          JSR   CRW
          LDX   #ERRMSG
          PSHU  X
          LDD   #ERRMSGL
          PSHU  D
          JSR   TYPE
          LDD   QTHROWCODE
          PSHU  D
          JSR   DOT
          BRA   QLOOP
```

```
QOK:     LDD   STATE
         BNE   QLOOP
         JSR   CRW
         LDX   #OKMSG
         PSHU  X
         LDD   #OKMSGL
         PSHU  D
         JSR   TYPE
         BRA   QLOOP

SIGNON:  FCC   "6809 FORTH v1.0"
SIGNONL  EQU   *-SIGNON
OKMSG:   FCC   "  ok"
OKMSGL   EQU   *-OKMSG
ERRMSG:  FCC   "  ERROR "
ERRMSGL  EQU   *-ERRMSG
```

8.6 Inner-Interpreter Support

```
; ============================================================
; SECTION 5: INNER-INTERPRETER SUPPORT (LIT, ZBRANCH, BRANCH,
; DODOES, DODEFER, EXECUTE)
; ============================================================
LIT:     PULS  X
         LDD   ,X++
         PSHU  D
         PSHS  X
         RTS

ZBRANCH: PULU  D
         PULS  X
         CMPD  #0
         BNE   ZSKIP
         LDD   ,X
         LEAX  D,X
         PSHS  X
         RTS
ZSKIP:   LEAX  2,X
         PSHS  X
         RTS

BRANCH:  PULS  X
         LDD   ,X
         LEAX  D,X
         PSHS  X
         RTS

DODOES:  PULS  X
         LDY   ,X++
         LDD   ,X
         PSHU  D
         JMP   ,Y

DOESRT0: RTS

; ------------------------------------------------------------
; SETDOES - compiled via JSR by DOES>'s immediate action.
; Patches LATEST's trampoline BEHAVIOR field, then returns
; two levels up - skipping the rest of the defining word's
; body entirely, straight back to whoever invoked it.
; ------------------------------------------------------------
SETDOES: PULS  X               ; X = addr right after "JSR SETDOES" - new BEHAVIOR
         STX   DOESBEH

         LDX   LATEST
         LDA   ,X
         STA   HDRFLAGS
```

```
         LEAX  1,X
         LDB   HDRFLAGS
         ANDB  #$1F
         CLRA
         LEAX  D,X              ; skip name -> LINK field
         LEAX  2,X               ; skip LINK -> CFA field
         LDD   ,X                 ; D = CFA (trampoline address)
         ADDD  #3                  ; +3 -> BEHAVIOR field (past JSR DODOES)
         TFR   D,X

         LDD   DOESBEH
         STD   ,X                   ; patch it

         PULS  X                     ; X = the OUTER defining word's own return addr
         JMP   ,X                    ; jump there directly - "double RTS"

DODEFER: PULU  X
         LDD   ,X
         TFR   D,X
         JMP   ,X

DOABORTUNDEF: LDD #-21
              PSHU D
              JMP  THROW

DOMARKER: PULU  X
          LDD   ,X
          STD   DPHERE
          LDD   2,X
          STD   CODEHERE
          LDD   4,X
          STD   VARHERE
          LDD   6,X
          STD   LATEST
          RTS

EXECUTE: PULU  X
         JSR   ,X
         RTS
```

8.7 Comma Family

```
; ============================================================
; SECTION 6: COMMA FAMILY (factored via APPENDCELL/APPENDBYTE)
; ============================================================
APPENDCELL: PULU D
            LDY   ,X
            STD   ,Y++
            STY   ,X
            RTS

APPENDBYTE: PULU D
            LDY   ,X
            STB   ,Y+
            STY   ,X
            RTS

COMMA:      LDX   #CODEHERE
            JMP   APPENDCELL

CODECOMMA:  LDX   #CODEHERE
            JMP   APPENDCELL

CCOMMA:     LDX   #CODEHERE
            JMP   APPENDBYTE

CCOMMA1:    LDX   #CODEHERE
            JMP   APPENDBYTE
```

```
VCOMMA:     LDX   #VARHERE
            JMP   APPENDCELL

VCCOMMA:    LDX   #VARHERE
            JMP   APPENDBYTE

ALLOT:    PULU  D
          LDX   CODEHERE
          LEAX  D,X
          STX   CODEHERE
          RTS

VALLOT:   PULU  D
          LDX   VARHERE
          LEAX  D,X
          STX   VARHERE
          RTS

HEREW:    LDD   CODEHERE
          PSHU  D
          RTS

VHEREW:   LDD   VARHERE
          PSHU  D
          RTS

PADW:     LDD   CODEHERE
          ADDD  #84
          PSHU  D
          RTS

UNUSEDW:  LDD   #CODETOP
          SUBD  CODEHERE
          PSHU  D
          RTS

VUNUSEDW: LDD  #APPVARSEND    ; was #APPCODE - a real bug, not just a
           SUBD VARHERE       ; missing check: computed a meaningless
           PSHU D             ; distance to an unrelated region instead
           RTS                ; of remaining APPVARS space
```

8.8 HEADER

```
; ============================================================
; SECTION 7: HEADER (factored from :/CREATE/VARIABLE)
; ============================================================
HEADER:   LDD   #32
          PSHU  D
          JSR   WORD
          PULU  X
          LDA   ,X
          STA   NAMELEN
          LEAX  1,X
          STX   NAMEP

          PULU  D
          STB   HDRSMUDGE

          LDD   DPHERE
          STD   NEWHDR
          LDX   DPHERE
          LDA   NAMELEN
          TST   HDRSMUDGE
          BEQ   HDNOSM
          ORA   #$40
HDNOSM:   STA   ,X+
          LDY   NAMEP
```

```
         LDB   NAMELEN
         BEQ   HDNONM
HDCPY:   LDA   ,Y+
         STA   ,X+
         DECB
         BNE   HDCPY
HDNONM:  LDD   LATEST
         STD   ,X++
         LDD   CODEHERE
         STD   ,X++
         STX   DPHERE
         LDD   NEWHDR
         STD   LATEST
         RTS
```

8.9 Defining Words

```
; ============================================================
; SECTION 8: DEFINING WORDS
; ============================================================
COLON:   LDD   #TRUEV
         PSHU  D
         JSR   HEADER
         TFR   U,D
         STD   CSP
         LDD   #-1
         STD   STATE
         RTS

SEMI:    LDD   #RTSOPC
         PSHU  D
         JSR   CCOMMA1
         TFR   U,D
         CMPD  CSP
         BEQ   SEMIOK
         JSR   CFERR
SEMIOK:  LDX   LATEST
         LDA   ,X
         ANDA  #$BF
         STA   ,X
         LDD   #0
         STD   STATE
         RTS

CREATE:  LDD   #0
         PSHU  D
         JSR   HEADER
         LDD   #DODOES
         PSHU  D
         JSR   CCALL
         LDD   #DOESRT0
         PSHU  D
         JSR   CODECOMMA
         LDD   CODEHERE
         PSHU  D
         JSR   CODECOMMA
         RTS

; ------------------------------------------------------------
; DOES> ( -- )  IMMEDIATE, compile-only. Compiles a call to
; SETDOES. Code label DOESGT, not "DOES>" - a literal ">" is
; not valid in a 6809 assembler label, same reason ?DUP/2DUP/
; etc. all use mnemonic labels rather than their literal names.
; ------------------------------------------------------------
DOESGT:  LDD   #SETDOES
         PSHU  D
         JSR   CCALL
         RTS
```

```
VARIABLE: LDD  #0
          PSHU D
          JSR  HEADER
          LDD  #DODOES
          PSHU D
          JSR  CCALL
          LDD  #DOESRT0
          PSHU D
          JSR  CODECOMMA
          LDD  VARHERE
          PSHU D
          JSR  CODECOMMA
          LDD  #0
          LDX  VARHERE
          STD  ,X++
          STX  VARHERE
          RTS

ATSIGN:  PULU  X
         LDD   ,X
         PSHU  D
         RTS

CONSTANT: LDD  #0
          PSHU D
          JSR  HEADER
          LDD  #DODOES
          PSHU D
          JSR  CCALL
          LDD  #ATSIGN
          PSHU D
          JSR  CODECOMMA
          LDD  CODEHERE
          PSHU D
          JSR  CODECOMMA
          JSR  COMMA
          RTS

DOVALUE: PULU  X
         LDD   ,X
         PSHU  D
         RTS

VALUEW:  LDD   #0
         PSHU  D
         JSR   HEADER          ; not smudged - immediately findable
         LDD   #DODOES
         PSHU  D
         JSR   CCALL
         LDD   #DOVALUE
         PSHU  D
         JSR   CODECOMMA         ; trampoline itself is still code
         LDD   VARHERE            ; PFA = VARHERE, mutable space - was
         PSHU  D                   ; CODEHERE; TO writes through this PFA,
         JSR   CODECOMMA            ; so it must live in mutable space
         JSR   VCOMMA                ; store x into VARHERE via VCOMMA,
                                       ; not COMMA (which targets CODEHERE)
         RTS

TOW:     LDD   #32
         PSHU  D
         JSR   WORD
         JSR   FIND
         PULU  D
         CMPD  #0
         BNE   TOFOUND
         PULU  D
         LDD   #-13
         PSHU  D
         JSR   THROW
```

```
TOFOUND: JSR   TOBODY
         LDD   STATE
         BEQ   TOIMMED
         JSR   LITERALW
         LDD   #STOREW
         PSHU  D
         JSR   CCALL
         RTS
TOIMMED: PULU  X
         PULU  D
         STD   ,X
         RTS

TWOVARIABLE: LDD #0
             PSHU D
             JSR HEADER
             LDD #DODOES
             PSHU D
             JSR CCALL
             LDD #DOESRT0
             PSHU D
             JSR CODECOMMA
             LDD VARHERE
             PSHU D
             JSR CODECOMMA
             LDD #0
             LDX VARHERE
             STD ,X++
             STD ,X++
             STX VARHERE
             RTS

TWOCONSTANT: LDD #0
             PSHU D
             JSR HEADER
             LDD #DODOES
             PSHU D
             JSR CCALL
             LDD #DFETCH
             PSHU D
             JSR CODECOMMA
             LDD CODEHERE
             PSHU D
             JSR CODECOMMA
             PULU D              ; x2, off the top
             STD  MSCR
             JSR  COMMA            ; x1 -> lower address
             LDD  MSCR
             PSHU D
             JSR  COMMA              ; x2 -> higher address
             RTS

BUFFERCOLON: PULU D
             STD  MSCR2
             LDD  #0
             PSHU D
             JSR  HEADER
             LDD  #DODOES
             PSHU D
             JSR  CCALL
             LDD  #DOESRT0
             PSHU D
             JSR  CODECOMMA
             LDD  VARHERE
             PSHU D
             JSR  CODECOMMA
             LDD  MSCR2
             PSHU D
             JSR  VALLOT
             RTS
```

```
DEFERW:  LDD   #0
         PSHU  D
         JSR   HEADER
         LDD   #DODOES
         PSHU  D
         JSR   CCALL
         LDD   #DODEFER
         PSHU  D
         JSR   CODECOMMA
         LDD   CODEHERE
         PSHU  D
         JSR   CODECOMMA
         LDD   #DOABORTUNDEF
         PSHU  D
         JSR   COMMA
         RTS

DEFERFETCH: JSR TOBODY
            PULU X
            LDD  ,X
            PSHU D
            RTS

DEFERSTORE: JSR TOBODY
            PULU X
            PULU D
            STD  ,X
            RTS

ISW:     LDD   #32
         PSHU  D
         JSR   WORD
         JSR   FIND
         PULU  D
         CMPD  #0
         BNE   ISFOUND
         PULU  D
         LDD   #-13
         PSHU  D
         JSR   THROW
ISFOUND: PULU  X
         LDD   STATE
         BEQ   ISIMMED
         PSHU  X
         JSR   LITERALW
         LDD   #DEFERSTORE
         PSHU  D
         JSR   CCALL
         RTS
ISIMMED: PSHU  X
         JSR   DEFERSTORE
         RTS

ACTIONOF: LDD  #32
          PSHU D
          JSR  WORD
          JSR  FIND
          PULU D
          CMPD #0
          BNE  AOFOUND
          PULU D
          LDD  #-13
          PSHU D
          JSR  THROW
AOFOUND:  PULU X
          LDD  STATE
          BEQ  AOIMMED
          PSHU X
          JSR  LITERALW
          LDD  #DEFERFETCH
          PSHU D
```

```
          JSR  CCALL
          RTS
AOIMMED:  PSHU X
          JSR  DEFERFETCH
          RTS

MARKERW: LDD   DPHERE
         STD   MKDP
         LDD   CODEHERE
         STD   MKCODE
         LDD   VARHERE
         STD   MKVAR
         LDD   LATEST
         STD   MKLATEST
         LDD   #0
         PSHU  D
         JSR   HEADER
         LDD   #DODOES
         PSHU  D
         JSR   CCALL
         LDD   #DOMARKER
         PSHU  D
         JSR   CODECOMMA
         LDD   CODEHERE
         PSHU  D
         JSR   CODECOMMA
         LDD   MKDP
         PSHU  D
         JSR   COMMA
         LDD   MKCODE
         PSHU  D
         JSR   COMMA
         LDD   MKVAR
         PSHU  D
         JSR   COMMA
         LDD   MKLATEST
         PSHU  D
         JSR   COMMA
         RTS
```

8.10 Outer Interpreter

```
; ============================================================
; SECTION 9: OUTER INTERPRETER (INTERPRET / WORD / FIND / NUMBER?)
; ============================================================
INTERPRET:
ILOOP:   JSR   WORD
         LDX   ,U
         LDA   ,X
         BEQ   IDONE

         JSR   FIND
         PULU  D
         TSTB
         LBEQ  TRYNUM

         LDA   STATE+1
         BEQ   DOEXEC
         TSTB
         BPL   DOEXEC
         JSR   CCALL
         BRA   ILOOP

DOEXEC:  JSR   EXECUTE
         BRA   ILOOP

TRYNUM:  JSR   NUMBERQ
         PULU  D
```

```
          CMPD  #0            ; was TSTD (6309-only) - PULU doesn't set CC on
                               ; genuine 6809, so compare D against 0 directly
          BEQ   BADWORD

          LDD   STATE
          BEQ   ILOOP
          LDD   #LIT
          PSHU  D
          JSR   CCALL
          JSR   CODECOMMA
          BRA   ILOOP

BADWORD:  JSR   COUNT
          JSR   TYPE
          LDD   #-13
          PSHU  D
          JSR   THROW

IDONE:    RTS

WORD:     PULU  D
          STB   DELIM
          LDD   TOIN
          LDX   SRCADDR
          LEAX  D,X
          LDD   SRCLEN
          SUBD  TOIN
          TFR   D,Y

SKIPLP:   CMPY  #0
          BEQ   EMPTY
          LDA   ,X
          CMPA  DELIM
          BNE   STARTW
          LEAX  1,X
          LEAY  -1,Y
          BRA   SKIPLP

STARTW:   STX   WSTART
          LDB   #0

SCANLP:   CMPY  #0
          BEQ   ENDW
          LDA   ,X
          CMPA  DELIM
          BEQ   CONSUME
          CMPB  #31
          BEQ   ENDW
          LEAX  1,X
          LEAY  -1,Y
          INCB
          BRA   SCANLP

CONSUME:  LEAX  1,X
          LEAY  -1,Y
ENDW:     TFR   X,D
          SUBD  SRCADDR
          STD   TOIN

          LDX   #WORDBUF
          STB   ,X+
          LDY   WSTART
COPYLP:   TSTB
          BEQ   COPYDONE
          LDA   ,Y+
          STA   ,X+
          DECB
          BRA   COPYLP
COPYDONE:  LDX  #WORDBUF
           PSHU X
           RTS
```

```
EMPTY:    LDX   #WORDBUF
          CLR   ,X
          PSHU  X
          RTS

FIND:     PULU  X
          LDA   ,X
          STA   SLEN
          LEAX  1,X
          STX   SNAMEP

          LDD   LATEST
          STD   FNDPTR

FFLOOP:   LDD   FNDPTR
          BEQ   NOTFOUND
          STD   HDRPTR
          TFR   D,X
          LDA   ,X
          STA   HDRFLAGS
          BITA  #$40
          BNE   FNEXT
          ANDA  #$1F
          CMPA  SLEN
          BNE   FNEXT
          LEAX  1,X
          LDY   SNAMEP
          LDB   SLEN
          BEQ   FMATCH
CMPLP:    LDA   ,X+
          CMPA  ,Y+
          BNE   FNEXT
          DECB
          BNE   CMPLP

FMATCH:   LDX   HDRPTR
          LEAX  1,X
          LDB   HDRFLAGS
          ANDB  #$1F
          CLRA
          LEAX  D,X
          LEAX  2,X
          LDD   ,X
          PSHU  D
          LDA   HDRFLAGS
          BITA  #$80
          BEQ   FISNORM
          LDD   #1
          BRA   FPUSH
FISNORM:  LDD   #-1
FPUSH:    PSHU  D
          RTS

FNEXT:    LDX   HDRPTR
          LEAX  1,X
          LDB   HDRFLAGS
          ANDB  #$1F
          CLRA
          LEAX  D,X
          LDD   ,X
          STD   FNDPTR
          BRA   FFLOOP

NOTFOUND: LDX   SNAMEP
           LEAX -1,X
           PSHU X
           LDD  #0
           PSHU D
           RTS
```

```
UDMULADD: STB  CARRY
          LDA  BASE+1
          STA  MULBASE
          LDA  UDLO+1
          LDB  MULBASE
          MUL
          ADDB CARRY
          BCC  UM0
          INCA
UM0:      STB  UDLO+1
          STA  CARRY
          LDA  UDLO
          LDB  MULBASE
          MUL
          ADDB CARRY
          BCC  UM1
          INCA
UM1:      STB  UDLO
          STA  CARRY
          LDA  UDHI+1
          LDB  MULBASE
          MUL
          ADDB CARRY
          BCC  UM2
          INCA
UM2:      STB  UDHI+1
          STA  CARRY
          LDA  UDHI
          LDB  MULBASE
          MUL
          ADDB CARRY
          BCC  UM3
          INCA
UM3:      STB  UDHI
          RTS

NUMLOOP: LDD   NCNT
         BEQ   NLDONE
         LDX   NADDR
         LDA   ,X
         CMPA  #'0'
         BLO   NLDONE
         CMPA  #'9'
         BHI   NLALPHA
         SUBA  #'0'
         BRA   NLGOT
NLALPHA: ANDA  #$DF
         CMPA  #'A'
         BLO   NLDONE
         CMPA  #'Z'
         BHI   NLDONE
         SUBA  #'A'-10
NLGOT:   CMPA  BASE+1
         BHS   NLDONE
         TFR   A,B
         JSR   UDMULADD
         LDX   NADDR
         LEAX  1,X
         STX   NADDR
         LDD   NCNT
         SUBD  #1
         STD   NCNT
         BRA   NUMLOOP
NLDONE:  RTS

TONUMBER: PULU D
          STD  NCNT
          PULU D
          STD  NADDR
          PULU D
          STD  UDHI
```

```
          PULU D
          STD  UDLO
          JSR  NUMLOOP
          LDD  UDLO
          PSHU D
          LDD  UDHI
          PSHU D
          LDX  NADDR
          PSHU X
          LDD  NCNT
          PSHU D
          RTS

NUMBERQ: PULU  X
         STX   CADDR
         LDA   ,X
         BEQ   NQBAD
         STA   CNTREM
         LEAX  1,X

         CLR   NUMNEG
         LDA   ,X
         CMPA  #'-'
         BNE   NQNOSIGN
         COM   NUMNEG
         LEAX  1,X
         DEC   CNTREM
         BEQ   NQBAD

NQNOSIGN: STX  NADDR
          CLRA
          LDB   CNTREM
          STD   NCNT
          LDD   #0
          STD   UDHI
          STD   UDLO

          JSR   NUMLOOP

          LDD   NCNT
          BNE   NQBAD

          LDD   UDLO
          TST   NUMNEG
          BEQ   NQPOS
          COMA
          COMB
          ADDD  #1
NQPOS:    PSHU  D
          LDD   #-1
          PSHU  D
          RTS

NQBAD:    LDX   CADDR
          PSHU  X
          LDD   #0
          PSHU  D
          RTS
```

8.11 QUERY / ACCEPT / EXPECT / KEY / KEY? / EMIT

```
; ============================================================
; SECTION 10: QUERY / ACCEPT / EXPECT / KEY / KEY? / EMIT
; ============================================================
         IFEQ SERIALPOLL  ; >>>>>>>>>>
KEY:     LDA   INHEAD
         CMPA  INTAIL
         BEQ   KEY
```

```
         LDX   #INBUF
         LDB   INTAIL
         LDA   B,X
         INCB
         ANDB  #INBUFSZ-1
         STB   INTAIL
         PSHS  A               ; stash the char on the return stack across
                                ; the call - JSR/RTS is self-balancing, so
                                ; this needs no dedicated scratch global
         JSR   RTSCHECKLO
         PULS  A
         TFR   A,B
         CLRA
         PSHU  D
         RTS

KEYQ:    LDA   INHEAD
         CMPA  INTAIL
         BNE   KQTRUE
         LDD   #FALSEV
         PSHU  D
         RTS
KQTRUE:  LDD   #TRUEV
         PSHU  D
         RTS

EMIT:    PULU  D
         STB   EMITCH
EMITWT:  LDB   OUTHEAD
         INCB
         ANDB  #OUTBUFSZ-1
         CMPB  OUTTAIL
         BEQ   EMITWT
         LDX   #OUTBUF
         LDB   OUTHEAD
         LDA   EMITCH
         STA   B,X
         INCB
         ANDB  #OUTBUFSZ-1
         STB   OUTHEAD
         TST   RTSSTATE
         BNE   EMITNORTS       ; RTS is asserted high - leave ACIACR alone;
                                ; output stays queued until RTS drops low,
                                ; at which point RTSCHECKLO re-enables TX
                                ; interrupt itself if OUTBUF still has data
         LDA   #CR_RXTX
         STA   ACIACR
EMITNORTS: RTS

         ELSE  ; <<<<<>>>>>
; ------------------------------------------------------------
; Polling versions of KEY/KEYQ/EMIT (SERIALPOLL=1) - no ring
; buffers, no interrupts, no RTS/CTS handshaking. Each blocks
; (KEY, EMIT) or checks once (KEYQ) directly against ACIASR.
; ------------------------------------------------------------
KEY:     LDA   ACIASR
         BITA  #SR_RDRF
         BEQ   KEY
         LDA   ACIADR
         TFR   A,B
         CLRA
         PSHU  D
         RTS

KEYQ:    LDA   ACIASR
         BITA  #SR_RDRF
         BEQ   KQFALSE
         LDD   #TRUEV
         PSHU  D
         RTS
KQFALSE: LDD   #FALSEV
```

```
          PSHU  D
          RTS

EMIT:     PULU  D
          STB   EMITCH
EMITWT:   LDA   ACIASR
          BITA  #SR_TDRE
          BEQ   EMITWT
          LDA   EMITCH
          STA   ACIADR
          RTS
          ENDC  ; <<<<<<<<<<

ACCEPT:   PULU  D
          STD   AMAX
          PULU  D
          STD   ABUFP
          LDD   #0
          STD   ACNT

ALOOP:    JSR   KEY
          PULU  D
          STB   ACH

          CMPB  #13
          BEQ   ADONE
          CMPB  #10
          BEQ   ALOOP
          CMPB  #8
          BEQ   ABKSP
          CMPB  #127
          BEQ   ABKSP

          LDD   ACNT
          CMPD  AMAX
          BEQ   ALOOP

          LDX   ABUFP
          LEAX  D,X
          LDA   ACH
          STA   ,X
          LDD   ACNT
          ADDD  #1
          STD   ACNT

          CLRA
          LDB   ACH
          PSHU  D
          JSR   EMIT
          BRA   ALOOP

ABKSP:    LDD   ACNT
          BEQ   ALOOP
          SUBD  #1
          STD   ACNT
          LDD   #8
          PSHU  D
          JSR   EMIT
          LDD   #32
          PSHU  D
          JSR   EMIT
          LDD   #8
          PSHU  D
          JSR   EMIT
          BRA   ALOOP

ADONE:    LDD   ACNT
          PSHU  D
          RTS

EXPECTW:  JSR   ACCEPT
```

```
         PULU  D
         STD   SPAN
         RTS

QUERY:   LDX   #TIBBUF
         PSHU  X
         LDD   #TIBBUFL
         PSHU  D
         JSR   ACCEPT
         PULU  D
         STD   NTIB
         STD   SRCLEN
         LDD   #TIBBUF
         STD   SRCADDR
         LDD   #0
         STD   SRCID
         STD   TOIN
         RTS
```

8.12 CATCH / THROW

```
; ============================================================
; SECTION 11: COLON / SEMICOLON support already in section 8
; (COLON/SEMI) - CATCH/THROW, CFERR
; ============================================================
CFERR:   LDD   #-22
         PSHU  D
         JSR   THROW
         RTS

CATCH:   PULU  X
         LDD   HANDLER
         PSHS  D
         PSHS  U
         TFR   S,D
         STD   HANDLER

         JSR   ,X

         LEAS  2,S
         PULS  D
         STD   HANDLER
         LDD   #0
         PSHU  D
         RTS

THROW:   PULU  D
         CMPD  #0
         BEQ   THDONE

         STD   THROWN
         LDX   HANDLER
         BEQ   THUNCAU

         TFR   X,S
         PULS  D
         TFR   D,U
         PULS  D
         STD   HANDLER

         LDD   THROWN
         PSHU  D
         RTS
THDONE:  RTS

THUNCAU: LDD   THROWN
         PSHU  D
         JMP   ABORT
```

8.13 Control Flow

```
; ============================================================
; SECTION 12: CONTROL FLOW (IF/THEN/ELSE, BEGIN family,
; DO/LOOP/+LOOP/I/J/LEAVE/UNLOOP/?DO, EXIT, CASE family)
; ============================================================
PATCH:   PULU  D
         STD   PFIELD
         PULU  D
         STD   PTARGET
         LDD   PTARGET
         SUBD  PFIELD
         LDX   PFIELD
         STD   ,X
         RTS

IF:      LDD   #ZBRANCH
         PSHU  D
         JSR   CCALL
         LDD   #0
         PSHU  D
         JSR   CODECOMMA
         LDD   CODEHERE
         SUBD  #2
         PSHU  D
         LDD   #TAGFWD
         PSHU  D
         RTS

THEN:    PULU  D
         CMPD  #TAGFWD
         BEQ   THOK
         JSR   CFERR
THOK:    PULU  X
         LDD   CODEHERE
         PSHU  D
         PSHU  X
         JSR   PATCH
         RTS

ELSE:    PULU  D
         CMPD  #TAGFWD
         BEQ   ELOK
         JSR   CFERR
ELOK:    PULU  D          ; BUG FIX: same class as LOOP/EOFOK - was PULU X
         STD   MSCR       ; then used via PSHU X after CCALL/CODECOMMA
                          ; below, both of which clobber X internally.
                          ; Saved to scratch (not parked on U, since
                          ; CODEHERE gets pushed onto U further down,
                          ; between the save and the retrieve).
         LDD   #BRANCH
         PSHU  D
         JSR   CCALL
         LDD   #0
         PSHU  D
         JSR   CODECOMMA
         LDD   CODEHERE
         SUBD  #2
         STD   NEWFLD
         LDD   CODEHERE
         PSHU  D
         LDD   MSCR
         PSHU  D
         JSR   PATCH
         LDD   NEWFLD
         PSHU  D
         LDD   #TAGFWD
```

```
          PSHU  D
          RTS

BEGIN:    LDD   CODEHERE
          PSHU  D
          LDD   #TAGBACK
          PSHU  D
          RTS

UNTIL:    PULU  D
          CMPD  #TAGBACK
          BEQ   UNOK
          JSR   CFERR
UNOK:     PULU  D           ; BUG FIX: same class as LOOP - was PULU X then
          PSHU  D           ; TFR X,D after CCALL/CODECOMMA, both of which
                            ; clobber X internally. Parked on U instead.
          LDD   #ZBRANCH
          PSHU  D
          JSR   CCALL
          LDD   #0
          PSHU  D
          JSR   CODECOMMA
          LDD   CODEHERE
          SUBD  #2
          STD   PFIELD
          PULU  D
          PSHU  D
          LDD   PFIELD
          PSHU  D
          JSR   PATCH
          RTS

AGAIN:    PULU  D
          CMPD  #TAGBACK
          BEQ   AGOK
          JSR   CFERR
AGOK:     PULU  D           ; BUG FIX: same class as LOOP - was PULU X then
          PSHU  D           ; TFR X,D after CCALL/CODECOMMA, both of which
                            ; clobber X internally. Parked on U instead.
          LDD   #BRANCH
          PSHU  D
          JSR   CCALL
          LDD   #0
          PSHU  D
          JSR   CODECOMMA
          LDD   CODEHERE
          SUBD  #2
          STD   PFIELD
          PULU  D
          PSHU  D
          LDD   PFIELD
          PSHU  D
          JSR   PATCH
          RTS

WHILE:    LDD   #ZBRANCH
          PSHU  D
          JSR   CCALL
          LDD   #0
          PSHU  D
          JSR   CODECOMMA
          LDD   CODEHERE
          SUBD  #2
          PSHU  D
          LDD   #TAGFWD
          PSHU  D
          RTS

REPEAT:   PULU  D
          CMPD  #TAGFWD
          BEQ   RPOK1
```

```
          JSR   CFERR
RPOK1:    PULU  X
          STX   NEWFLD
          PULU  D
          CMPD  #TAGBACK
          BEQ   RPOK2
          JSR   CFERR
RPOK2:    PULU  D           ; BUG FIX: same class as LOOP - was PULU X then
          PSHU  D           ; TFR X,D after CCALL/CODECOMMA, both of which
                            ; clobber X internally. Parked on U instead.
          LDD   #BRANCH
          PSHU  D
          JSR   CCALL
          LDD   #0
          PSHU  D
          JSR   CODECOMMA
          LDD   CODEHERE
          SUBD  #2
          STD   PFIELD
          PULU  D
          PSHU  D
          LDD   PFIELD
          PSHU  D
          JSR   PATCH
          LDD   CODEHERE
          PSHU  D
          LDD   NEWFLD
          PSHU  D
          JSR   PATCH
          RTS

RECURSE:  LDX   LATEST
          LDA   ,X
          STA   HDRFLAGS
          LEAX  1,X
          LDB   HDRFLAGS
          ANDB  #$1F
          CLRA
          LEAX  D,X
          LEAX  2,X
          LDD   ,X
          PSHU  D
          JSR   CCALL
          RTS

DO:       LDD   #DOSETUP
          PSHU  D
          JSR   CCALL
          LDD   CODEHERE
          PSHU  D
          LDD   #TAGDO
          PSHU  D
          RTS

DOSETUP:  PULU  D
          STD   MSCR
          PULU  D
          STD   MSCR2
          PULS  X
          LDD   #0
          PSHS  D
          LDD   MSCR2
          PSHS  D
          LDD   MSCR
          PSHS  D
          PSHS  X
          RTS

IWORD:    LDD   2,S         ; BUG FIX: PULS X/PSHS X used to bracket this
          PSHU  D           ; read, shifting S by 2 first - so "2,S" landed
          RTS               ; on the limit (what's really at 4,S unshifted)
```

```
                        ; instead of the index. The pair served no
                        ; purpose (nothing needed offset 0 for anything
                        ; here) - removed, so 2,S directly and
                        ; correctly targets the index DOSETUP pushed
                        ; there. Confirmed via MAME debugger: "I"
                        ; returned the loop limit on every iteration.

JWORD:  LDD   10,S      ; BUG FIX: same class as IWORD above - the
        PSHU  D         ; PULS X/PSHS X pair shifted S by 2 first, so
        RTS             ; "10,S" landed on the outer loop's limit
                        ; instead of its index. Removed for the same
                        ; reason - 10,S unshifted already correctly
                        ; targets the outer index across a nested nest
                        ; of nested DOSETUP frames.

LEAVE:  LDD   #TRUEV    ; BUG FIX (earlier): was PULS X first, shifting
        STD   6,S       ; S by 2 before this write, landing 2 bytes
        RTS             ; past the LEAVE flag DOSETUP actually pushed.
                        ; Fixed by deferring PULS X to after the
                        ; write - but that left a PULS X/PSHS X pair
                        ; that had become a genuine no-op (nothing
                        ; between them touches S or X, and this
                        ; routine never uses X's value for anything,
                        ; unlike DOTEST's own deferred PULS X, which
                        ; feeds LDD ,X/LEAX D,X afterward). Removed,
                        ; per the parallel 68000 port's observation,
                        ; confirmed by inspection.

LOOP:   PULU  D
        CMPD  #TAGDO
        BEQ   LOOPOK
        JSR   CFERR
LOOPOK: PULU  D         ; BUG FIX: was PULU X, then TFR X,D to retrieve
        PSHU  D         ; it after CCALL/CODECOMMA below - but both of
                        ; those clobber X internally (CCALL's own LDX
                        ; CODEHERE, CODECOMMA's LDX #CODEHERE via
                        ; APPENDCELL), so X no longer held the target
                        ; by the time TFR X,D ran - it held the address
                        ; of the CODEHERE variable itself, corrupting
                        ; the branch displacement PATCH computed.
                        ; Parking the target on U instead (untouched by
                        ; CCALL/CODECOMMA, which only use D/X/A) keeps
                        ; it safe across both calls; retrieved below via
                        ; PULU D in place of the old TFR X,D. Confirmed
                        ; via MAME debugger: the compiled displacement
                        ; came out as +8, executing DOTEST's return to
                        ; a near-zero, invalid address.
        LDD   #DOTEST
        PSHU  D
        JSR   CCALL
        LDD   #0
        PSHU  D
        JSR   CODECOMMA
        LDD   CODEHERE
        SUBD  #2
        STD   PFIELD
        PULU  D
        PSHU  D
        LDD   PFIELD
        PSHU  D
        JSR   PATCH

        LDD   ,U
        CMPD  #TAGFWD
        BNE   LOOPDONE
        PULU  D
        PULU  X
        LDD   CODEHERE
        PSHU  D
        PSHU  X
        JSR   PATCH
```

```
LOOPDONE: RTS

DOTEST:  LDD   6,S       ; BUG FIX: PULS X used to run FIRST, shifting S
         BNE   DTEXIT    ; by 2 before these offset reads - "6,S" ended
         LDD   2,S       ; up reading past the 3 cells DOSETUP pushed,
         ADDD  #1        ; into whatever return address was already on
         STD   2,S       ; S below them (e.g. EXECUTE's own), which is
         CMPD  4,S       ; almost always nonzero - so BNE DTEXIT fired
         BEQ   DTEXIT    ; on the very first pass, exiting immediately.
         PULS  X         ; X (the return address to the FDB <offset>
         LDD   ,X        ; field, needed for the back-branch) is only
         LEAX  D,X       ; actually needed here and in DTEXIT below -
         PSHS  X         ; deferred to each path separately instead of
         RTS             ; popped once up front. Confirmed via MAME
                          ; debugger: DOTEST was tested at $E184, a
                          ; leftover EXECUTE return address, instead of
                          ; the intended $0000 LEAVE flag.
DTEXIT:  PULS  X
         LEAX  2,X
         LEAS  6,S
         PSHS  X
         RTS

PLUSLOOP: PULU D
          CMPD #TAGDO
          BEQ  PLOOPOK
          JSR  CFERR
PLOOPOK:  PULU D          ; BUG FIX: same class as LOOP above - was PULU X
          PSHU D          ; then TFR X,D after CCALL/CODECOMMA, both of
                          ; which clobber X internally. Parked on U
                          ; instead, retrieved below via PULU D.
          LDD  #DOPLUSTEST
          PSHU D
          JSR  CCALL
          LDD  #0
          PSHU D
          JSR  CODECOMMA
          LDD  CODEHERE
          SUBD #2
          STD  PFIELD
          PULU D
          PSHU D
          LDD  PFIELD
          PSHU D
          JSR  PATCH

          LDD  ,U
          CMPD #TAGFWD
          BNE  PLOOPDONE
          PULU D
          PULU X
          LDD  CODEHERE
          PSHU D
          PSHU X
          JSR  PATCH
PLOOPDONE: RTS

DOPLUSTEST: LDD  6,S      ; BUG FIX: same class as DOTEST - PULS X used to
            BNE  DPTEXIT  ; run first, shifting S by 2 before every one of
            PULU D        ; these offset reads (6,S/2,S/4,S), landing past
            STD  MSCR     ; the 3 cells DOSETUP pushed. Deferred PULS X to
            LDD  2,S      ; each path separately below, same fix as DOTEST.
            SUBD 4,S
            STD  MSCR2
            LDD  2,S
            ADDD MSCR
            STD  2,S
            SUBD 4,S
            STD  MSCR3
            LDA  MSCR2
            LDB  MSCR3
```

```
            PSHS B
            EORA ,S+          ; was "EORA B" - not valid 6809 syntax (no
                              ; register-to-register EORA); push B, then
                              ; operate through ,S+ - the standard 6809
                              ; idiom for adding/combining two registers
            BMI  DPTEXIT
            LDD  MSCR3
            BEQ  DPTEXIT
            PULS X
            LDD  ,X
            LEAX D,X
            PSHS X
            RTS
DPTEXIT:    PULS X
            LEAX 2,X
            LEAS 6,S
            PSHS X
            RTS

QDO:      LDD   #QDOSETUP
          PSHU  D
          JSR   CCALL
          LDD   #0
          PSHU  D
          JSR   CODECOMMA
          LDD   CODEHERE
          SUBD  #2
          PSHU  D
          LDD   #TAGFWD
          PSHU  D
          LDD   CODEHERE
          PSHU  D
          LDD   #TAGDO
          PSHU  D
          RTS

QDOSETUP: PULU D
           STD  MSCR
           PULU D
           STD  MSCR2
           PULS X
           LDD  MSCR2
           CMPD MSCR
           BNE  QDBUILD
           LDD  ,X
           LEAX D,X
           PSHS X
           RTS
QDBUILD:   LEAX 2,X
           LDD  #0
           PSHS D
           LDD  MSCR2
           PSHS D
           LDD  MSCR
           PSHS D
           PSHS X
           RTS

UNLOOP:  RTS             ; DESIGN CHANGE: was PULS X/LEAS 6,S/PSHS X/RTS
                         ; (a real discard of the loop-control frame).
                         ; EXIT's own EXITUNLOOP mechanism already
                         ; discards the frame automatically and
                         ; correctly on every path - traced and
                         ; confirmed: with no enclosing DO (compiled
                         ; count 0), with one enclosing DO and no
                         ; prior UNLOOP (discards the full 8-byte
                         ; frame correctly). The one combination that
                         ; broke was UNLOOP immediately before EXIT -
                         ; UNLOOP discarding 6 of the 8 bytes itself,
                         ; then EXITUNLOOP unconditionally discarding
                         ; a full 8 more, overshooting into whatever
```

```
                        ; sat below (typically the caller's own
                        ; return address) and branching into random
                        ; memory on return. Since UNLOOP has no
                        ; other legitimate use than immediately
                        ; preceding an exit from the definition, and
                        ; EXIT already handles that correctly on its
                        ; own, UNLOOP is now a true no-op rather than
                        ; a second, conflicting discard mechanism.

EXIT:    LDD   #0
         STD   EXITCNT
         TFR   U,D
         STD   EXITPTR
EXSCAN:  LDD   EXITPTR
         CMPD  CSP
         BEQ   EXSCANDONE
         LDX   EXITPTR
         LDD   ,X
         CMPD  #TAGDO
         BNE   EXNOTDO
         LDD   EXITCNT
         ADDD  #1
         STD   EXITCNT
EXNOTDO: LDD   EXITPTR
         ADDD  #4
         STD   EXITPTR
         BRA   EXSCAN
EXSCANDONE:
         LDD   #EXITUNLOOP
         PSHU  D
         JSR   CCALL
         LDD   EXITCNT
         PSHU  D
         JSR   CODECOMMA
         RTS

EXITUNLOOP: PULS X
            LDD  ,X
            TFR  D,Y
EULOOP:     CMPY #0
            BEQ  EUDONE
            LEAS 8,S
            LEAY -1,Y
            BRA  EULOOP
EUDONE:     PULS Y
            JMP  ,Y

CASEW:   LDD   #0
         PSHU  D
         LDD   #TAGCASE
         PSHU  D
         RTS

OF:      LDD   #OVER
         PSHU  D
         JSR   CCALL
         LDD   #EQUALW
         PSHU  D
         JSR   CCALL
         LDD   #ZBRANCH
         PSHU  D
         JSR   CCALL
         LDD   #0
         PSHU  D
         JSR   CODECOMMA
         LDD   CODEHERE
         SUBD  #2
         PSHU  D
         LDD   #DROP
         PSHU  D
         JSR   CCALL
```

```
          LDD   #TAGOF
          PSHU  D
          RTS

ENDOF:    PULU  D
          CMPD  #TAGOF
          BEQ   EOFOK
          JSR   CFERR
EOFOK:    PULU  D           ; BUG FIX: same class as LOOP - was PULU X then
          STD   MSCR        ; used via PSHU X after CCALL/CODECOMMA below,
                            ; both of which clobber X internally. Saved to
                            ; scratch instead (not parked on U, since
                            ; CODEHERE gets pushed onto U further down,
                            ; between the save and the retrieve below -
                            ; parking on U would retrieve that instead).
          LDD   #BRANCH
          PSHU  D
          JSR   CCALL
          LDD   #0
          PSHU  D
          JSR   CODECOMMA
          LDD   CODEHERE
          SUBD  #2
          STD   NEWFLD
          LDD   CODEHERE
          PSHU  D
          LDD   MSCR
          PSHU  D
          JSR   PATCH
          LDD   NEWFLD
          PSHU  D
          LDD   #TAGENDOF
          PSHU  D
          RTS

ENDCASE:  LDD   #DROP
          PSHU  D
          JSR   CCALL
ECLOOP:   PULU  D
          CMPD  #TAGCASE
          BEQ   ECDONE
          CMPD  #TAGENDOF
          BEQ   ECPATCH
          JSR   CFERR
ECPATCH:  PULU  X
          LDD   CODEHERE
          PSHU  D
          PSHU  X
          JSR   PATCH
          BRA   ECLOOP
ECDONE:   RTS
```

8.14 Compiling Words

```
; ============================================================
; SECTION 13: COMPILING WORDS (IMMEDIATE/[/]/'/COMPILE,/
; LITERAL/[']/POSTPONE/>BODY, SLITERAL, ABORT")
; ============================================================
STATEW:     LDD  #STATE
            PSHU D
            RTS

IMMEDIATE:  LDX  LATEST
            LDA  ,X
            ORA  #$80
            STA  ,X
            RTS
```

```
LBRACKET: LDD  #0
          STD  STATE
          RTS

RBRACKET: LDD  #-1
          STD  STATE
          RTS

TICK:     LDD  #32
          PSHU D
          JSR  WORD
          JSR  FIND
          PULU D
          CMPD #0
          BNE  TICKOK
          PULU D
          LDD  #-13
          PSHU D
          JSR  THROW
TICKOK:   RTS

COMPILECOMMA: JMP  CCALL

CCALL:    PULU  D
          LDX   CODEHERE
          LDA   #OPJSR
          STA   ,X+
          STD   ,X++
          STX   CODEHERE
          RTS

LITERALW: LDD  #LIT
          PSHU D
          JSR  CCALL
          JSR  CODECOMMA
          RTS

BRACKTICK: LDD  STATE
           BNE  BTSTOK
           LDD  #-14
           PSHU D
           JSR  THROW
BTSTOK:    LDD  #32
           PSHU D
           JSR  WORD
           JSR  FIND
           PULU D
           CMPD #0
           BNE  BTOK
           PULU D
           LDD  #-13
           PSHU D
           JSR  THROW
BTOK:      JSR  LITERALW
           RTS

POSTPONEW: LDD  STATE
           BNE  PPSTOK
           LDD  #-14
           PSHU D
           JSR  THROW
PPSTOK:    LDD  #32
           PSHU D
           JSR  WORD
           JSR  FIND
           PULU D
           CMPD #0
           BNE  PPFOUND
           PULU D
           LDD  #-13
           PSHU D
```

```
           JSR  THROW
PPFOUND:   CMPD #1
           BEQ  PPIMM
           JSR  LITERALW
           LDD  #COMPILECOMMA
           PSHU D
           JSR  CCALL
           RTS
PPIMM:     JSR  COMPILECOMMA
           RTS

TOBODY:  PULU D
         ADDD #5
         TFR  D,X
         LDD  ,X
         PSHU D
         RTS

SLITERALW: LDD  STATE
           BNE  SLSTOK
           LDD  #-14
           PSHU D
           JSR  THROW
SLSTOK:    PULU D
           STD  SCNT
           PULU D
           STD  SPTR
           LDD  #DOSTR
           PSHU D
           JSR  CCALL
           LDX  CODEHERE
           LDB  SCNT+1
           STB  ,X+
           LDY  SPTR
           LDB  SCNT+1
           BEQ  SLEND
SLCPY:     LDA  ,Y+
           STA  ,X+
           DECB
           BNE  SLCPY
SLEND:     STX  CODEHERE
           RTS

DOABORTQUOTE: PULS X
              LDB  ,X
              LEAX 1,X
              STX  SPTR
              CLRA
              STD  SCNT
              LDX  SPTR
              LDB  SCNT+1
              LEAX B,X
              PULU D
              CMPD #0
              BNE  AQTHROW
              PSHS X
              RTS
AQTHROW:      LDD  SPTR
              PSHU D
              LDD  SCNT
              PSHU D
              JSR  TYPE
              PSHS X
              LDD  #-2
              PSHU D
              JMP  THROW

ABORTQUOTE: LDD  STATE
            BNE  AQSTOK
            LDD  #-14
            PSHU D
```

```
            JSR  THROW
AQSTOK:     LDD  #34
            PSHU D
            JSR  WORD
            PULU X
            LDA  ,X
            STA  SCNT
            LEAX 1,X
            STX  SPTR
            LDD  #DOABORTQUOTE
            PSHU D
            JSR  CCALL
            LDX  CODEHERE
            LDA  SCNT
            STA  ,X+
            LDY  SPTR
            LDB  SCNT
            BEQ  AQEND
AQCPY:      LDA  ,Y+
            STA  ,X+
            DECB
            BNE  AQCPY
AQEND:      STX  CODEHERE
            RTS

BLW:     LDD   #32
         PSHU  D
         RTS

TOINW:   LDD   #TOIN
         PSHU  D
         RTS

SPANW:   LDD   #SPAN
         PSHU  D
         RTS

TIBW:    LDD   #TIBBUF
         PSHU  D
         RTS

NTIBW:   LDD   #NTIB
         PSHU  D
         RTS
```

8.15 Stack Manipulation

```
; ============================================================
; SECTION 14: STACK MANIPULATION (Core + Core Ext + return stack)
; ============================================================
DUP:     LDD   ,U
         PSHU  D
         RTS

DROP:    LEAU  2,U
         RTS

SWAP:    LDD   ,U
         LDX   2,U
         STX   ,U
         STD   2,U
         RTS

OVER:    LDD   2,U
         PSHU  D
         RTS

ROT:     LDD   ,U
```

```
         LDX   2,U
         LDY   4,U
         STY   ,U
         STD   2,U
         STX   4,U
         RTS

QDUP:    LDD   ,U
         CMPD  #0
         BEQ   QDUPDONE
         PSHU  D
QDUPDONE: RTS

DEPTH:   TFR   U,D
         STD   DEPTHTMP
         LDD   #SP0
         SUBD  DEPTHTMP
         LSRA
         RORB
         PSHU  D
         RTS

DDUP:    LDD   2,U
         LDX   ,U
         PSHU  D
         PSHU  X
         RTS

DDROP:   LEAU  4,U
         RTS

DSWAP:   LDD   ,U
         STD   MSCR
         LDD   2,U
         LDX   4,U
         LDY   6,U
         STD   6,U
         STX   ,U
         STY   2,U
         LDD   MSCR
         STD   4,U
         RTS

DOVER:   LDD   6,U
         LDX   4,U
         PSHU  D
         PSHU  X
         RTS

NIP:     LDD   ,U
         STD   2,U
         LEAU  2,U
         RTS

TUCK:    LDD   ,U
         LDX   2,U
         PSHU  D
         STX   2,U
         STD   4,U
         RTS

PICK:    PULU  D
         LSLB
         ROLA
         LDD   D,U
         PSHU  D
         RTS

ROLL:    PULU  D
         CMPD  #0
         BEQ   ROLLDONE
```

```
         LSLB
         ROLA
         STD   RDST
         LEAX  D,U
         LDD   ,X
         STD   RVAL
RLOOP:   LDD   RDST
         CMPD  #2
         BLT   RSTORE
         LEAY  D,U
         SUBD  #2
         LEAX  D,U
         LDD   ,X
         STD   ,Y
         LDD   RDST
         SUBD  #2
         STD   RDST
         BRA   RLOOP
RSTORE:  LDD   RVAL
         STD   ,U
ROLLDONE: RTS

DROT:    LDD   10,U
         STD   TR1
         LDD   8,U
         STD   TR2
         LDD   6,U
         STD   10,U
         LDD   4,U
         STD   8,U
         LDD   2,U
         STD   6,U
         LDD   0,U
         STD   4,U
         LDD   TR2
         STD   ,U
         LDD   TR1
         STD   2,U
         RTS

TOR:     PULU  D
         PULS  X
         PSHS  D
         PSHS  X
         RTS

FROMR:   PULS  X
         PULS  D
         PSHS  X
         PSHU  D
         RTS

RFETCH:  PULS  X
         LDD   ,S
         PSHS  X
         PSHU  D
         RTS

TWOTOR:  PULU  D
         STD   R2A
         PULU  D
         STD   R2B
         PULS  X
         LDD   R2B
         PSHS  D
         LDD   R2A
         PSHS  D
         PSHS  X
         RTS

TWOFROMR: PULS X
```

```
          PULS D
          STD  R2A
          PULS D
          STD  R2B
          PSHS X
          LDD  R2B
          PSHU D
          LDD  R2A
          PSHU D
          RTS

TWORFETCH: PULS X
           LDD  ,S
           STD  R2A
           LDD  2,S
           STD  R2B
           PSHS X
           LDD  R2B
           PSHU D
           LDD  R2A
           PSHU D
           RTS
```

8.16 Arithmetic

```
; ============================================================
; SECTION 15: ARITHMETIC (single + double + mixed precision)
; ============================================================
PLUS:    PULU  D
         ADDD  ,U
         STD   ,U
         RTS

; MSCR is declared once in the GLOBALS layout above - no local
; redeclaration needed here.
MINUS:   PULU  D
         STD   MSCR
         LDD   ,U
         SUBD  MSCR
         STD   ,U
         RTS

NEGATE:  LDD   ,U
         COMA
         COMB
         ADDD  #1
         STD   ,U
         RTS

ABSW:    LDD   ,U
         BPL   ABSDONE
         COMA
         COMB
         ADDD  #1
         STD   ,U
ABSDONE: RTS

MIN:     PULU  D
         CMPD  ,U
         BLT   MINISN2
         RTS
MINISN2: STD   ,U
         RTS

MAX:     PULU  D
         CMPD  ,U
         BGT   MAXISN2
         RTS
```

```
MAXISN2: STD   ,U
         RTS

ONEPLUS: LDD   ,U
         ADDD  #1
         STD   ,U
         RTS

ONEMINUS: LDD  ,U
          SUBD #1
          STD  ,U
          RTS

TWOPLUS: LDD   ,U
         ADDD  #2
         STD   ,U
         RTS

STAR:    PULU  D
         STD   MSCR
         LDD   ,U
         CLR   MSIGN
         BPL   SNOFLIP1
         COM   MSIGN
         COMA
         COMB
         ADDD  #1
SNOFLIP1: STA  MAHI
          STB  MALO
          LDD  MSCR
          BPL  SNOFLIP2
          COM  MSIGN
          COMA
          COMB
          ADDD #1
SNOFLIP2: STA  MBHI
          STB  MBLO
          LDA  MALO
          LDB  MBLO
          MUL
          STD  MRESULT
          LDA  MAHI
          LDB  MBLO
          MUL
          LDA  MRESULT
          PSHS B
          ADDA ,S+          ; was "ADDA B" - not valid 6809 syntax
          STA  MRESULT
          LDA  MALO
          LDB  MBHI
          MUL
          LDA  MRESULT
          PSHS B
          ADDA ,S+          ; was "ADDA B" - not valid 6809 syntax
          STA  MRESULT
          LDD  MRESULT
          TST  MSIGN
          BEQ  SDONE
          COMA
          COMB
          ADDD #1
SDONE:    STD  ,U
          RTS

TWOSTAR: LDD   ,U
         ASLB
         ROLA
         STD   ,U
         RTS

UDIV16:  CLR   DIVREM
```

```
         CLR   DIVREM+1
         LDB   #16
         STB   DIVCNT
UD16LOOP: ASL   DIVNUM+1
         ROL   DIVNUM
         ROL   DIVREM+1
         ROL   DIVREM
         LDD   DIVREM
         SUBD  DIVDEN
         BLO   UDSKIP
         STD   DIVREM
         INC   DIVNUM+1
UDSKIP:  DEC   DIVCNT
         BNE   UD16LOOP
         RTS

DIVCOMMON: PULU  D
           STD   DIVDEN
           CMPD  #0
           BNE   DCOK
           LDD   #-10
           PSHU  D
           JSR   THROW
DCOK:      PULU  D
           STD   DIVNUM
           CLR   DVSIGN
           CLR   DNSIGN
           TST   DIVNUM
           BPL   DNPOS
           COM   DNSIGN
           COM   DVSIGN
           LDD   DIVNUM
           COMA
           COMB
           ADDD  #1
           STD   DIVNUM
DNPOS:     TST   DIVDEN
           BPL   DVPOS
           COM   DVSIGN
           LDD   DIVDEN
           COMA
           COMB
           ADDD  #1
           STD   DIVDEN
DVPOS:     JSR   UDIV16
           LDD   DIVNUM
           TST   DVSIGN
           BEQ   DQPOS
           COMA
           COMB
           ADDD  #1
           STD   DIVNUM
DQPOS:     LDD   DIVREM
           TST   DNSIGN
           BEQ   DCRPOS
           COMA
           COMB
           ADDD  #1
           STD   DIVREM
DCRPOS:    RTS

SLASH:   JSR   DIVCOMMON
         LDD   DIVNUM
         PSHU  D
         RTS

MODW:    JSR   DIVCOMMON
         LDD   DIVREM
         PSHU  D
         RTS
```

```
SLASHMOD: JSR  DIVCOMMON
          LDD  DIVREM
          PSHU D
          LDD  DIVNUM
          PSHU D
          RTS

TWOSLASH: LDD  ,U
          ASRA
          RORB
          STD  ,U
          RTS

UMUL32:  LDA   MALO
         LDB   MBLO
         MUL
         STD   PRODLO
         CLR   PRODHI
         CLR   PRODHI+1
         LDA   MAHI
         LDB   MBLO
         MUL
         ADDB  PRODLO
         STB   PRODLO
         ADCA  #0
         ADDA  PRODHI+1
         STA   PRODHI+1
         BCC   UM32A
         INC   PRODHI
UM32A:   LDA   MALO
         LDB   MBHI
         MUL
         ADDB  PRODLO
         STB   PRODLO
         ADCA  #0
         ADDA  PRODHI+1
         STA   PRODHI+1
         BCC   UM32B
         INC   PRODHI
UM32B:   LDA   MAHI
         LDB   MBHI
         MUL
         ADDD  PRODHI
         STD   PRODHI
         RTS

UDIV32:  CLR   DIVREM
         CLR   DIVREM+1
         LDB   #32
         STB   DIVCNT
UD32LP:  ASL   PRODLO+1
         ROL   PRODLO
         ROL   PRODHI+1
         ROL   PRODHI
         ROL   DIVREM+1
         ROL   DIVREM
         LDD   DIVREM
         SUBD  DIVDEN
         BLO   UD32SKIP
         STD   DIVREM
         INC   PRODLO+1
UD32SKIP: DEC  DIVCNT
          BNE  UD32LP
          RTS

MNEG32:  LDD   PRODLO
         COMA
         COMB
         STD   PRODLO
         LDD   PRODHI
         COMA
```

```
          COMB
          STD   PRODHI
          LDD   PRODLO
          ADDD  #1
          STD   PRODLO
          BCC   MN32DONE
          LDD   PRODHI
          ADDD  #1
          STD   PRODHI
MN32DONE: RTS

STARSLASHCOMMON:
          PULU  D
          STD   DIVDEN
          CMPD  #0
          BNE   SSOK
          LDD   #-10
          PSHU  D
          JSR   THROW
SSOK:     CLR   PSIGN
          TST   DIVDEN
          BPL   SSN3POS
          COM   PSIGN
          LDD   DIVDEN
          COMA
          COMB
          ADDD  #1
          STD   DIVDEN
SSN3POS:  LDA   #0
          STA   PRSIGN
          PULU  D
          STD   MSCR
          TST   MSCR
          BPL   SSN2POS
          COM   PSIGN
          COM   PRSIGN
          LDD   MSCR
          COMA
          COMB
          ADDD  #1
          STD   MSCR
SSN2POS:  LDA   MSCR
          STA   MBHI
          LDA   MSCR+1
          STA   MBLO
          PULU  D
          STD   MSCR
          TST   MSCR
          BPL   SSN1POS
          COM   PSIGN
          COM   PRSIGN
          LDD   MSCR
          COMA
          COMB
          ADDD  #1
          STD   MSCR
SSN1POS:  LDA   MSCR
          STA   MAHI
          LDA   MSCR+1
          STA   MALO
          JSR   UMUL32
          JSR   UDIV32
          LDD   PRODLO
          TST   PSIGN
          BEQ   SSQPOS
          COMA
          COMB
          ADDD  #1
          STD   PRODLO
SSQPOS:   LDD   DIVREM
          TST   PRSIGN
```

```
         BEQ   SSRPOS
         COMA
         COMB
         ADDD  #1
         STD   DIVREM
SSRPOS:  RTS

STARSLASH: JSR  STARSLASHCOMMON
           LDD  PRODLO
           PSHU D
           RTS

STARSLASHMOD: JSR  STARSLASHCOMMON
              LDD  DIVREM
              PSHU D
              LDD  PRODLO
              PSHU D
              RTS

UMSTAR:  PULU  D
         STD   MSCR
         PULU  D
         STA   MAHI
         STB   MALO
         LDD   MSCR
         STA   MBHI
         STB   MBLO
         JSR   UMUL32
         LDD   PRODLO
         PSHU  D
         LDD   PRODHI
         PSHU  D
         RTS

UMSLASHMOD: PULU D
            STD  DIVDEN
            CMPD #0
            BNE  UMOK
            LDD  #-10
            PSHU D
            JSR  THROW
UMOK:       PULU D
            STD  PRODHI
            PULU D
            STD  PRODLO
            JSR  UDIV32
            LDD  DIVREM
            PSHU D
            LDD  PRODLO
            PSHU D
            RTS

MSTAR:   PULU  D
         STD   MSCR
         PULU  D
         CLR   MSIGN
         BPL   MSN1POS
         COM   MSIGN
         COMA
         COMB
         ADDD  #1
MSN1POS: STA   MAHI
         STB   MALO
         LDD   MSCR
         BPL   MSN2POS
         COM   MSIGN
         COMA
         COMB
         ADDD  #1
MSN2POS: STA   MBHI
         STB   MBLO
```

```
          JSR   UMUL32
          TST   MSIGN
          BEQ   MSDONE
          JSR   MNEG32
MSDONE:   LDD   PRODLO
          PSHU  D
          LDD   PRODHI
          PSHU  D
          RTS

SMSLASHREM: PULU D
            STD  DIVDEN
            CMPD #0
            BNE  SMOK
            LDD  #-10
            PSHU D
            JSR  THROW
SMOK:       PULU D
            STD  PRODHI
            PULU D
            STD  PRODLO
            CLR  DNSIGN
            CLR  DVSIGN
            TST  PRODHI
            BPL  SMDPOS
            COM  DNSIGN
            COM  DVSIGN
            JSR  MNEG32
SMDPOS:     LDD  DIVDEN
            BPL  SMDVPOS
            COM  DVSIGN
            LDD  DIVDEN
            COMA
            COMB
            ADDD #1
            STD  DIVDEN
SMDVPOS:    JSR  UDIV32
            LDD  DIVREM
            TST  DNSIGN
            BEQ  SMRPOS
            COMA
            COMB
            ADDD #1
SMRPOS:     PSHU D
            LDD  PRODLO
            TST  DVSIGN
            BEQ  SMQPOS
            COMA
            COMB
            ADDD #1
SMQPOS:     PSHU D
            RTS

FMSLASHMOD: PULU D
            STD  DIVDEN
            CMPD #0
            BNE  FMOK
            LDD  #-10
            PSHU D
            JSR  THROW
FMOK:       PULU D
            STD  PRODHI
            PULU D
            STD  PRODLO
            CLR  DNSIGN
            CLR  DVSIGN
            CLR  DVOWNSIGN
            TST  PRODHI
            BPL  FMDPOS
            COM  DNSIGN
            COM  DVSIGN
```

```
            JSR  MNEG32
FMDPOS:     LDD  DIVDEN
            BPL  FMDVPOS
            COM  DVSIGN
            COM  DVOWNSIGN
            LDD  DIVDEN
            COMA
            COMB
            ADDD #1
            STD  DIVDEN
FMDVPOS:    JSR  UDIV32
            TST  DVSIGN
            BEQ  FMNOFLOOR
            LDD  DIVREM
            BEQ  FMNOFLOOR
            LDD  PRODLO
            ADDD #1
            STD  PRODLO
            LDD  DIVDEN
            SUBD DIVREM
            STD  DIVREM
FMNOFLOOR:  LDD  DIVREM
            TST  DVOWNSIGN
            BEQ  FMRPOS
            COMA
            COMB
            ADDD #1
FMRPOS:     PSHU D
            LDD  PRODLO
            TST  DVSIGN
            BEQ  FMQPOS
            COMA
            COMB
            ADDD #1
FMQPOS:     PSHU D
            RTS

DPLUS:   PULU  D
         STD   MSCR
         PULU  D
         STD   MSCR2
         PULU  D
         STD   MSCR3
         PULU  D
         ADDD  MSCR2
         STD   MSCR4
         BCC   DPNOCY
         LDD   MSCR3
         ADDD  MSCR
         ADDD  #1
         BRA   DPHIDONE
DPNOCY:  LDD   MSCR3
         ADDD  MSCR
DPHIDONE: STD  MSCR3
         LDD   MSCR4
         PSHU  D
         LDD   MSCR3
         PSHU  D
         RTS

DMINUS:  PULU  D
         STD   MSCR
         PULU  D
         STD   MSCR2
         PULU  D
         STD   MSCR3
         PULU  D
         SUBD  MSCR2
         STD   MSCR4
         BCC   DMNOBOR
         LDD   MSCR3
```

```
         SUBD  MSCR
         SUBD  #1
         BRA   DMHIDONE
DMNOBOR: LDD   MSCR3
         SUBD  MSCR
DMHIDONE: STD  MSCR3
         LDD   MSCR4
         PSHU  D
         LDD   MSCR3
         PSHU  D
         RTS

DNEGATEW: PULU D
          STD  PRODHI
          PULU D
          STD  PRODLO
          JSR  MNEG32
          LDD  PRODLO
          PSHU D
          LDD  PRODHI
          PSHU D
          RTS

DABSW:   PULU  D
         STD   PRODHI
         PULU  D
         STD   PRODLO
         TST   PRODHI
         BPL   DABSDONE
         JSR   MNEG32
DABSDONE: LDD  PRODLO
          PSHU D
          LDD  PRODHI
          PSHU D
          RTS

MPLUS:   PULU  D
         STD   MSCR2
         BPL   MPPOSN
         LDD   #-1
         BRA   MPSIGNED
MPPOSN:  LDD   #0
MPSIGNED: STD  MSCR
          PULU D
          STD  MSCR3
          PULU D
          ADDD MSCR2
          STD  MSCR4
          BCC  MPNOCY
          LDD  MSCR3
          ADDD MSCR
          ADDD #1
          BRA  MPHIDONE
MPNOCY:   LDD  MSCR3
          ADDD MSCR
MPHIDONE: STD MSCR3
          LDD  MSCR4
          PSHU D
          LDD  MSCR3
          PSHU D
          RTS

STOD:    PULU  D
         PSHU  D
         BPL   SDPOS
         LDD   #-1
         BRA   SDPUSH
SDPOS:   LDD   #0
SDPUSH:  PSHU  D
         RTS
```

```
DTOS:    PULU  D
         RTS

DMAXW:   PULU  D
         STD   MSCR
         PULU  D
         STD   MSCR2
         PULU  D
         STD   MSCR3
         PULU  D
         STD   MSCR4
         LDD   MSCR3
         CMPD  MSCR
         BGT   DMXD1
         BLT   DMXD2
         LDD   MSCR4
         CMPD  MSCR2
         BHS   DMXD1
DMXD2:   LDD   MSCR2
         PSHU  D
         LDD   MSCR
         PSHU  D
         RTS
DMXD1:   LDD   MSCR4
         PSHU  D
         LDD   MSCR3
         PSHU  D
         RTS

DMINW:   PULU  D
         STD   MSCR
         PULU  D
         STD   MSCR2
         PULU  D
         STD   MSCR3
         PULU  D
         STD   MSCR4
         LDD   MSCR3
         CMPD  MSCR
         BLT   DMND1
         BGT   DMND2
         LDD   MSCR4
         CMPD  MSCR2
         BLS   DMND1
DMND2:   LDD   MSCR2
         PSHU  D
         LDD   MSCR
         PSHU  D
         RTS
DMND1:   LDD   MSCR4
         PSHU  D
         LDD   MSCR3
         PSHU  D
         RTS
```

8.17 Logic, Shifts & Address Arithmetic

```
; ============================================================
; SECTION 16: LOGIC / SHIFTS / ADDRESS ARITHMETIC
; ============================================================
ANDW:    PULU  D
         ANDA  ,U
         ANDB  1,U
         STD   ,U
         RTS

ORW:     PULU  D
         ORA   ,U
```

```
          ORB   1,U
          STD   ,U
          RTS

XORW:     PULU  D
          EORA  ,U
          EORB  1,U
          STD   ,U
          RTS

INVERT:   LDD   ,U
          COMA
          COMB
          STD   ,U
          RTS

LSHIFT:   PULU  D
          STB   SHCNT
          LDD   ,U
LSLOOP:   LDB   SHCNT
          BEQ   LSDONE
          ASL   1,U
          ROL   ,U
          DEC   SHCNT
          BRA   LSLOOP
LSDONE:   RTS

RSHIFT:   PULU  D
          STB   SHCNT
RSLOOP:   LDB   SHCNT
          BEQ   RSDONE
          LSR   ,U
          ROR   1,U
          DEC   SHCNT
          BRA   RSLOOP
RSDONE:   RTS

CELLSW:   LDD   ,U
          ASLB
          ROLA
          STD   ,U
          RTS

CELLPLUS: LDD  ,U
           ADDD #2
           STD  ,U
           RTS

CHARSW:   RTS

CHARPLUS: LDD  ,U
           ADDD #1
           STD  ,U
           RTS

ALIGNW:   RTS
ALIGNEDW: RTS
```

8.18 Comparison

```
; ============================================================
; SECTION 17: COMPARISON
; ============================================================
EQUALW:   PULU  D
          CMPD  ,U
          BEQ   EQTRUE
          LDD   #FALSEV
          STD   ,U
```

```
         RTS
EQTRUE:  LDD   #TRUEV
         STD   ,U
         RTS

LESSW:   PULU  D
         STD   MSCR
         LDD   ,U
         CMPD  MSCR
         BLT   LTTRUE
         LDD   #FALSEV
         STD   ,U
         RTS
LTTRUE:  LDD   #TRUEV
         STD   ,U
         RTS

GREATERW: PULU D
          STD  MSCR
          LDD  ,U
          CMPD MSCR
          BGT  GTTRUE
          LDD  #FALSEV
          STD  ,U
          RTS
GTTRUE:   LDD  #TRUEV
          STD  ,U
          RTS

ZEROEQ:  LDD   ,U
         BEQ   ZEQTRUE
         LDD   #FALSEV
         STD   ,U
         RTS
ZEQTRUE: LDD   #TRUEV
         STD   ,U
         RTS

ZEROLT:  LDD   ,U
         BMI   ZLTTRUE
         LDD   #FALSEV
         STD   ,U
         RTS
ZLTTRUE: LDD   #TRUEV
         STD   ,U
         RTS

ULESSW:  PULU  D
         STD   MSCR
         LDD   ,U
         CMPD  MSCR
         BLO   ULTRUE
         LDD   #FALSEV
         STD   ,U
         RTS
ULTRUE:  LDD   #TRUEV
         STD   ,U
         RTS

NOTEQUAL: JSR  EQUALW
          LDD  ,U
          COMA
          COMB
          STD  ,U
          RTS

ZERONE:  JSR   ZEROEQ
         LDD   ,U
         COMA
         COMB
         STD   ,U
```

```
         RTS

ZEROGT:  LDD   ,U
         BEQ   ZGTFALSE
         BMI   ZGTFALSE
         LDD   #TRUEV
         STD   ,U
         RTS
ZGTFALSE: LDD  #FALSEV
          STD  ,U
          RTS

UGREATER: PULU D
          STD  MSCR
          LDD  ,U
          CMPD MSCR
          BLO  UGFALSE
          BEQ  UGFALSE
          LDD  #TRUEV
          STD  ,U
          RTS
UGFALSE:  LDD  #FALSEV
          STD  ,U
          RTS

WITHINW: PULU  D
         STD   MSCR
         PULU  D
         STD   MSCR2
         LDD   ,U
         SUBD  MSCR2
         STD   MSCR3
         LDD   MSCR
         SUBD  MSCR2
         STD   MSCR
         LDD   MSCR3
         CMPD  MSCR
         BLO   WITHTRUE
         LDD   #FALSEV
         STD   ,U
         RTS
WITHTRUE: LDD  #TRUEV
          STD  ,U
          RTS

DEQUAL:  PULU  D
         STD   MSCR
         PULU  D
         STD   MSCR2
         PULU  D
         STD   MSCR3
         PULU  D
         CMPD  MSCR2
         BNE   DEQFALSE
         LDD   MSCR3
         CMPD  MSCR
         BNE   DEQFALSE
         LDD   #TRUEV
         PSHU  D
         RTS
DEQFALSE: LDD  #FALSEV
          PSHU D
          RTS

DLESSW:  PULU  D
         STD   MSCR
         PULU  D
         STD   MSCR2
         PULU  D
         STD   MSCR3
         PULU  D
```

```
         STD   MSCR4
         LDD   MSCR3
         CMPD  MSCR
         BLT   DLTRUE
         BGT   DLFALSE
         LDD   MSCR4
         CMPD  MSCR2
         BLO   DLTRUE
DLFALSE: LDD   #FALSEV
         PSHU  D
         RTS
DLTRUE:  LDD   #TRUEV
         PSHU  D
         RTS

DULESSW: PULU  D
         STD   MSCR
         PULU  D
         STD   MSCR2
         PULU  D
         STD   MSCR3
         PULU  D
         STD   MSCR4
         LDD   MSCR3
         CMPD  MSCR
         BLO   DULTRUE
         BHI   DULFALSE
         LDD   MSCR4
         CMPD  MSCR2
         BLO   DULTRUE
DULFALSE: LDD  #FALSEV
          PSHU D
          RTS
DULTRUE:  LDD  #TRUEV
          PSHU D
          RTS
```

8.19 Memory

```
; ============================================================
; SECTION 18: MEMORY (fetch/store, block ops)
; ============================================================
STOREW:  PULU  X
         PULU  D
         STD   ,X
         RTS

CFETCH:  PULU  X
         LDB   ,X
         CLRA
         PSHU  D
         RTS

CSTOREW: PULU  X
         PULU  D
         STB   ,X
         RTS

PLUSSTORE: PULU X
           PULU D
           ADDD ,X
           STD  ,X
           RTS

DFETCH:  PULU  X
         LDD   2,X
         PSHU  D
         LDD   ,X
```

```
          PSHU  D
          RTS

DSTORE:   PULU  X
          PULU  D
          STD   2,X
          PULU  D
          STD   ,X
          RTS

CMOVEW:   PULU  D
          STD   MVCNT
          PULU  D
          STD   MVDST
          PULU  D
          STD   MVSRC
          LDX   MVSRC
          LDY   MVDST
CMVLOOP:  LDD   MVCNT
          BEQ   CMDONE
          LDA   ,X+
          STA   ,Y+
          SUBD  #1
          STD   MVCNT
          BRA   CMVLOOP
CMDONE:   RTS

CMOVEGT:  PULU  D
          STD   MVCNT
          PULU  D
          STD   MVDST
          PULU  D
          STD   MVSRC
          LDD   MVCNT
          BEQ   CGDONE
          LDX   MVSRC
          LEAX  D,X
          LEAX  -1,X
          LDY   MVDST
          LEAY  D,Y
          LEAY  -1,Y
CGLOOP:   LDA   ,X
          STA   ,Y
          LEAX  -1,X
          LEAY  -1,Y
          LDD   MVCNT
          SUBD  #1
          STD   MVCNT
          BNE   CGLOOP
CGDONE:   RTS

MOVEW:    PULU  D
          STD   MVCNT
          PULU  D
          STD   MVDST
          PULU  D
          STD   MVSRC
          LDD   MVDST
          CMPD  MVSRC
          BLS   MVLOW
          LDD   MVSRC
          PSHU  D
          LDD   MVDST
          PSHU  D
          LDD   MVCNT
          PSHU  D
          JMP   CMOVEGT
MVLOW:    LDD   MVSRC
          PSHU  D
          LDD   MVDST
          PSHU  D
```

```
         LDD   MVCNT
         PSHU  D
         JMP   CMOVEW

FILLW:   PULU  D
         STB   FILLCHR
         PULU  D
         STD   FILLCNT
         PULU  D
         STD   FILLADDR
         LDX   FILLADDR
FILLOOP: LDD   FILLCNT
         BEQ   FDONE
         LDA   FILLCHR
         STA   ,X+
         SUBD  #1
         STD   FILLCNT
         BRA   FILLOOP
FDONE:   RTS

ERASEW:  LDD   #0
         PSHU  D
         JMP   FILLW
```

8.20 String Words

```
; ============================================================
; SECTION 19: STRING WORDS
; ============================================================
DOSTR:   PULS  X
         LDB   ,X
         LEAX  1,X
         PSHU  X
         CLRA
         PSHU  D
         LEAX  B,X
         PSHS  X
         RTS

SQUOTE:  LDD   #34
         PSHU  D
         JSR   WORD
         PULU  X
         LDA   ,X
         STA   SCNT
         LEAX  1,X
         STX   SPTR
         LDD   STATE
         BEQ   SQINTERP

         LDD   #DOSTR
         PSHU  D
         JSR   CCALL
         LDX   CODEHERE
         LDA   SCNT
         STA   ,X+
         LDY   SPTR
         LDB   SCNT
         BEQ   SQEND
SQCPY:   LDA   ,Y+
         STA   ,X+
         DECB
         BNE   SQCPY
SQEND:   STX   CODEHERE
         RTS

SQINTERP: ; REDESIGN: was a fixed "LDX #SIBUF" here and at SQIEND -
                      ; SIBUF was a dedicated, fixed 32-byte buffer,
```

```
                        ; capping every interpreted S" string at 32
                        ; characters and (worse) shared identically
                        ; by every S" call, so a second S" call before
                        ; the first string was actually used (e.g.
                        ; registering two REPLACES strings, or a
                        ; SUBSTITUTE template argument) silently
                        ; overwrote the first - confirmed via MAME
                        ; testing and traced precisely earlier this
                        ; session. Retired per user's own follow-up
                        ; testing/design decision: rather than give
                        ; REPLACES its own dedicated copy-on-register
                        ; storage, wrap each string in its own colon
                        ; definition for stable storage instead - but
                        ; that still leaves interpreted S" itself
                        ; capped at SIBUF's 32 characters for any
                        ; single string. Now computes PAD's current
                        ; address fresh via PADW (dynamic - depends
                        ; on CODEHERE, per ANS's own transient-region
                        ; semantics) and writes there instead, giving
                        ; interpreted S" access to PAD's full
                        ; PADMINSIZE-character region (84, comfortably
                        ; above the old 32-character SIBUF limit) -
                        ; SIBUF itself is now unused and retired (see
                        ; the GLOBALS layout notes above).
          JSR  PADW
          PULU X
          STX  MSCR4        ; save PAD's own address - X gets advanced
                             ; by the copy loop below, need the
                             ; original back for the return value
          LDY  SPTR
          LDB  SCNT
          BEQ  SQIEND
SQICPY:   LDA  ,Y+
          STA  ,X+
          DECB
          BNE  SQICPY
SQIEND:   LDX  MSCR4
          PSHU X
          CLRA
          LDB  SCNT
          PSHU D
          RTS

DOTSTR:  PULS  X
         LDB   ,X
         LEAX  1,X
         STX   SPTR
         CLRA
         STD   SCNT
         LEAX  B,X
         PSHS  X
         LDX   SPTR
         PSHU  X
         LDD   SCNT
         PSHU  D
         JSR   TYPE
         RTS

DOTQUOTE: LDD  #34
          PSHU D
          JSR  WORD
          PULU X
          LDA  ,X
          STA  SCNT
          LEAX 1,X
          STX  SPTR
          LDD  #DOTSTR
          PSHU D
          JSR  CCALL
          LDX  CODEHERE
          LDA  SCNT
```

```
          STA  ,X+
          LDY  SPTR
          LDB  SCNT
          BEQ  DQEND
DQCPY:    LDA  ,Y+
          STA  ,X+
          DECB
          BNE  DQCPY
DQEND:    STX  CODEHERE
          RTS

TYPE:    PULU  D
         STD   TYPECNT
         PULU  D
         STD   TYPEADDR
TYLOOP:  LDD   TYPECNT
         BEQ   TYDONE
         LDX   TYPEADDR
         LDA   ,X+
         STX   TYPEADDR
         TFR   A,B
         CLRA
         PSHU  D
         JSR   EMIT
         LDD   TYPECNT
         SUBD  #1
         STD   TYPECNT
         BRA   TYLOOP
TYDONE:  RTS

COUNT:   PULU  X
         LDB   ,X
         CLRA
         STD   MSCR
         LEAX  1,X
         PSHU  X
         LDD   MSCR
         PSHU  D
         RTS

CHARW:   LDD   #32
         PSHU  D
         JSR   WORD
         PULU  X
         LDB   1,X
         CLRA
         PSHU  D
         RTS

BRACKCHAR: LDD  STATE
           BNE  BCSTOK
           LDD  #-14
           PSHU D
           JSR  THROW
BCSTOK:    LDD  #32
           PSHU D
           JSR  WORD
           PULU X
           LDB  1,X
           CLRA
           PSHU D
           JSR  LITERALW
           RTS

PARSEW:  PULU  D
         STB   PDELIM
         LDD   TOIN
         LDX   SRCADDR
         LEAX  D,X
         STX   PSTART
         LDD   SRCLEN
```

```
          SUBD  TOIN
          TFR   D,Y
          LDD   #0
          STD   PLEN
PSCAN:    CMPY  #0
          BEQ   PDONE
          LDA   ,X
          CMPA  PDELIM
          BEQ   PFOUND
          LEAX  1,X
          LEAY  -1,Y
          LDD   PLEN
          ADDD  #1
          STD   PLEN
          BRA   PSCAN
PFOUND:   LEAX  1,X
          LEAY  -1,Y
PDONE:    TFR   X,D
          SUBD  SRCADDR
          STD   TOIN
          LDX   PSTART
          PSHU  X
          LDD   PLEN
          PSHU  D
          RTS

PARSENAME: LDD  TOIN
           LDX  SRCADDR
           LEAX D,X
           LDD  SRCLEN
           SUBD TOIN
           TFR  D,Y
PNSKIP:    CMPY #0
           BEQ  PNEMPTY
           LDA  ,X
           CMPA #32
           BNE  PNSTART
           LEAX 1,X
           LEAY -1,Y
           BRA  PNSKIP
PNSTART:   STX  PSTART
           LDD  #0
           STD  PLEN
PNSCAN:    CMPY #0
           BEQ  PNDONE
           LDA  ,X
           CMPA #32
           BEQ  PNFOUND
           LEAX 1,X
           LEAY -1,Y
           LDD  PLEN
           ADDD #1
           STD  PLEN
           BRA  PNSCAN
PNFOUND:   LEAX 1,X
           LEAY -1,Y
PNDONE:    TFR  X,D
           SUBD SRCADDR
           STD  TOIN
           LDX  PSTART
           PSHU X
           LDD  PLEN
           PSHU D
           RTS
PNEMPTY:   LDD  SRCLEN
           LDX  SRCADDR
           LEAX D,X
           STD  TOIN
           PSHU X
           LDD  #0
           PSHU D
```

```
           RTS

SLASHSTRING: PULU D
             STD  MSCR
             PULU D
             SUBD MSCR
             STD  MSCR2
             PULU D
             ADDD MSCR
             PSHU D
             LDD  MSCR2
             PSHU D
             RTS

DASHTRAILING: LDD  ,U
              STD  PLEN
DTLOOP:       LDD  PLEN
              BEQ  DTDONE
              LDX  2,U
              LEAX D,X
              LEAX -1,X
              LDA  ,X
              CMPA #32
              BNE  DTDONE
              LDD  PLEN
              SUBD #1
              STD  PLEN
              BRA  DTLOOP
DTDONE:       LDD  PLEN
              STD  ,U
              RTS

COMPAREW: PULU D
          STD  CMPL2
          PULU D
          STD  CMPA2
          PULU D
          STD  CMPL1
          PULU D
          STD  CMPA1
          LDD  CMPL1
          CMPD CMPL2
          BLS  CMMINIS1
          LDD  CMPL2
          BRA  CMMINSET
CMMINIS1: LDD  CMPL1
CMMINSET: STD  CMPMIN
          LDX  CMPA1
          LDY  CMPA2
CMPLOOP:  LDD  CMPMIN
          BEQ  CMTIEBREAK
          LDA  ,X+
          CMPA ,Y
          BLO  CMLT
          BHI  CMGT
          LEAY 1,Y
          LDD  CMPMIN
          SUBD #1
          STD  CMPMIN
          BRA  CMPLOOP
CMTIEBREAK: LDD CMPL1
            CMPD CMPL2
            BLO  CMLT
            BHI  CMGT
            LDD  #0
            PSHU D
            RTS
CMLT:     LDD  #-1
          PSHU D
          RTS
CMGT:     LDD  #1
```

```
          PSHU D
          RTS

SEARCHW: PULU  D
         STD   SRCH2L
         PULU  D
         STD   SRCH2
         PULU  D
         STD   SRCH1L
         PULU  D
         STD   SRCH1
         LDD   SRCH2L
         BEQ   SRCHNOTFOUND
         LDD   SRCH1L
         SUBD  SRCH2L
         BLT   SRCHNOTFOUND
         ADDD  #1
         STD   SRCHPOS
         LDD   #0
         STD   SRCHI
SPOSLOOP: LDD  SRCHI
          CMPD SRCHPOS
          BEQ  SRCHNOTFOUND
          LDX  SRCH1
          LDD  SRCHI
          LEAX D,X
          LDY  SRCH2
          LDD  SRCH2L
          STD  MSCR3
SMATCH:   LDD  MSCR3
          BEQ  SFOUND
          LDA  ,X+
          CMPA ,Y+
          BNE  SNOMATCH
          LDD  MSCR3
          SUBD #1
          STD  MSCR3
          BRA  SMATCH
SNOMATCH: LDD  SRCHI
          ADDD #1
          STD  SRCHI
          BRA  SPOSLOOP
SFOUND:   LDX  SRCH1
          LDD  SRCHI
          LEAX D,X
          PSHU X
          LDD  SRCH2L
          PSHU D
          LDD  #TRUEV
          PSHU D
          RTS
SRCHNOTFOUND: LDD SRCH1
              PSHU D
              LDD  SRCH1L
              PSHU D
              LDD  #FALSEV
              PSHU D
              RTS

SNAMEW:  PULU  D
         STD   SNTARGET
         LDD   LATEST
         STD   SNXT
SNLOOP:  LDD   SNXT
         BEQ   SNNOTFOUND
         LDX   SNXT
         LDA   ,X
         STA   HDRFLAGS
         LEAX  1,X
         LDB   HDRFLAGS
         ANDB  #$1F
```

```
         CLRA
         LEAX  D,X
         LEAX  2,X
         LDD   ,X
         CMPD  SNTARGET
         BEQ   SNFOUND
         LDX   SNXT
         LEAX  1,X
         LDB   HDRFLAGS
         ANDB  #$1F
         CLRA
         LEAX  D,X
         LDD   ,X
         STD   SNXT
         BRA   SNLOOP
SNFOUND: LDX   SNXT
         LEAX  1,X
         PSHU  X
         LDX   SNXT
         LDA   ,X
         ANDA  #$1F
         CLRB
         TFR   A,B
         CLRA
         PSHU  D
         RTS
SNNOTFOUND: LDD #0
            PSHU D
            PSHU D
            RTS

UNESCAPEW: PULU D
           STD  UESRCLEN
           LDD  ,U
           STD  UEADDR
           STD  UEDST
           LDD  #0
           STD  UEOUTLEN
           LDX  UEADDR
           LDY  UEDST
UELOOP:    LDD  UESRCLEN
           BEQ  UEDONE
           LDA  ,X+
           CMPA #'\'
           BNE  UEPLAIN
           LDD  UESRCLEN
           SUBD #1
           STD  UESRCLEN
           BEQ  UELASTBS   ; BUG FIX: was BEQ UEPLAIN - lone trailing
                           ; backslash. UESRCLEN is already correctly
                           ; 0 here (just decremented for the
                           ; backslash itself); falling into UEPLAIN's
                           ; shared decrement would double-count it,
                           ; underflowing to $FFFF (nonzero), so the
                           ; next UELOOP pass's BEQ UEDONE would never
                           ; fire - the loop would run past the end
                           ; of the intended input. Flagged by the
                           ; parallel 68000 port, confirmed by
                           ; inspection.
           LDA  ,X+
           CMPA #'n'
           BNE  UECKT
           LDA  #10
           BRA  UEEMIT
UECKT:     CMPA #'t'
           BNE  UECKBS
           LDA  #9
           BRA  UEEMIT
UECKBS:    CMPA #'\'
           BEQ  UEEMIT
           CMPA #'"'
```

```
            BEQ  UEEMIT
            PSHS A
            LDA  #'\'
            STA  ,Y+
            LDD  UEOUTLEN
            ADDD #1
            STD  UEOUTLEN
            PULS A
UEEMIT:     STA  ,Y+
            LDD  UEOUTLEN
            ADDD #1
            STD  UEOUTLEN
            LDD  UESRCLEN
            SUBD #1
            STD  UESRCLEN
            BRA  UELOOP
UELASTBS:   STA  ,Y+       ; A already holds the '\' from the earlier
            LDD  UEOUTLEN  ; LDA ,X+ - emit it literally (nothing to
            ADDD #1        ; interpret it with), without re-decrementing
            STD  UEOUTLEN  ; UESRCLEN (already correctly 0 above)
            BRA  UELOOP
UEPLAIN:    STA  ,Y+
            LDD  UEOUTLEN
            ADDD #1
            STD  UEOUTLEN
            LDD  UESRCLEN
            SUBD #1
            STD  UESRCLEN
            BRA  UELOOP
UEDONE:     LDD  UEOUTLEN
            STD  ,U
            RTS

; REPLACES/SUBSTITUTE - single-slot simplified version, per
; the explicit scoping-down discussed in the source conversation
REPLACESW:  PULU D
            STD  REPLNLEN
            PULU D
            STD  REPLNAME
            PULU D
            STD  REPLVLEN
            PULU D
            STD  REPLVAL
            RTS

SUBCOPY:  STD  SUBCOPYCNT
          STX  SUBCOPYSRC
SUBCPLP:  LDD  SUBCOPYCNT
          BEQ  SUBCPDONE
          LDD  SUBOUTLEN
          CMPD SUBDESTCAP
          LBHS SUBOVERFLOW      ; was BHS - out of short-branch range
          LDX  SUBCOPYSRC
          LDA  ,X+
          STX  SUBCOPYSRC
          LDY  SUBWPTR
          STA  ,Y+
          STY  SUBWPTR
          LDD  SUBOUTLEN
          ADDD #1
          STD  SUBOUTLEN
          LDD  SUBCOPYCNT
          SUBD #1
          STD  SUBCOPYCNT
          BRA  SUBCPLP
SUBCPDONE: RTS

SUBSTITUTEW: ; REWRITE: was a plain substring search-and-replace on
             ; the bare registered name (via SEARCHW), which found
             ; "girl" and replaced only that span, leaving surrounding
             ; "%...%" delimiters untouched in the output - and never
```

```
                ; returned the substitution count ANS requires as a
                ; third stack item. Per the ANS spec (forth-standard.org/
                ; standard/string/SUBSTITUTE), SUBSTITUTE must scan for
                ; text between '%' (ASCII $25) delimiter pairs
                ; specifically: "%%" collapses to a single '%' (count
                ; unchanged); a name matching the REPLACES registration
                ; has the ENTIRE "%name%" span - delimiters included -
                ; replaced by the substitution text (count incremented);
                ; a non-matching name is passed through unchanged,
                ; delimiters and all (count unchanged); an unpaired
                ; trailing '%' with no closing delimiter passes the
                ; residue through unchanged. Confirmed via MAME testing
                ; against a real template. GLOBALS is fully packed
                ; (256/256), so this reuses MSCR/MSCR2/MSCR3/MSCR4
                ; (confirmed untouched by SUBCOPY) rather than adding
                ; dedicated cells: MSCR = current read position, MSCR2 =
                ; running substitution count, MSCR3/MSCR4 = local scratch
                ; per %-pair found. Reuses the existing COMPAREW for name
                ; matching rather than a new comparison loop.
                PULU D
                STD  SUBDESTCAP
                PULU D
                STD  SUBDESTADR
                PULU D
                STD  SUBSRCLEN
                PULU D
                STD  SUBSRCADR

                LDY  SUBDESTADR
                STY  SUBWPTR
                LDD  #0
                STD  SUBOUTLEN
                STD  MSCR        ; read position, starts at 0
                STD  MSCR2       ; substitution count, starts at 0

SUBSCAN:        LDD  MSCR
                CMPD SUBSRCLEN
                LBHS SUBSDONE
                LDX  SUBSRCADR
                LEAX D,X
                LDA  ,X
                CMPA #$25         ; '%' delimiter
                BEQ  SUBPCT
                LDD  #1
                JSR  SUBCOPY
                LDD  MSCR
                ADDD #1
                STD  MSCR
                LBRA SUBSCAN

SUBPCT:         LDD  MSCR
                ADDD #1
                STD  MSCR3        ; scan for the closing '%'
SUBFINDCL:      LDD  MSCR3
                CMPD SUBSRCLEN
                BLO  SUBFC2
                ; no closing delimiter found - residue passed unchanged
                LDD  MSCR
                LDX  SUBSRCADR
                LEAX D,X
                LDD  SUBSRCLEN
                SUBD MSCR
                JSR  SUBCOPY
                LDD  SUBSRCLEN
                STD  MSCR
                LBRA SUBSCAN
SUBFC2:         LDD  MSCR3
                LDX  SUBSRCADR
                LEAX D,X
                LDA  ,X
                CMPA #$25
```

```
            BEQ  SUBFOUNDCL
            LDD  MSCR3
            ADDD #1
            STD  MSCR3
            LBRA SUBFINDCL

SUBFOUNDCL: LDD  MSCR3
            SUBD MSCR
            SUBD #1
            STD  MSCR4         ; enclosed name length
            LBNE SUBHASNAME
            ; namelen 0: "%%" -> single '%' to output, count unchanged
            LDD  MSCR
            LDX  SUBSRCADR
            LEAX D,X
            LDD  #1
            JSR  SUBCOPY
            LDD  MSCR3
            ADDD #1
            STD  MSCR
            LBRA SUBSCAN

SUBHASNAME: LDD  MSCR
            ADDD #1
            LDX  SUBSRCADR
            LEAX D,X
            PSHU X
            LDD  MSCR4
            PSHU D
            LDD  REPLNAME
            PSHU D
            LDD  REPLNLEN
            PSHU D
            JSR  COMPAREW
            PULU D
            CMPD #0
            BNE  SUBNOMATCH
            ; valid name - replace the whole %name% span
            LDX  REPLVAL
            LDD  REPLVLEN
            JSR  SUBCOPY
            LDD  MSCR2
            ADDD #1
            STD  MSCR2
            LDD  MSCR3
            ADDD #1
            STD  MSCR
            LBRA SUBSCAN

SUBNOMATCH: ; not a registered name - pass %name% through unchanged
            LDD  MSCR
            LDX  SUBSRCADR
            LEAX D,X
            LDD  MSCR3
            SUBD MSCR
            ADDD #1
            JSR  SUBCOPY
            LDD  MSCR3
            ADDD #1
            STD  MSCR
            LBRA SUBSCAN

SUBSDONE:   LDX  SUBDESTADR
            PSHU X
            LDD  SUBOUTLEN
            PSHU D
            LDD  MSCR2
            PSHU D
            RTS

SUBOVERFLOW: LDD #-1
```

```
            PSHU D
            JSR  THROW
```

8.21 Numeric Output

```
; ============================================================
; SECTION 20: NUMERIC OUTPUT (pictured + direct)
; ============================================================
LTNUM:   JSR   PADW
         PULU  D
         STD   HLD
         RTS

HOLD:    PULU  D
         LDX   HLD
         LEAX  -1,X
         STX   HLD
         STB   ,X
         RTS

HOLDS:   PULU  D
         STD   HSLEN
         PULU  D
         STD   HSADDR
HSLOOP:  LDD   HSLEN
         BEQ   HSDONE
         SUBD  #1
         STD   HSLEN
         LDX   HSADDR
         LDD   HSLEN
         LEAX  D,X
         LDA   ,X
         TFR   A,B
         CLRA
         PSHU  D
         JSR   HOLD
         BRA   HSLOOP
HSDONE:  RTS

NUMSIGN: PULU  D
         STD   UDHI
         PULU  D
         STD   UDLO
         JSR   UDDIGIT
         LDA   REM
         CMPA  #10
         BLO   NDIGIT
         ADDA  #'A'-10
         BRA   NHOLD
NDIGIT:  ADDA  #'0'
NHOLD:   TFR   A,B
         CLRA
         PSHU  D
         JSR   HOLD
         LDD   UDLO
         PSHU  D
         LDD   UDHI
         PSHU  D
         RTS

UDDIGIT: CLR   REM
         LDB   #32
         STB   DCNT
UDDLOOP: ASL   UDLO+1
         ROL   UDLO
         ROL   UDHI+1
         ROL   UDHI
         ROL   REM
```

```
         LDA   REM
         CMPA  BASE+1
         BLO   UDNEXT
         SUBA  BASE+1
         STA   REM
         INC   UDLO+1
UDNEXT:  DEC   DCNT
         BNE   UDDLOOP
         RTS

NUMSIGNS: JSR  NUMSIGN
          LDD  UDHI
          BNE  NUMSIGNS
          LDD  UDLO
          BNE  NUMSIGNS
          RTS

SIGN:    PULU  D
         BPL   SIGNDONE
         LDD   #'-'
         PSHU  D
         JSR   HOLD
SIGNDONE: RTS

NUMGT:   PULU  D
         PULU  D
         LDX   HLD
         PSHU  X
         JSR   PADW
         PULU  D
         SUBD  HLD
         PSHU  D
         RTS

DOT:     PULU  D
         STD   SAVEN
         BPL   DABSOK
         COMA
         COMB
         ADDD  #1
DABSOK:  PSHU  D
         LDD   #0
         PSHU  D
         JSR   LTNUM
         JSR   NUMSIGNS
         LDD   SAVEN
         PSHU  D
         JSR   SIGN
         JSR   NUMGT
         JSR   TYPE
         LDD   #32
         PSHU  D
         JSR   EMIT
         RTS

UDOT:    PULU  D
         PSHU  D
         LDD   #0
         PSHU  D
         JSR   LTNUM
         JSR   NUMSIGNS
         JSR   NUMGT
         JSR   TYPE
         LDD   #32
         PSHU  D
         JSR   EMIT
         RTS

DOTR:    PULU  D
         STD   DRWIDTH
         PULU  D
```

```
         STD   SAVEN
         BPL   DRABSOK
         COMA
         COMB
         ADDD  #1
DRABSOK: PSHU  D
         LDD   #0
         PSHU  D
         JSR   LTNUM
         JSR   NUMSIGNS
         LDD   SAVEN
         PSHU  D
         JSR   SIGN
         JSR   NUMGT
         PULU  D
         STD   DRLEN
         PULU  D
         STD   DRADDR
         LDD   DRWIDTH
         SUBD  DRLEN
         BLE   DRNOPAD
         STD   DRPAD
DRPADLP: LDD   DRPAD
         BEQ   DRNOPAD
         SUBD  #1
         STD   DRPAD
         LDD   #32
         PSHU  D
         JSR   EMIT
         BRA   DRPADLP
DRNOPAD: LDX   DRADDR
         PSHU  X
         LDD   DRLEN
         PSHU  D
         JSR   TYPE
         RTS

UDOTR:   PULU  D
         STD   DRWIDTH
         PULU  D
         PSHU  D
         LDD   #0
         PSHU  D
         JSR   LTNUM
         JSR   NUMSIGNS
         JSR   NUMGT
         PULU  D
         STD   DRLEN
         PULU  D
         STD   DRADDR
         LDD   DRWIDTH
         SUBD  DRLEN
         BLE   UDRNOPAD
         STD   DRPAD
UDRPADLP: LDD  DRPAD
          BEQ  UDRNOPAD
          SUBD #1
          STD  DRPAD
          LDD  #32
          PSHU D
          JSR  EMIT
          BRA  UDRPADLP
UDRNOPAD: LDX  DRADDR
          PSHU X
          LDD  DRLEN
          PSHU D
          JSR  TYPE
          RTS

QMARK:   PULU  X
         LDD   ,X
```

```
          PSHU  D
          JSR   DOT
          RTS

DDOT:     PULU  D
          STD   PRODHI
          PULU  D
          STD   PRODLO
          LDD   PRODHI
          STD   SAVEN
          BPL   DDPOS
          JSR   MNEG32
DDPOS:    LDD   PRODLO
          PSHU  D
          LDD   PRODHI
          PSHU  D
          JSR   LTNUM
          JSR   NUMSIGNS
          LDD   SAVEN
          PSHU  D
          JSR   SIGN
          JSR   NUMGT
          JSR   TYPE
          LDD   #32
          PSHU  D
          JSR   EMIT
          RTS

DDOTR:    PULU  D
          STD   DRWIDTH
          PULU  D
          STD   PRODHI
          PULU  D
          STD   PRODLO
          LDD   PRODHI
          STD   SAVEN
          BPL   DDRPOS
          JSR   MNEG32
DDRPOS:   LDD   PRODLO
          PSHU  D
          LDD   PRODHI
          PSHU  D
          JSR   LTNUM
          JSR   NUMSIGNS
          LDD   SAVEN
          PSHU  D
          JSR   SIGN
          JSR   NUMGT
          PULU  D
          STD   DRLEN
          PULU  D
          STD   DRADDR
          LDD   DRWIDTH
          SUBD  DRLEN
          BLE   DRDNOPAD
          STD   DRPAD
DRDPADLP: LDD  DRPAD
           BEQ  DRDNOPAD
           SUBD #1
           STD  DRPAD
           LDD  #32
           PSHU D
           JSR  EMIT
           BRA  DRDPADLP
DRDNOPAD: LDX  DRADDR
           PSHU X
           LDD  DRLEN
           PSHU D
           JSR  TYPE
           RTS
```

8.22 Base / Radix Control

```
; ============================================================
; SECTION 21: BASE / RADIX CONTROL
; ============================================================
BASEW:   LDD  #BASE
         PSHU D
         RTS

DECIMAL: LDD  #10
         STD  BASE
         RTS

HEXW:    LDD  #16
         STD  BASE
         RTS

BINARYW: LDD  #2
         STD  BASE
         RTS
```

8.23 Output Formatting

```
; ============================================================
; SECTION 22: OUTPUT FORMATTING (CR/SPACE/SPACES)
; ============================================================
CRW:     LDD   #13
         PSHU  D
         JSR   EMIT
         LDD   #10
         PSHU  D
         JSR   EMIT
         RTS

SPACEW:  LDD   #32
         PSHU  D
         JSR   EMIT
         RTS

SPACESW: PULU  D
         STD   SHCNT2
SPLOOP:  LDD   SHCNT2
         BLE   SPDONE
         LDD   #32
         PSHU  D
         JSR   EMIT
         LDD   SHCNT2
         SUBD  #1
         STD   SHCNT2
         BRA   SPLOOP
SPDONE:  RTS
```

8.24 Comment Words

```
; ============================================================
; SECTION 23: COMMENT WORDS
; ============================================================
LPAREN:  LDD   #')'
         PSHU  D
         JSR   WORD
         RTS
```

```
BACKSLASH: LDD  SRCLEN
           STD  TOIN
           RTS
```

8.25 Environmental Query / SOURCE / REFILL / EVALUATE

```
; ============================================================
; SECTION 24: ENVIRONMENTAL QUERY / SOURCE / REFILL / EVALUATE
; ============================================================
SOURCEW: LDD   SRCADDR
         PSHU  D
         LDD   SRCLEN
         PSHU  D
         RTS

SOURCEID: LDD  SRCID
          PSHU D
          RTS

REFILLW: LDD   SRCID
         BEQ   RFTERM
         LDD   #FALSEV
         PSHU  D
         RTS
RFTERM:  JSR   QUERY
         LDD   #TRUEV
         PSHU  D
         RTS

EVALUATEW: LDD  SRCADDR
           STD  EVSAVEA
           LDD  SRCLEN
           STD  EVSAVEL
           LDD  SRCID
           STD  EVSAVEI
           LDD  TOIN
           STD  EVSAVET
           PULU D
           STD  SRCLEN
           PULU D
           STD  SRCADDR
           LDD  #-1
           STD  SRCID
           LDD  #0
           STD  TOIN
           JSR  INTERPRET
           LDD  EVSAVEA
           STD  SRCADDR
           LDD  EVSAVEL
           STD  SRCLEN
           LDD  EVSAVEI
           STD  SRCID
           LDD  EVSAVET
           STD  TOIN
           RTS

; ENVIRONMENT? - dispatcher complete; table has only the
; entries that could be derived without fabricating unfixed
; capacities (/HOLD, /PAD were explicitly left out - see the
; source conversation's ENVTABLE discussion). MAX-D/MAX-UD
; and WORDLISTS/FLOORED need dispatcher extensions not yet
; built (single-cell-only ENVFOUND path).
ENVQUERY: PULU D
          STD  ENVLEN
          PULU D
          STD  ENVADDR
          LDX  #ENVTABLE
ENVLOOP:  LDD  ,X
```

```
          CMPD #0
          BEQ  ENVNOTFOUND
          PSHU D
          LDD  2,X
          PSHU D
          LDD  ENVADDR
          PSHU D
          LDD  ENVLEN
          PSHU D
          JSR  COMPAREW
          PULU D
          CMPD #0
          BEQ  ENVFOUND
          LEAX 6,X
          BRA  ENVLOOP
ENVFOUND: LDD  4,X
          PSHU D
          LDD  #TRUEV
          PSHU D
          RTS
ENVNOTFOUND: LDD #FALSEV
             PSHU D
             RTS

ENVTABLE:
         FDB   EN1,EN1L,31
         FDB   EN2,EN2L,32767
         FDB   EN3,EN3L,65535
         FDB   EN6,EN6L,8
         FDB   0
EN1:     FCC   "/COUNTED-STRING"
EN1L     EQU   *-EN1
EN2:     FCC   "MAX-N"
EN2L     EQU   *-EN2
EN3:     FCC   "MAX-U"
EN3L     EQU   *-EN3
EN6:     FCC   "ADDRESS-UNIT-BITS"
EN6L     EQU   *-EN6
```

8.26 Tools Word Set

```
; ============================================================
; SECTION 25: TOOLS WORD SET (.S / WORDS / DUMP)
; ============================================================
DOTS:    TFR   U,D
         STD   DSPTMP
DSLOOP:  LDD   DSPTMP
         CMPD  #SP0
         BEQ   DSDONE
         LDX   DSPTMP
         LDD   ,X
         PSHU  D
         JSR   DOT
         LDD   DSPTMP
         ADDD  #2
         STD   DSPTMP
         BRA   DSLOOP
DSDONE:  RTS

WORDSW:  LDD   LATEST
         STD   WWALK
WWLOOP:  LDD   WWALK
         BEQ   WWDONE
         LDX   WWALK
         LDA   ,X
         STA   HDRFLAGS
         LEAX  1,X
         PSHU  X
```

```
         LDB   HDRFLAGS
         ANDB  #$1F
         CLRA
         PSHU  D
         JSR   TYPE
         JSR   SPACEW
         LDX   WWALK
         LEAX  1,X
         LDB   HDRFLAGS
         ANDB  #$1F
         CLRA
         LEAX  D,X
         LDD   ,X
         STD   WWALK
         BRA   WWLOOP
WWDONE:  JSR   CRW
         RTS

HEXDIGIT: PULU D
          CMPD #10
          BLO  HDDIGIT
          ADDD #'A'-10
          BRA  HDEMIT
HDDIGIT:  ADDD #'0'
HDEMIT:   PSHU D
          JSR  EMIT
          RTS

HEXBYTE: PULU  D
         STB   MSCR
         LDB   MSCR+1
         LSRB
         LSRB
         LSRB
         LSRB
         CLRA
         PSHU  D
         JSR   HEXDIGIT
         LDB   MSCR+1
         ANDB  #$0F
         CLRA
         PSHU  D
         JSR   HEXDIGIT
         RTS

; DUMP - includes the partial-final-line ASCII fix
DUMPW:   PULU  D
         STD   DUMPCNT
         PULU  D
         STD   DUMPADDR
DULINE:  LDD   DUMPCNT
         LBEQ  DUDONE          ; was BEQ - out of short-branch range
         LDD   DUMPADDR
         STD   HEXBUF
         CLR   DUMPCOL
         LDA   #16
         STA   DUVALID
DUHEX:   LDB   DUMPCOL
         CMPB  #16
         BEQ   DUASCII
         LDD   DUMPCNT
         BNE   DUHEXBYTE
         LDA   DUMPCOL
         STA   DUVALID
         BRA   DUHEXPAD
DUHEXBYTE: LDX  DUMPADDR
           LDB  ,X
           CLRA
           PSHU D
           JSR  HEXBYTE
           LDD  #32
```

```
           PSHU D
           JSR  EMIT
           LDX  DUMPADDR
           LEAX 1,X
           STX  DUMPADDR
           LDD  DUMPCNT
           SUBD #1
           STD  DUMPCNT
           INC  DUMPCOL
           BRA  DUHEX
DUHEXPAD: LDD  #32
          PSHU D
          JSR  EMIT
          PSHU D
          JSR  EMIT
          PSHU D
          JSR  EMIT
          INC  DUMPCOL
          LDB  DUMPCOL
          CMPB #16
          BNE  DUHEXPAD
          BRA  DUASCII
DUASCII: LDD  #32
         PSHU D
         JSR  EMIT
         CLR  DUMPCOL
DUACHAR: LDB  DUMPCOL
         CMPB #16
         BEQ  DULEND
         CMPB DUVALID
         BHS  DUABLANK
         LDX  HEXBUF
         LDB  DUMPCOL
         CLRA
         LEAX D,X
         LDA  ,X
         CMPA #32
         BLO  DUDOT
         CMPA #127
         BHS  DUDOT
         BRA  DUPRINT
DUDOT:   LDA  #'.'
DUPRINT: TFR  A,B
         CLRA
         PSHU D
         JSR  EMIT
         INC  DUMPCOL
         BRA  DUACHAR
DUABLANK: LDD #32
          PSHU D
          JSR EMIT
          INC DUMPCOL
          BRA DUACHAR
DULEND:  JSR  CRW
         LDD  DUMPCNT
         LBNE DULINE           ; was BNE - out of short-branch range
DUDONE:  RTS
```

8.27 ABORT / QUIT Hand-Built Headers

```
; ============================================================
; SECTION 26: TRUE / FALSE (CONSTANT-pattern ROM words). This
; section used to also hold ABORT/QUIT's hand-built dictionary
; headers (ABORTHDR/QUITHDR) - moved into BASEDICT and renamed to
; H_ABORT/H_QUIT (see the end of SECTION 27) so every header in
; this ROM lives in one contiguous block, rather than header data
; sitting apart from the rest of the dictionary inside BASECODE.
; ============================================================
```

```
; ------------------------------------------------------------
; TRUE / FALSE - CONSTANT TRUE -1 / CONSTANT FALSE 0. The first
; CONSTANT-pattern ROM-resident words in this system: every
; other ROM word's CFA is a plain code label (a real routine),
; but a CONSTANT's CFA is the DODOES trampoline pattern (JSR
; DODOES + FDB ATSIGN + FDB <value-cell-address>), matching
; exactly what interactive CONSTANT compiles at runtime - the
; only difference is the value-cell address is a fixed, known-
; at-assemble-time label here (TRUEVAL/FALSEVAL) rather than a
; CODEHERE snapshot captured dynamically. DODOES pushes the
; value stored at the PFA field (the address in TRUEVAL/
; FALSEVAL); ATSIGN then fetches through that address, yielding
; the actual -1 / 0 - the same indirection interactive CONSTANT
; relies on, just with fixed addresses instead of a runtime one.
; ------------------------------------------------------------
TRUEBODY:  JSR   DODOES
           FDB   ATSIGN
           FDB   TRUEVAL
TRUEVAL:   FDB   -1

FALSEBODY: JSR   DODOES
           FDB   ATSIGN
           FDB   FALSEVAL
FALSEVAL:  FDB   0

; Verify no collision with init code,
; value should match ORG INITCODE.
BASECODEEND  EQU   *
BASECODESIZE EQU   BASECODEEND-BASECODE

; Prevent the assembler from extinguishing the gap between the
; BASECODE block and the INITCODE block when it generates the
; .bin file.
BASEND:
         FILL $FF,INITCODE-BASEND
```

8.28 Forth Dictionary (ROM Base Dictionary Headers)

```
; ============================================================
; SECTION 27: FORTH DICTIONARY (ROM base dictionary headers)
; Every primitive word in the Glossary gets a real header here,
; chained via LINK, living in BASEDICT ($D83F-$E006, an exact fit
; for this dictionary's 1992 bytes - was $E000-$E7FF). CFA points
; directly at each primitive's own code label for almost every
; entry - these are raw code entries, not DODOES trampolines, so
; CFA = the label itself. The two exceptions are TRUE and FALSE
; (added in a later pass, chain's newest entries before H_ABORT/
; H_QUIT below): their CFA is the DODOES-trampoline pattern
; instead, matching what CONSTANT would compile interactively -
; see TRUEBODY/FALSEBODY, section 26, for why and how.
;
; DOES> is included (H_DOESGT) - added in a follow-up pass after
; the original 214-entry generation flagged it as missing, then
; moved again so it sits immediately after CREATE in the chain
; (H_CREATE -> H_DOESGT -> H_VARIABLE) rather than at the chain's
; newest end, since the two words are tightly coupled and read
; better adjacent. SETDOES (the runtime it compiles a call to)
; lives beside DODOES/DOESRT0; DOESGT is DOES>'s code label,
; since a literal ">" is not valid in a 6809 assembler label.
;
; H_ABORT and H_QUIT (formerly ABORTHDR/QUITHDR, renamed to match
; this file's H_ naming convention) are hand-built, not produced
; by the same generation pass as the other 217 entries - see the
; note where they're defined, at the end of this section, for why
; HEADER/CREATE couldn't be used directly for these two. They used
; to live physically apart from the rest of the dictionary, inside
; BASECODE rather than BASEDICT, despite being header data rather
; than code - moved here so every header lives in one place and
```

```
; the chain (H_QUIT -> H_ABORT -> (newest entry below) -> ... ->
; (oldest entry) -> 0) is visible in one block. BASELATEST remains
; H_QUIT, the chain's true overall head.
;
; H_ABORT's LINK field, a placeholder 0 since it was first built,
; is resolved here: it now points to this chain's newest entry.
;
; Names containing a literal double-quote (S", ."", ABORT") have
; that character split into a standalone FCB $22 rather than
; escaped inside FCC - see emit_name's comment for why.
; ============================================================

          ORG   BASEDICT       ; BASEDICT is $D83F

H_KEY:
          FCB   $03
          FCC   "KEY"
          FDB   0
          FDB   KEY
H_KEYQ:
          FCB   $04
          FCC   "KEY?"
          FDB   H_KEY
          FDB   KEYQ
H_EMIT:
          FCB   $04
          FCC   "EMIT"
          FDB   H_KEYQ
          FDB   EMIT
H_ACCEPT:
          FCB   $06
          FCC   "ACCEPT"
          FDB   H_EMIT
          FDB   ACCEPT
H_EXPECTW:
          FCB   $06
          FCC   "EXPECT"
          FDB   H_ACCEPT
          FDB   EXPECTW
H_QUERY:
          FCB   $05
          FCC   "QUERY"
          FDB   H_EXPECTW
          FDB   QUERY
H_TYPE:
          FCB   $04
          FCC   "TYPE"
          FDB   H_QUERY
          FDB   TYPE
H_CRW:
          FCB   $02
          FCC   "CR"
          FDB   H_TYPE
          FDB   CRW
H_SPACEW:
          FCB   $05
          FCC   "SPACE"
          FDB   H_CRW
          FDB   SPACEW
H_SPACESW:
          FCB   $06
          FCC   "SPACES"
          FDB   H_SPACEW
          FDB   SPACESW
H_DUP:
          FCB   $03
          FCC   "DUP"
          FDB   H_SPACESW
          FDB   DUP
H_DROP:
          FCB   $04
```

```
          FCC   "DROP"
          FDB   H_DUP
          FDB   DROP
H_SWAP:
          FCB   $04
          FCC   "SWAP"
          FDB   H_DROP
          FDB   SWAP
H_OVER:
          FCB   $04
          FCC   "OVER"
          FDB   H_SWAP
          FDB   OVER
H_ROT:
          FCB   $03
          FCC   "ROT"
          FDB   H_OVER
          FDB   ROT
H_QDUP:
          FCB   $04
          FCC   "?DUP"
          FDB   H_ROT
          FDB   QDUP
H_DEPTH:
          FCB   $05
          FCC   "DEPTH"
          FDB   H_QDUP
          FDB   DEPTH
H_DDUP:
          FCB   $04
          FCC   "2DUP"
          FDB   H_DEPTH
          FDB   DDUP
H_DDROP:
          FCB   $05
          FCC   "2DROP"
          FDB   H_DDUP
          FDB   DDROP
H_DSWAP:
          FCB   $05
          FCC   "2SWAP"
          FDB   H_DDROP
          FDB   DSWAP
H_DOVER:
          FCB   $05
          FCC   "2OVER"
          FDB   H_DSWAP
          FDB   DOVER
H_NIP:
          FCB   $03
          FCC   "NIP"
          FDB   H_DOVER
          FDB   NIP
H_TUCK:
          FCB   $04
          FCC   "TUCK"
          FDB   H_NIP
          FDB   TUCK
H_PICK:
          FCB   $04
          FCC   "PICK"
          FDB   H_TUCK
          FDB   PICK
H_ROLL:
          FCB   $04
          FCC   "ROLL"
          FDB   H_PICK
          FDB   ROLL
H_DROT:
          FCB   $04
          FCC   "2ROT"
```

```
          FDB   H_ROLL
          FDB   DROT
H_TOR:
          FCB   $02
          FCC   ">R"
          FDB   H_DROT
          FDB   TOR
H_FROMR:
          FCB   $02
          FCC   "R>"
          FDB   H_TOR
          FDB   FROMR
H_RFETCH:
          FCB   $02
          FCC   "R@"
          FDB   H_FROMR
          FDB   RFETCH
H_TWOTOR:
          FCB   $03
          FCC   "2>R"
          FDB   H_RFETCH
          FDB   TWOTOR
H_TWOFROMR:
          FCB   $03
          FCC   "2R>"
          FDB   H_TWOTOR
          FDB   TWOFROMR
H_TWORFETCH:
          FCB   $03
          FCC   "2R@"
          FDB   H_TWOFROMR
          FDB   TWORFETCH
H_PLUS:
          FCB   $01
          FCC   "+"
          FDB   H_TWORFETCH
          FDB   PLUS
H_MINUS:
          FCB   $01
          FCC   "-"
          FDB   H_PLUS
          FDB   MINUS
H_STAR:
          FCB   $01
          FCC   "*"
          FDB   H_MINUS
          FDB   STAR
H_SLASH:
          FCB   $01
          FCC   "/"
          FDB   H_STAR
          FDB   SLASH
H_MODW:
          FCB   $03
          FCC   "MOD"
          FDB   H_SLASH
          FDB   MODW
H_SLASHMOD:
          FCB   $04
          FCC   "/MOD"
          FDB   H_MODW
          FDB   SLASHMOD
H_NEGATE:
          FCB   $06
          FCC   "NEGATE"
          FDB   H_SLASHMOD
          FDB   NEGATE
H_ABSW:
          FCB   $03
          FCC   "ABS"
          FDB   H_NEGATE
```

```
          FDB   ABSW
H_MIN:
          FCB   $03
          FCC   "MIN"
          FDB   H_ABSW
          FDB   MIN
H_MAX:
          FCB   $03
          FCC   "MAX"
          FDB   H_MIN
          FDB   MAX
H_ONEPLUS:
          FCB   $02
          FCC   "1+"
          FDB   H_MAX
          FDB   ONEPLUS
H_ONEMINUS:
          FCB   $02
          FCC   "1-"
          FDB   H_ONEPLUS
          FDB   ONEMINUS
H_TWOPLUS:
          FCB   $02
          FCC   "2+"
          FDB   H_ONEMINUS
          FDB   TWOPLUS
H_TWOSTAR:
          FCB   $02
          FCC   "2*"
          FDB   H_TWOPLUS
          FDB   TWOSTAR
H_TWOSLASH:
          FCB   $02
          FCC   "2/"
          FDB   H_TWOSTAR
          FDB   TWOSLASH
H_STARSLASH:
          FCB   $02
          FCC   "*/"
          FDB   H_TWOSLASH
          FDB   STARSLASH
H_STARSLASHMOD:
          FCB   $05
          FCC   "*/MOD"
          FDB   H_STARSLASH
          FDB   STARSLASHMOD
H_UMSTAR:
          FCB   $03
          FCC   "UM*"
          FDB   H_STARSLASHMOD
          FDB   UMSTAR
H_UMSLASHMOD:
          FCB   $06
          FCC   "UM/MOD"
          FDB   H_UMSTAR
          FDB   UMSLASHMOD
H_MSTAR:
          FCB   $02
          FCC   "M*"
          FDB   H_UMSLASHMOD
          FDB   MSTAR
H_FMSLASHMOD:
          FCB   $06
          FCC   "FM/MOD"
          FDB   H_MSTAR
          FDB   FMSLASHMOD
H_SMSLASHREM:
          FCB   $06
          FCC   "SM/REM"
          FDB   H_FMSLASHMOD
          FDB   SMSLASHREM
```

```
H_DPLUS:
         FCB   $02
         FCC   "D+"
         FDB   H_SMSLASHREM
         FDB   DPLUS
H_DMINUS:
         FCB   $02
         FCC   "D-"
         FDB   H_DPLUS
         FDB   DMINUS
H_DNEGATEW:
         FCB   $07
         FCC   "DNEGATE"
         FDB   H_DMINUS
         FDB   DNEGATEW
H_DABSW:
         FCB   $04
         FCC   "DABS"
         FDB   H_DNEGATEW
         FDB   DABSW
H_MPLUS:
         FCB   $02
         FCC   "M+"
         FDB   H_DABSW
         FDB   MPLUS
H_STOD:
         FCB   $03
         FCC   "S>D"
         FDB   H_MPLUS
         FDB   STOD
H_DTOS:
         FCB   $03
         FCC   "D>S"
         FDB   H_STOD
         FDB   DTOS
H_DMAXW:
         FCB   $04
         FCC   "DMAX"
         FDB   H_DTOS
         FDB   DMAXW
H_DMINW:
         FCB   $04
         FCC   "DMIN"
         FDB   H_DMAXW
         FDB   DMINW
H_ANDW:
         FCB   $03
         FCC   "AND"
         FDB   H_DMINW
         FDB   ANDW
H_ORW:
         FCB   $02
         FCC   "OR"
         FDB   H_ANDW
         FDB   ORW
H_XORW:
         FCB   $03
         FCC   "XOR"
         FDB   H_ORW
         FDB   XORW
H_INVERT:
         FCB   $06
         FCC   "INVERT"
         FDB   H_XORW
         FDB   INVERT
H_LSHIFT:
         FCB   $06
         FCC   "LSHIFT"
         FDB   H_INVERT
         FDB   LSHIFT
H_RSHIFT:
```

```
          FCB   $06
          FCC   "RSHIFT"
          FDB   H_LSHIFT
          FDB   RSHIFT
H_CELLSW:
          FCB   $05
          FCC   "CELLS"
          FDB   H_RSHIFT
          FDB   CELLSW
H_CELLPLUS:
          FCB   $05
          FCC   "CELL+"
          FDB   H_CELLSW
          FDB   CELLPLUS
H_CHARSW:
          FCB   $05
          FCC   "CHARS"
          FDB   H_CELLPLUS
          FDB   CHARSW
H_CHARPLUS:
          FCB   $05
          FCC   "CHAR+"
          FDB   H_CHARSW
          FDB   CHARPLUS
H_ALIGNW:
          FCB   $05
          FCC   "ALIGN"
          FDB   H_CHARPLUS
          FDB   ALIGNW
H_ALIGNEDW:
          FCB   $07
          FCC   "ALIGNED"
          FDB   H_ALIGNW
          FDB   ALIGNEDW
H_EQUALW:
          FCB   $01
          FCC   "="
          FDB   H_ALIGNEDW
          FDB   EQUALW
H_LESSW:
          FCB   $01
          FCC   "<"
          FDB   H_EQUALW
          FDB   LESSW
H_GREATERW:
          FCB   $01
          FCC   ">"
          FDB   H_LESSW
          FDB   GREATERW
H_ZEROEQ:
          FCB   $02
          FCC   "0="
          FDB   H_GREATERW
          FDB   ZEROEQ
H_ZEROLT:
          FCB   $02
          FCC   "0<"
          FDB   H_ZEROEQ
          FDB   ZEROLT
H_ULESSW:
          FCB   $02
          FCC   "U<"
          FDB   H_ZEROLT
          FDB   ULESSW
H_NOTEQUAL:
          FCB   $02
          FCC   "<>"
          FDB   H_ULESSW
          FDB   NOTEQUAL
H_ZERONE:
          FCB   $03
```

```
         FCC   "0<>"
         FDB   H_NOTEQUAL
         FDB   ZERONE
H_ZEROGT:
         FCB   $02
         FCC   "0>"
         FDB   H_ZERONE
         FDB   ZEROGT
H_UGREATER:
         FCB   $02
         FCC   "U>"
         FDB   H_ZEROGT
         FDB   UGREATER
H_WITHINW:
         FCB   $06
         FCC   "WITHIN"
         FDB   H_UGREATER
         FDB   WITHINW
H_DEQUAL:
         FCB   $02
         FCC   "D="
         FDB   H_WITHINW
         FDB   DEQUAL
H_DLESSW:
         FCB   $02
         FCC   "D<"
         FDB   H_DEQUAL
         FDB   DLESSW
H_DULESSW:
         FCB   $03
         FCC   "DU<"
         FDB   H_DLESSW
         FDB   DULESSW
H_IF:
         FCB   $82
         FCC   "IF"
         FDB   H_DULESSW
         FDB   IF
H_THEN:
         FCB   $84
         FCC   "THEN"
         FDB   H_IF
         FDB   THEN
H_ELSE:
         FCB   $84
         FCC   "ELSE"
         FDB   H_THEN
         FDB   ELSE
H_BEGIN:
         FCB   $85
         FCC   "BEGIN"
         FDB   H_ELSE
         FDB   BEGIN
H_UNTIL:
         FCB   $85
         FCC   "UNTIL"
         FDB   H_BEGIN
         FDB   UNTIL
H_AGAIN:
         FCB   $85
         FCC   "AGAIN"
         FDB   H_UNTIL
         FDB   AGAIN
H_WHILE:
         FCB   $85
         FCC   "WHILE"
         FDB   H_AGAIN
         FDB   WHILE
H_REPEAT:
         FCB   $86
         FCC   "REPEAT"
```

```
          FDB   H_WHILE
          FDB   REPEAT
H_RECURSE:
          FCB   $87
          FCC   "RECURSE"
          FDB   H_REPEAT
          FDB   RECURSE
H_DO:
          FCB   $82
          FCC   "DO"
          FDB   H_RECURSE
          FDB   DO
H_QDO:
          FCB   $83
          FCC   "?DO"
          FDB   H_DO
          FDB   QDO
H_LOOP:
          FCB   $84
          FCC   "LOOP"
          FDB   H_QDO
          FDB   LOOP
H_PLUSLOOP:
          FCB   $85
          FCC   "+LOOP"
          FDB   H_LOOP
          FDB   PLUSLOOP
H_IWORD:
          FCB   $01
          FCC   "I"
          FDB   H_PLUSLOOP
          FDB   IWORD
H_JWORD:
          FCB   $01
          FCC   "J"
          FDB   H_IWORD
          FDB   JWORD
H_LEAVE:
          FCB   $05
          FCC   "LEAVE"
          FDB   H_JWORD
          FDB   LEAVE
H_UNLOOP:
          FCB   $06
          FCC   "UNLOOP"
          FDB   H_LEAVE
          FDB   UNLOOP
H_EXIT:
          FCB   $84
          FCC   "EXIT"
          FDB   H_UNLOOP
          FDB   EXIT
H_CASEW:
          FCB   $84
          FCC   "CASE"
          FDB   H_EXIT
          FDB   CASEW
H_OF:
          FCB   $82
          FCC   "OF"
          FDB   H_CASEW
          FDB   OF
H_ENDOF:
          FCB   $85
          FCC   "ENDOF"
          FDB   H_OF
          FDB   ENDOF
H_ENDCASE:
          FCB   $87
          FCC   "ENDCASE"
          FDB   H_ENDOF
```

```
          FDB   ENDCASE
H_COLON:
          FCB   $01
          FCC   ":"
          FDB   H_ENDCASE
          FDB   COLON
H_SEMI:
          FCB   $81
          FCC   ";"
          FDB   H_COLON
          FDB   SEMI
H_CREATE:
          FCB   $06
          FCC   "CREATE"
          FDB   H_SEMI
          FDB   CREATE
H_DOESGT:
          FCB   $85           ; $80 IMMEDIATE | 5 (length of "DOES>")
          FCC   "DOES>"
          FDB   H_CREATE
          FDB   DOESGT
H_VARIABLE:
          FCB   $08
          FCC   "VARIABLE"
          FDB   H_DOESGT
          FDB   VARIABLE
H_CONSTANT:
          FCB   $08
          FCC   "CONSTANT"
          FDB   H_VARIABLE
          FDB   CONSTANT
H_VALUEW:
          FCB   $05
          FCC   "VALUE"
          FDB   H_CONSTANT
          FDB   VALUEW
H_TOW:
          FCB   $82
          FCC   "TO"
          FDB   H_VALUEW
          FDB   TOW
H_TWOVARIABLE:
          FCB   $09
          FCC   "2VARIABLE"
          FDB   H_TOW
          FDB   TWOVARIABLE
H_TWOCONSTANT:
          FCB   $09
          FCC   "2CONSTANT"
          FDB   H_TWOVARIABLE
          FDB   TWOCONSTANT
H_BUFFERCOLON:
          FCB   $07
          FCC   "BUFFER:"
          FDB   H_TWOCONSTANT
          FDB   BUFFERCOLON
H_DEFERW:
          FCB   $05
          FCC   "DEFER"
          FDB   H_BUFFERCOLON
          FDB   DEFERW
H_DEFERFETCH:
          FCB   $06
          FCC   "DEFER@"
          FDB   H_DEFERW
          FDB   DEFERFETCH
H_DEFERSTORE:
          FCB   $06
          FCC   "DEFER!"
          FDB   H_DEFERFETCH
          FDB   DEFERSTORE
```

```
H_ISW:
         FCB   $82
         FCC   "IS"
         FDB   H_DEFERSTORE
         FDB   ISW
H_ACTIONOF:
         FCB   $89
         FCC   "ACTION-OF"
         FDB   H_ISW
         FDB   ACTIONOF
H_MARKERW:
         FCB   $06
         FCC   "MARKER"
         FDB   H_ACTIONOF
         FDB   MARKERW
H_IMMEDIATE:
         FCB   $09
         FCC   "IMMEDIATE"
         FDB   H_MARKERW
         FDB   IMMEDIATE
H_STATEW:
         FCB   $05
         FCC   "STATE"
         FDB   H_IMMEDIATE
         FDB   STATEW
H_LBRACKET:
         FCB   $81
         FCC   "["
         FDB   H_STATEW
         FDB   LBRACKET
H_RBRACKET:
         FCB   $81
         FCC   "]"
         FDB   H_LBRACKET
         FDB   RBRACKET
H_TICK:
         FCB   $01
         FCC   "'"
         FDB   H_RBRACKET
         FDB   TICK
H_COMPILECOMMA:
         FCB   $08
         FCC   "COMPILE,"
         FDB   H_TICK
         FDB   COMPILECOMMA
H_LITERALW:
         FCB   $87
         FCC   "LITERAL"
         FDB   H_COMPILECOMMA
         FDB   LITERALW
H_BRACKTICK:
         FCB   $83
         FCC   "[']"
         FDB   H_LITERALW
         FDB   BRACKTICK
H_POSTPONEW:
         FCB   $88
         FCC   "POSTPONE"
         FDB   H_BRACKTICK
         FDB   POSTPONEW
H_TOBODY:
         FCB   $05
         FCC   ">BODY"
         FDB   H_POSTPONEW
         FDB   TOBODY
H_EXECUTE:
         FCB   $07
         FCC   "EXECUTE"
         FDB   H_TOBODY
         FDB   EXECUTE
H_SLITERALW:
```

```
          FCB   $88
          FCC   "SLITERAL"
          FDB   H_EXECUTE
          FDB   SLITERALW
H_ABORTQUOTE:
          FCB   $86
          FCC   "ABORT"
          FCB   $22        ; '"' - split out of FCC, not escaped within it
          FDB   H_SLITERALW
          FDB   ABORTQUOTE
H_ATSIGN:
          FCB   $01
          FCC   "@"
          FDB   H_ABORTQUOTE
          FDB   ATSIGN
H_STOREW:
          FCB   $01
          FCC   "!"
          FDB   H_ATSIGN
          FDB   STOREW
H_CFETCH:
          FCB   $02
          FCC   "C@"
          FDB   H_STOREW
          FDB   CFETCH
H_CSTOREW:
          FCB   $02
          FCC   "C!"
          FDB   H_CFETCH
          FDB   CSTOREW
H_PLUSSTORE:
          FCB   $02
          FCC   "+!"
          FDB   H_CSTOREW
          FDB   PLUSSTORE
H_DFETCH:
          FCB   $02
          FCC   "2@"
          FDB   H_PLUSSTORE
          FDB   DFETCH
H_DSTORE:
          FCB   $02
          FCC   "2!"
          FDB   H_DFETCH
          FDB   DSTORE
H_COMMA:
          FCB   $01
          FCC   ","
          FDB   H_DSTORE
          FDB   COMMA
H_CCOMMA:
          FCB   $02
          FCC   "C,"
          FDB   H_COMMA
          FDB   CCOMMA
H_ALLOT:
          FCB   $05
          FCC   "ALLOT"
          FDB   H_CCOMMA
          FDB   ALLOT
H_HEREW:
          FCB   $04
          FCC   "HERE"
          FDB   H_ALLOT
          FDB   HEREW
H_VCOMMA:
          FCB   $02
          FCC   "V,"
          FDB   H_HEREW
          FDB   VCOMMA
H_VCCOMMA:
```

```
          FCB   $03
          FCC   "VC,"
          FDB   H_VCOMMA
          FDB   VCCOMMA
H_VALLOT:
          FCB   $06
          FCC   "VALLOT"
          FDB   H_VCCOMMA
          FDB   VALLOT
H_VHEREW:
          FCB   $05
          FCC   "VHERE"
          FDB   H_VALLOT
          FDB   VHEREW
H_PADW:
          FCB   $03
          FCC   "PAD"
          FDB   H_VHEREW
          FDB   PADW
H_UNUSEDW:
          FCB   $06
          FCC   "UNUSED"
          FDB   H_PADW
          FDB   UNUSEDW
H_VUNUSEDW:
          FCB   $07
          FCC   "VUNUSED"
          FDB   H_UNUSEDW
          FDB   VUNUSEDW
H_MOVEW:
          FCB   $04
          FCC   "MOVE"
          FDB   H_VUNUSEDW
          FDB   MOVEW
H_FILLW:
          FCB   $04
          FCC   "FILL"
          FDB   H_MOVEW
          FDB   FILLW
H_ERASEW:
          FCB   $05
          FCC   "ERASE"
          FDB   H_FILLW
          FDB   ERASEW
H_CMOVEW:
          FCB   $05
          FCC   "CMOVE"
          FDB   H_ERASEW
          FDB   CMOVEW
H_CMOVEGT:
          FCB   $06
          FCC   "CMOVE>"
          FDB   H_CMOVEW
          FDB   CMOVEGT
H_COUNT:
          FCB   $05
          FCC   "COUNT"
          FDB   H_CMOVEGT
          FDB   COUNT
H_WORD:
          FCB   $04
          FCC   "WORD"
          FDB   H_COUNT
          FDB   WORD
H_CHARW:
          FCB   $04
          FCC   "CHAR"
          FDB   H_WORD
          FDB   CHARW
H_BRACKCHAR:
          FCB   $86
```

```
         FCC   "[CHAR]"
         FDB   H_CHARW
         FDB   BRACKCHAR
H_PARSEW:
         FCB   $05
         FCC   "PARSE"
         FDB   H_BRACKCHAR
         FDB   PARSEW
H_PARSENAME:
         FCB   $0A
         FCC   "PARSE-NAME"
         FDB   H_PARSEW
         FDB   PARSENAME
H_SQUOTE:
         FCB   $82
         FCC   "S"
         FCB   $22        ; '"' - split out of FCC, not escaped within it
         FDB   H_PARSENAME
         FDB   SQUOTE
H_DOTQUOTE:
         FCB   $82
         FCC   "."
         FCB   $22        ; '"' - split out of FCC, not escaped within it
         FDB   H_SQUOTE
         FDB   DOTQUOTE
H_COMPAREW:
         FCB   $07
         FCC   "COMPARE"
         FDB   H_DOTQUOTE
         FDB   COMPAREW
H_SEARCHW:
         FCB   $06
         FCC   "SEARCH"
         FDB   H_COMPAREW
         FDB   SEARCHW
H_DASHTRAILING:
         FCB   $09
         FCC   "-TRAILING"
         FDB   H_SEARCHW
         FDB   DASHTRAILING
H_SLASHSTRING:
         FCB   $07
         FCC   "/STRING"
         FDB   H_DASHTRAILING
         FDB   SLASHSTRING
H_REPLACESW:
         FCB   $08
         FCC   "REPLACES"
         FDB   H_SLASHSTRING
         FDB   REPLACESW
H_SUBSTITUTEW:
         FCB   $0A
         FCC   "SUBSTITUTE"
         FDB   H_REPLACESW
         FDB   SUBSTITUTEW
H_SNAMEW:
         FCB   $05
         FCC   "SNAME"
         FDB   H_SUBSTITUTEW
         FDB   SNAMEW
H_UNESCAPEW:
         FCB   $08
         FCC   "UNESCAPE"
         FDB   H_SNAMEW
         FDB   UNESCAPEW
H_LTNUM:
         FCB   $02
         FCC   "<#"
         FDB   H_UNESCAPEW
         FDB   LTNUM
H_NUMSIGN:
```

```
          FCB   $01
          FCC   "#"
          FDB   H_LTNUM
          FDB   NUMSIGN
H_NUMSIGNS:
          FCB   $02
          FCC   "#S"
          FDB   H_NUMSIGN
          FDB   NUMSIGNS
H_NUMGT:
          FCB   $02
          FCC   "#>"
          FDB   H_NUMSIGNS
          FDB   NUMGT
H_HOLD:
          FCB   $04
          FCC   "HOLD"
          FDB   H_NUMGT
          FDB   HOLD
H_HOLDS:
          FCB   $05
          FCC   "HOLDS"
          FDB   H_HOLD
          FDB   HOLDS
H_SIGN:
          FCB   $04
          FCC   "SIGN"
          FDB   H_HOLDS
          FDB   SIGN
H_DOT:
          FCB   $01
          FCC   "."
          FDB   H_SIGN
          FDB   DOT
H_UDOT:
          FCB   $02
          FCC   "U."
          FDB   H_DOT
          FDB   UDOT
H_DOTR:
          FCB   $02
          FCC   ".R"
          FDB   H_UDOT
          FDB   DOTR
H_UDOTR:
          FCB   $03
          FCC   "U.R"
          FDB   H_DOTR
          FDB   UDOTR
H_QMARK:
          FCB   $01
          FCC   "?"
          FDB   H_UDOTR
          FDB   QMARK
H_DDOT:
          FCB   $02
          FCC   "D."
          FDB   H_QMARK
          FDB   DDOT
H_DDOTR:
          FCB   $03
          FCC   "D.R"
          FDB   H_DDOT
          FDB   DDOTR
H_BASEW:
          FCB   $04
          FCC   "BASE"
          FDB   H_DDOTR
          FDB   BASEW
H_DECIMAL:
          FCB   $07
```

```
          FCC   "DECIMAL"
          FDB   H_BASEW
          FDB   DECIMAL
H_HEXW:
          FCB   $03
          FCC   "HEX"
          FDB   H_DECIMAL
          FDB   HEXW
H_BINARYW:
          FCB   $06
          FCC   "BINARY"
          FDB   H_HEXW
          FDB   BINARYW
H_CATCH:
          FCB   $05
          FCC   "CATCH"
          FDB   H_BINARYW
          FDB   CATCH
H_THROW:
          FCB   $05
          FCC   "THROW"
          FDB   H_CATCH
          FDB   THROW
H_LPAREN:
          FCB   $81
          FCC   "("
          FDB   H_THROW
          FDB   LPAREN
H_BACKSLASH:
          FCB   $81
          FCC   "\"
          FDB   H_LPAREN
          FDB   BACKSLASH
H_ENVQUERY:
          FCB   $0C
          FCC   "ENVIRONMENT?"
          FDB   H_BACKSLASH
          FDB   ENVQUERY
H_SOURCEW:
          FCB   $06
          FCC   "SOURCE"
          FDB   H_ENVQUERY
          FDB   SOURCEW
H_SOURCEID:
          FCB   $09
          FCC   "SOURCE-ID"
          FDB   H_SOURCEW
          FDB   SOURCEID
H_REFILLW:
          FCB   $06
          FCC   "REFILL"
          FDB   H_SOURCEID
          FDB   REFILLW
H_EVALUATEW:
          FCB   $08
          FCC   "EVALUATE"
          FDB   H_REFILLW
          FDB   EVALUATEW
H_TIBW:
          FCB   $03
          FCC   "TIB"
          FDB   H_EVALUATEW
          FDB   TIBW
H_NTIBW:
          FCB   $04
          FCC   "#TIB"
          FDB   H_TIBW
          FDB   NTIBW
H_TOINW:
          FCB   $03
          FCC   ">IN"
```

```
          FDB   H_NTIBW
          FDB   TOINW
H_SPANW:
          FCB   $04
          FCC   "SPAN"
          FDB   H_TOINW
          FDB   SPANW
H_BLW:
          FCB   $02
          FCC   "BL"
          FDB   H_SPANW
          FDB   BLW
H_DOTS:
          FCB   $02
          FCC   ".S"
          FDB   H_BLW
          FDB   DOTS
H_WORDSW:
          FCB   $05
          FCC   "WORDS"
          FDB   H_DOTS
          FDB   WORDSW
H_DUMPW:
          FCB   $04
          FCC   "DUMP"
          FDB   H_WORDSW
          FDB   DUMPW

H_TRUE:
          FCB   $04
          FCC   "TRUE"
          FDB   H_DUMPW
          FDB   TRUEBODY

H_FALSE:
          FCB   $05
          FCC   "FALSE"
          FDB   H_TRUE
          FDB   FALSEBODY

; H_ABORT and H_QUIT are hand-built, not produced by the same
; generation pass as the 217 entries above - ABORT and QUIT need
; to be findable at the prompt, but HEADER/CREATE couldn't be used
; directly for these two (see the source conversation this file
; was derived from for why). Moved here from their own separate
; section so every header in this ROM lives in one contiguous
; block, with the chain visible end to end: H_QUIT -> H_ABORT ->
; H_FALSE (the newest of the 217 generated entries) -> ... ->
; H_KEY -> 0.
H_ABORT: FCB   5
          FCC   "ABORT"
          FDB   H_FALSE        ; resolved - was placeholder 0, then H_DUMPW,
                                ; then H_DOESGT, then H_DUMPW again once
                                ; DOES> moved out of the chain's newest slot;
                                ; now H_FALSE, the chain's newest entry
          FDB   ABORT

H_QUIT:  FCB   4
          FCC   "QUIT"
          FDB   H_ABORT
          FDB   QUIT

BASELATEST EQU  H_QUIT   ; the ROM dictionary's true head - referenced by
                         ; COLD to initialize LATEST. Was QUITHDR before
                         ; the H_QUIT rename; before that, undefined
                         ; entirely - a real bug, not a placeholder,
                         ; found and fixed several turns before this one.

; Verify no collision with base code.
; Value should match ORG BASECODE
BASEDICTEND  EQU   *
```

```
BASEDICTSIZE EQU   BASEDICTEND-BASEDICT
```

9. Alphabetical Index

Every word implemented, alphabetically, with the page in the Glossary (Section 3) where its full entry appears. Page numbers are live Word fields — update fields (Ctrl+A, F9) if they show as blank or "?" after edits.